

FACULTY OF ENGINEERING

DEGREE OF BACHELOR OF ENGINEERING IN

BIOMEDICAL ENGINEERING

DEPARTMENT OF BIOMEDICAL ENGINEERING
(REGULAR PROGRAMME)

CURRICULUM & SYLLABI
(2022-2023)



KARPAGAM ACADEMY OF HIGHER EDUCATION

(Deemed to be University)

(Established Under Section 3 of UGC Act 1956)

(Accredited with A⁺ Grade by NAAC in the Second Cycle)

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FACULTY OF ENGINEERING
DEGREE OF BACHELOR OF ENGINEERING /TECHNOLOGY REGULAR PROGRAMME
REGULATIONS 2022
CHOICE BASED CREDIT SYSTEM

These regulations are effective from the academic year 2022–2023 and applicable to the candidates admitted to B. E. /B. Tech. during 2022-2023 and onwards.

1. ADMISSION

1.1 Candidates seeking admission to the first semester of the eight semesters B.E./B. Tech Degree Programme:

Should have passed the Higher Secondary Examination (10+2) prescribed by the State Government /Central Government with Mathematics/ Physics / Chemistry / Computer Science /Electronics / Information Technology / Biology / Informatics Practices / Biotechnology / Technical Vocational subject / Agriculture / Engineering Graphics/ Business Studies/ Entrepreneurship. (Any of the above three subjects) or any similar Examination of any other institution/ University or authority accepted by the Karpagam Academy of Higher Education as equivalent there to).

Should obtained at least 45% marks (40% marks in case of candidates belonging to reserved category) in the above subjects taken together.

(OR)

Passed min. 3 years Diploma examination with at least 45%marks (40% marks in case of candidates belonging to reserved category) subject to vacancies in the First Year, in case the vacancies at lateral entry are exhausted. (The Universities will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to achieve desired learning outcomes of the programme)

1.2 Lateral Entry Admission

Candidates who possess Diploma in Engineering / Technology (10+3 or 10+2+2) awarded by the Directorate of Technical Education with passed minimum THREE years / TWO years (Lateral Entry) Diploma examination with at least 45% marks (40% marks in case of candidates belonging to reserved category) in ANY branch of Engineering and Technology are eligible to apply for admission to the third semester of B.E. / B.Tech.. Such candidates shall undergo two additional engineering subjects in the 3rd and 4th semester as prescribed by the University.

OR

Passed B.Sc. Degree from a recognized University as defined by UGC, with at least 45% marks (40% marks in case of candidates belonging to reserved category) and passed 10+2 examination with Mathematics as a subject.

OR

Passed D. Voc. Stream in the same or allied sector.

(The Universities will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to achieve desired learning outcomes of the programme)

Eligibility criteria for admission in the third semester is given in the table below.

S.No.	Programme	Eligibility criteria
1.	B.E Bio Medical Engineering	Passed Minimum THREE years / TWO years (Lateral Entry) Diploma examination with atleast 45% marks (40% marks in case of candidates belonging to reserved category) in ANY branch of Engineering and Technology. OR Passed B.Sc. Degree from a recognized University as defined by UGC, with at least 45% marks (40% marks in case of candidates belonging to reserved category) and passed 10+2 examination with Mathematics as a subject. OR Passed D. Voc. Stream in the same or allied sector. (The Universities will offer suitable bridge courses such as Mathematics, Physics, Engineering drawing, etc., for the students coming from diverse backgrounds to achieve desired learning outcomes of the programme)
2	B.E. Civil Engineering	
3.	B.E. Computer Science and Design	
4.	B.E. Computer Science and Engineering	
5.	B.E. Electrical and Electronics Engineering	
6.	B. E. Electronics and Communications Engineering	
7.	B.E. Mechanical Engineering	
8.	B.Tech. Artificial Intelligence and Data Science	
9.	B. Tech Bio-Technology	
10.	B. Tech Food Technology	

1.3 Migration from other University

Candidates who have completed their first to sixth semesters of B. E./B. Tech. study in any University are eligible to apply for admission to their next semester of B. E./B. Tech. in the branch corresponding to their branch of study. The student will be exempted from appearing for Examination of the equivalent courses passed in the earlier programme and will have to appear for courses which he/she has not done during the period of his/her earlier programme. Along with the request letter and mark sheets, he/she has to submit a copy of syllabus of the programme duly attested by the Registrar, Competent authority, he / she has undergone. Equivalence Certificate shall be provided by the “Students’ Affairs Committee” of Karpagam Academy of Higher Education. Students’ Affairs Committee comprises all the Heads of the Departments and Dean of the Faculty of Engineering and a nominee of the Registrar.

2.PROGRAMMES OFFERED

A candidate may undergo a programme in any one of the branches of study approved by the University as given below.

List of B.E. and B.Tech. Degree Programmes

1. B.E Bio Medical Engineering
2. B.E. Civil Engineering

3. B.E. Computer Science and Design
4. B.E. Computer Science and Engineering
5. B.E. Electrical and Electronics Engineering
6. B.E. Electronics and Communications Engineering
7. B.E. Mechanical Engineering
8. B.Tech. Artificial Intelligence and Data Science
9. B.Tech. Bio-Technology
10. B.Tech. Food Technology

3. MODE OF STUDY

3.1 Full-Time:

In this mode of study, the candidates are required to attend classes regularly on the specified working days of the University.

3.2 Conversion from full time mode of study to part time is not permitted.

3.3 Change from one programme to another is not permitted.

4. STRUCTURE OF PROGRAMMES

4.1 Every programme will have a curriculum with syllabus consisting of theory and practical courses such as:

- (i) General core courses comprising Mathematics, Basic Sciences, Engineering Sciences and Humanities.
- (ii) Core courses of Engineering/Technology.
- (iii) Elective courses for specialization in related fields.
- (iv) Workshop practice, computer practice, engineering graphics, laboratory work, in-plant training, seminar presentation, project work, industrial visits, camps, etc.

Every student is encouraged to participate in at least any one of the following programmes

- NSS/Sports/Physical exercise/NCC/YRC/Red Ribbon club/Environment club and Energy club
- Other Co-Curricular and Extra Curricular activities

(v) Choice Based Credit System

CBCS is introduced for students admitted in the academic year 2017-18. As per AICTE guidelines, CBCS is an approach in which students opt for courses of their choice. CBCS provides greater flexibility with multiple courses and enable students to undergo additional courses. CBCS is applicable to Full Time Undergraduate & Post Graduate Programmes of study. It provides a choice for students to select from the prescribed courses (Professional soft core, Professional Hard core, Professional Electives, Open Electives, Value added courses, Humanity Sciences, Basic sciences & Engineering sciences). A course designated as hard core for a particular programme of study must

invariably be completed by the student to receive the degree in the programme. The Hard core courses cannot be substituted by another courses. Students can exercise their choice among a set of Soft core courses from the list of Soft core courses specified for each Programme of study. The student should meet the criteria for prerequisites to become eligible to register for that course. The student should request for the course for every semester within the first week of semester. Maximum no of students to be registered in each course shall depend on availability of physical facilities, classroom availability and lab capacity. Registration of already requested courses by students in previous semester is not allowed.

4.2 Each course is normally assigned certain number of credits.

No. of credits per lecture period per week	1
No. of credits per tutorial period per week	1
No. of credits for 3 periods of laboratory course per week	2
No. of credits for 3 periods of project work per week	2
No. of credits for 2 periods of Value added course per week	1
No. of credits for 3 weeks of in-plant training during	1
Semester vacations	1

4.3 In every semester, the curriculum shall normally have a blend of theory courses not exceeding 6 and practical courses not exceeding 3. However, the total number of courses per semester shall not exceed 8.

4.4 The prescribed credits required for the award of the degree shall be within the limits specified below.

PROGRAMME	PRESCRIBED CREDIT RANGE
B.E./B.Tech.	160– 165

4.5 The medium of instruction for all Courses, Examinations, Seminar presentations and Project/Thesis reports is English.

4.6 Value Added Course

Besides core courses and elective courses, value added course is introduced. The blend of different courses is so designed that the student would be trained not only in his / her relevant professional field but also as a socially conscious human being.

4.7 Evaluation in the courses comprises two parts, one is the Continuous Internal Assessment (CIA) and the other one is the End Semester Examination (ESE). Evaluation in few courses may be by Internal Assessment only.

5. DURATION OF THE PROGRAMME

5.1 The prescribed duration of the programme shall be

Programme	Min. No. of semesters	Max. No. of semesters
B.E./B. Tech.(H. Sc. Candidates)	8	14
B.E./B.Tech. (Lateral Entry Candidates)	6	12

5.2 Each semester shall normally consist of 90 working days or 540 hours.

5.3 Additional classes for improvement, conduct of model test, etc., over and above the specified periods shall be arranged, if required. But for the purpose of calculation of attendance requirement for eligibility to appear for the end semester Examinations (as per Clause 11) by the students, 540 hours conducted within the specified academic schedule alone shall be taken into account and the overall percentage of attendance shall be calculated accordingly.

6. REQUIREMENTS FOR COMPLETION OF THE SEMESTER

6.1 Ideally every student is expected to attend all classes and secure 100% attendance. However, in order to allow for certain unavoidable circumstances, the student is expected to attend at least 75% of the classes and the conduct of the candidate has been satisfactory during the course.

6.2 A candidate who has secured attendance between 65% and 74% (both included), due to medical reasons (Hospitalization / Accident / Specific Illness) or due to participation in University / District / State / National / International level sports or due to participation in Seminar / Conference / Workshop / Training Programme / Voluntary Service / Extension activities or similar programmes with prior permission from the Registrar shall be given exemption from prescribed attendance requirements and shall be permitted to appear for the Examination on the recommendation of the Head of the Department concerned and Dean to condone the lack of attendance. The Head of the Department has to verify and certify the genuineness of the case before recommending to the Dean. However, the candidate has to pay prescribed condonation fees.

6.3 Candidates who are not recommended for condonation and those who have less than 65% attendance will not be permitted to proceed to the next semester and have to redo the course. However, they are permitted to write the arrear Examinations, if any.

7. CLASS ADVISOR

To help the students in planning their courses of study and for general advice on the academic programme, the Head of the Department will attach a certain number of students to a teacher of the Department who shall function as Class Advisor for those students throughout their period of study. Such Class Advisors shall advise the students and monitor the courses undergone by the students, check the attendance and progress of the students and counsel them periodically. If necessary, the Class Advisor may display the cumulative attendance particulars in the Department notice board and also discuss with or inform the Parents/Guardian about the progress of the students. Each student shall be provided with course plan for each course at the beginning of each semester.

8. CLASS COMMITTEE

8.1. Every class shall have a class committee consisting of teachers of the class concerned, Maximum of six student representatives [boys and girls] and the concerned Head of the Department. It is like the 'Quality Circle' with the overall goal of improving the teaching-learning process. The functions of the class committee include

- Clarifying the regulations of the degree programme and the details of rules there in particularly Clause 4 and 5 which should be displayed on Department Notice-Board.
- Informing the student representatives, the details of Regulations regarding weight age used for each assessment. In the case of practical courses (laboratory / drawing / project work /seminar, etc.) the breakup of marks for each experiment / exercise /module of work, should be clearly discussed in the class committee meeting and informed to the students.
- Solving problems experienced by students in the classroom and in the laboratories.
- Informing the student representatives, the academic schedule, including the dates of assessments and the syllabus coverage for each assessment.
- Analyzing the performance of the students of the class after each test and finding the ways and means of solving problems, if any.
- Identifying the weak students, if any and requesting the teachers concerned to provide some additional academic support.

8.2 The class committee for a class under a particular branch is normally constituted by the Head of the Department. However, if the students of different branches are mixed in a class (like the first semester which is generally common to all branches), the class committee is to be constituted by the Dean.

8.3 The class committee shall be constituted within the first week of each semester.

8.4 The Chairperson of the Class Committee may convene the meeting of the class committee.

8.5 The Dean may participate in any Class Committee of the Faculty.

8.6 The Chairperson is required to prepare the minutes of every meeting, submit the same to Dean through the HOD within two days of the meeting and arrange to circulate it among the students and teachers concerned. If there are some points in the minutes requiring action by the Management, the same shall be brought to the notice of the Registrar by the HOD through Dean.

8.7 The first meeting of the Class Committee shall be held within one week from the date of commencement of the semester, in order to inform the students about the nature and weightage of assessments within the framework of the regulations. Two subsequent meetings may be held in a semester at suitable intervals. During these meetings the student members representing the entire class, shall meaning fully interact and express their opinions and suggestions of the other students of the class in order to improve the effectiveness of the teaching-learning process.

9. COURSE COMMITTEE FOR COMMON COURSES

Each common theory course offered to more than one discipline or group, shall have a “Course Committee” comprising all the teachers handling the common course with one of the nominated as Course Coordinator. The nomination of the Course Coordinator shall be made by the Dean depending upon whether all the teachers teaching the common course belong to a single department or to several departments. The ‘Course Committee’ shall meet to arrive at a common scheme of evaluation for the test and shall ensure a uniform evaluation of the tests. Where ever feasible, the Course Committee may also prepare a common question paper for the Internal Assessment test (s).

10. PROCEDURE FOR AWARDING MARKS FOR INTERNAL ASSESSMENT

10.1 Every teacher is required to maintain an 'ATTENDANCE AND ASSESSMENT RECORD'(Log book) which consists of attendance marked in each theory or practical or project work class, the test marks and the record of classwork (topic covered), separately for each course.

10.2 Continuous Internal Assessment (CIA): The performance of students in each subject will be continuously assessed by the respective teachers as per the guidelines given below:

THEORY COURSES:

S.No.	CATEGORY	MAXIMUM MARKS
1.	Assignment	5
2.	Seminar*	5
3.	Attendance	5
4.	Test –I	8
5.	Test –II	8
6.	Test–III	9
Continuous Internal Assessment: TOTAL		40

*Evaluation shall be made by a committee.

PATTERN OF TEST QUESTION PAPER (Test I & II)

INSTRUCTION	REMARKS
Maximum Marks	60
Duration	2Hours
Part-A	1to9 Two Mark Questions, uniformly covering the two units of the syllabus. All the 9 Questions are to be answered. (9x2=18Marks).
Part-B	Question10to12willbeofeitherortype, covering two units of the syllabus. Each Question may have subdivision. (3x14 =42Marks).

PATTERN OF TEST QUESTION PAPER (Test III)

INSTRUCTION	REMARKS
Maximum Marks	100
Duration	3Hours
Part-A	Part A will be online Examination. 20 Objective type Questions, Covering all the 5 units. (20x1=20Marks) (Online Examination).
Part-B	21 to 25 Two Mark Questions, uniformly covering the Five units of the syllabus. All the 5 Questions are to be answered. (5x2=10Marks).
Part-C	Question 26 to 30 will be of either or type, covering Five units of the syllabus. Each Question may have subdivision. (5x14=70Marks).

PRACTICAL COURSES:

S.No	CATEGORY	MAXIMUM MARKS
1.	Attendance	5
2.	Observation work	5
3.	Record work	5
4.	Model Examination	15
5.	Viva-Voce[Comprehensive]	10
Continuous Internal Assessment: TOTAL		40

Every practical exercise / experiment shall be evaluated based on the conduct of exercise/experiment and records maintained.

INTEGRATED THEORY AND PRACTICAL COURSES:

The Continuous Internal Assessment for Integrated Theory Course is awarded for 40 Marks with mark split up similar to regular theory course.

The external evaluation of integrated practical component from End semester Examination by internal mode is awarded for 50 Marks and later scaled down to 15 Marks and similarly the external evaluation for integrated theory from End semester Examination is awarded for 100 Marks and later scaled down to 45 Marks. Hence the external assessment for Integrated theory and practical components contribute to 60 Marks.

10.3 ATTENDANCE

Attendance carries a maximum of 5 marks and the distribution is as under:

S.No.	Attendance%	Marks
1	91andabove	5.0
2	81-90	4.0
3	76-80	3.0

10.4 PROJECTWORK / INTERNSHIPS:

Final year project work will be always in-house. However, as a special case, if a student is able to get a project from a government organization or private or public sector company, the student may be permitted to do his/her project work in reputed institution/research organization/industry. Hence, final year students may have commencement of eighth semester classes for 30 days in fast track mode and complete their final semester and are made eligible for undergoing Internships in Industry and also interested students are permitted for doing projects in Industries.

10.5 CERTIFICATION COURSES:

Students have to undergo a minimum of one value added course beyond curriculum as a certified course per semester for duration not less than 30 hours.

11. REQUIREMENTS FOR APPEARING FOR END SEMESTER EXAMINATION(ESE)

A candidate shall normally be permitted to appear for the ESE of any semester commencing from I semester if he/she has satisfied the semester completion requirements (Subject to Clause 5) and has registered for Examination in all courses of the semester. Registration is mandatory for Semester Examinations as well as arrear Examinations failing which the candidate will not be permitted to attend the next semester. A candidate already appeared for a subject in a semester and passed the Examination is not entitled to reappear in the same subject of the semester for improvement of grade.

12. END SEMESTER EXAMINATION

ESE will be held at the end of each semester for each subject, for 100 marks, later scaled down to 60 marks.

PATTERN OF ESE QUESTION PAPER:

INSTRUCTION	REMARKS
Maximum Marks	100
Duration	3Hours
Part-A	Part A will be online Examination. 200 objective type Questions. Covering all the 5 units. 20*1= 20 Marks (Online Examination)
Part-B	21 to 25 Two Mark Questions, uniformly covering the Five units of the syllabus. All the 5 Questions are to be answered. (5*2=10 Marks).
Part-C	Question 26 to 30 will be of either or type covering Five units of the syllabus. Each Question may have subdivision. (5*14=70 Marks)

13. PASSING REQUIREMENTS

13.1 Passing minimum: The passing minimum for CIA is 20 (i.e. out of 40 marks). The passing minimum for ESE is 30 (i.e. out of 60 marks). The overall passing minimum for theory/laboratory course is 50 (Sum of his/her score in CIA and ESE) out of 100 marks.

13.1.1 The passing minimum for value added course is 50 marks out of 100 marks. There will be two tests, the first covering 50% of syllabus for 50 marks and the other for 50 marks.

13.2 If the candidate fails to secure a pass in a particular course ESE, it is mandatory that candidate shall register and reappear for the Examination in that course during the subsequent semester when Examination is conducted in that course. Further the candidate should continue to register and reappear for the Examination till a pass is secured in such supplementary Examination within the stipulated maximum duration of the programme (Clause 5.1).

The CIA marks obtained by the candidate in his/her first or subsequent appearance where he/she secures a pass shall be retained by the office of the Controller of Examinations and considered valid for all remaining attempts till the candidate secures a pass in his/her ESE.

13.3 If the candidate fails to secure a pass in a particular course in CIA, it is mandatory that candidate shall register and reappear for the CIA in that course during the subsequent semester when CIA is conducted in that course by the faculty member assigned for that particular course during that semester by the concerned HOD. Further, the candidate should continue to register and reappear for the CIA till a pass is secured in such subsequent Examination within the stipulated maximum duration of the programme (Clause 5.1).

13.3.1 If a candidate fails to secure a pass in value added course, he/she has to appear for the tests when course is conducted subsequently.

13.4 ONLINE COURSE (MOOC) COORDINATOR

To help students in planning the iron line courses and for general advice on online courses, the HOD shall nominate a MOOC coordinator for the online courses. The Online course MOOC coordinator shall identify the courses which students can select for their programme from the available online courses offered by the different agencies periodically and inform the same to the students. Further, the coordinator shall advice the students regarding the online courses and monitor their course.

13.4.1 Student Shall study atleast one online course from Swayam /NPTEL in anyone of the first seven semesters for which examination shall be conducted at the end of the course by the respective organization body. The student can register to the course which are approved by the department. The student shall produce a pass certificate from the respective body before the end of the seventh semester.

14. AWARD OF LETTER GRADES

14.1 All assessments of a course will be done on absolute mark basis. However, for the purpose of reporting the performance of a candidate letter grades, each carrying certain number of points will be awarded as per the range of total marks (out of 100) obtained by the candidate in each subject as detailed below:

Letter grade	Marks Range	Grade Point	Description
O	91 -100	10	OUTSTANDING
A+	81-90	9	EXCELLENT
A	71-80	8	VERYGOOD
B+	66-70	7	GOOD
B	61 – 65	6	ABOVEAVERAGE
C	55 -60	5	AVERAGE
D	50 -54	4	PASS
RA	<50	-	REAPPEARANCE
AB		0	ABSENT

14.2 GRADE SHEET

After results are declared, Grade sheet will be issued to each student which will contain the following details:

- The list of courses enrolled during the semester and the grade scored,
- The Grade Point Average (**GPA**)for the semester

- iii. The Cumulative Grade Point Average (**CGPA**) of all courses enrolled from first semester onwards.

GPA is the ratio of the sum of the products of the number of Credits (**C**) of courses enrolled and the Grade Points (**GP**) corresponding to the grades scored in those courses, taken for all the courses to the sum of the number of credits of all the courses in the semester.

$$\text{GPA} = \frac{\text{Sum of [C*GP]}}{\text{Sum of C}}$$

CGPA will be calculated in a similar manner, considering all the courses enrolled from First semester. **RA** grade and value added course will be excluded for calculating **GPA** and **CGPA**.

14.3 REVALUATION

Revaluation and Re-totaling is allowed on representation. A candidate can apply for revaluation of his/her semester Examination answer paper in a theory course, within 2 weeks from the declaration of results, on payment of a prescribed fee through proper application to the Controller of Examinations through the Head of the Department and Dean. A candidate can apply for revaluation of answer scripts for not exceeding 5 subjects at a time. The Controller of Examinations will arrange for the revaluation and the results will be intimated to the candidate through the Head of the Department and Dean. Revaluation is not permitted for Supplementary Examinations, Practical Examinations, Technical Seminars, In-plant Training and Project Work.

14.4 TRANSPARENCY AND GRIEVANCE COMMITTEE

A student may get the Photostat copy of the answer script on payment of prescribed fee, if he/she wishes. The students can represent the grievance, if any, to the Grievance Committee, which consists of Dean of the Faculty, (if Dean is HOD, the Dean of another Faculty nominated by the University), HOD of the Department concerned, the faculty of the course and Dean from other discipline nominated by the University and the COE. If the Committee feels that the grievance is genuine, the script may be sent for external valuation; the marks awarded by the External Examiner will be final. The student has to pay prescribed fee for the same.

15. ELIGIBILITY FOR AWARD OF DEGREE

A student shall be declared to be eligible for award of Degree if he/she has

- Successfully gained the required number of total credits as specified in the curriculum corresponding to his/her programme within the stipulated time.
- No disciplinary action is pending against him/her.

The award of the degree must be approved by the Board of Management of Karpagam Academy of Higher Education.

16. CLASSIFICATION OF THE DEGREE AWARDED

- 16.1** A candidate who qualifies for the award of the Degree (vide Clause 15) having passed the Examination in all the courses in his/her first appearance within the specified minimum number of semesters (vide Clause 5.1) securing a CGPA of not less than **8** shall be declared to have passed the Examination in First Class with Distinction.
- 16.2** A regular candidate or a lateral entrant is eligible to register for BE (Honors), B.Tech. (Honors). If, he/she has passed all the courses in the first appearance and holds / maintains a CGPA of 7.5 at VI Semester. He / she has to take an additional 20 credits by studying online courses through Swayam/ NPTEL. Such a candidate is eligible for the award of BE(Honor), B.Tech.(Honor). However, if he/she fails in securing 20 additional credits but maintains CGPA of 7.5 and above is not eligible for Honors degree but eligible for First class with Distinction.
- 16.3** A candidate who qualifies for the award of the Degree (vide Clause 15) having passed the Examination in all the courses within the specified minimum number of semesters (vide Clause 5.1) plus one year (two semesters), securing CGPA of not less than **6.5** shall be declared to have passed the Examination in First Class.
- 16.3** All other candidates (not covered in Clauses 17.1 and 17.2) who qualify for the award of the degree (vide Clause 15) shall be declared to have passed the Examination in Second Class.

17. PROVISION FOR WITHDRAWAL FROM END-SEMESTER EXAMINATION

- 17.1** A candidate may for valid reasons and on prior application, be granted permission to Withdraw from appearing for the examination of any one course or consecutive examinations of more than one course in a semester examination.
- 17.2** Such withdrawal shall be permitted only once during the entire duration of the degree programme. Withdrawal application shall be valid only if the candidate is otherwise eligible to write the Examination
- 17.3** Withdrawal application is valid only if it is made within 10 days prior to the commencement of the Examination in that course or courses and recommended by the Head of the Department, Dean and approved by the Registrar.
- 17.3.1** Notwithstanding the requirement of mandatory TEN days' notice, applications for withdrawal for special cases under extraordinary conditions may be considered on the merit of the case.
- 17.4** Withdrawal shall not be construed as an appearance for the eligibility of a candidate for First Class with Distinction. This provision is not applicable to those who seek withdrawal during III semester.
- 17.5** Withdrawal from the ESE is NOT applicable to arrear Examinations.
- 17.6** The candidate shall reappear for the withdrawn courses during the Examination conducted in the subsequent semester.

18. PROVISION FOR AUTHORISED BREAK OF STUDY

18.1 Break of Study shall be granted only once for valid reasons for a maximum of one year during the entire period of study of the degree programme. However, in extraordinary situation the candidate may apply for additional break of study not exceeding another one year by paying prescribed fee for break of study. If a candidate intends to temporarily discontinue the programme in the middle of the semester for valid reasons and to rejoin the programme in a subsequent year, permission may be granted based on the merits of the case provided he/she applies to the Registrar, through the Head of the Department and Dean stating reasons thereof and the probable date of rejoining the programme.

18.2 The total number of semesters for completion of the programme from the commencement of the first semester to which the candidate was admitted shall not exceed the maximum no. of semesters specified in Clause 5.1 irrespective of the period of break of study (vide Clause 18) in order that he/she may be eligible for the award of the degree (vide Clause 15). The candidate thus permitted to rejoin the programme at the commencement of the semester after the break shall be governed by the curriculum and regulations in force at the time of rejoining. Such candidates may have to do additional courses as per the curriculum and regulations in force at that period of time.

18.3 The authorized break of study (for a maximum of one year) will not be counted for the duration specified for passing all the courses for the purpose of classification (vide Clause 17). However, additional break of study granted will be counted for the purpose of classification.

18.4 The total period for completion of the programme reckoned from, the commencement of the first semester to which the candidate was admitted shall not exceed the maximum period specified in Clause 5.1 irrespective of the period of break of study (vide Clause 18.3) in order that he/she may be eligible for the award of the degree.

18.5 If any student is detained for want of requisite attendance, progress and good conduct, the period spent in that semester shall not be considered as permitted 'Withdrawal' or 'Break of Study' (Clause 18 and 18 respectively).

19. SUPPLEMENTARY ESE: After the publication of VIII semester results, if a student has **ONE** arrear in any theory course of the entire programme, he/she will be permitted to apply within 15 days of the publication of results, and appear for supplementary Examination.

20. INDUSTRIAL VISIT

Every student is required to undergo one industrial visit for every semester, starting from the third semester of the programme.

21. DISCIPLINE

Every student is required to observe discipline and decorous behavior both inside and outside the University and not to indulge in any activity which will tend to bring down the prestige of the University. The erring student will be referred to the Disciplinary Committee constituted by the University, to enquire into acts of indiscipline and recommend to the University about the disciplinary action to be taken.

If a student indulges in malpractice in any of the ESE/ CIA he/ she shall be liable for punitive action as prescribed by the University from time to time.

22. REVISION OF REGULATION AND CURRICULUM

The University may from time to time revise, amend or change the Regulations, Scheme of Examinations and syllabi, if found necessary on there commendations of Board of Studies, Academic Council and Board of Management of Karpagam Academy of Higher Education.

B.E. BIOMEDICAL ENGINEERING

PEOs	PROGRAM EDUCATIONAL OBJECTIVES
PEO 1	• Develop the ability to work in a multidisciplinary environment by acquiring knowledge in mathematics, engineering and medical science.
PEO 2	• Establish as professionals in design, development, implementation and testing biomedical devices ensuring reliability and patient safety.
PEO 3	• Engage in lifelong learning and adhere to ethical values through professional development and to pursue advance degree program in engineering and business.

POs	PROGRAM OUTCOMES (POs)
PO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.
PO4	Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7	Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PSOs	PROGRAM SPECIFIC OUTCOMES (PSOs)
PSO1	Design and development of bio-electronic systems for biosignal & medical image acquisition, diagnostics and therapeutics
PSO2	To Apply information and communication technologies and software tools for analyzing and interpreting data to generate report.



KARPAGAM ACADEMY OF HIGHER EDUCATION
(Deemed to be University Established Under Section 3 of UGC Act 1956)
FACULTY OF ENGINEERING
B.E (BIOMEDICAL ENGINEERING)
COURSE OF STUDY AND SCHEME OF EXAMINATION
(2022 BATCH ONWARDS)

SEMESTER-I

Course Code	Course Title	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BECC101	English	HS	2	0	2	3	40	60	100	1
22BECC102	Mathematics –I	BS	3	1	0	4	40	60	100	3
22BECC141	Engineering physics	BS	3	1	2	5	40	60	100	5
22BECC142	Engineering chemistry	BS	3	0	4	5	40	60	100	8
22BECC143	Python programming	ES	2	0	2	3	40	60	100	12
22BEBME144	Basic Electrical & Electronics Engineering	ES	3	1	2	5	40	60	100	16
SEMESTER TOTAL			16	3	12	25	240	360	600	

SEMESTER-II

Course Code	Course Title	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BECC201	Communicative English	HS	2	0	2	3	40	60	100	19
22BECC202	Mathematics-II	BS	3	1	0	4	40	60	100	21
22BEBME203	Medical Physics	HS	3	0	0	3	40	60	100	24
22BECC204	Environmental studies	ES	3	0	0	3	40	60	100	27
22BEBME205	Introduction to Biomedical Engineering	PC	3	0	0	3	40	60	100	30

22BEBME206	Electronic Devices and Circuits	ES	3	0	0	3	40	60	100	32
22BEBME211	Engineering Graphics	ES	1	0	4	3	40	60	100	34
MANDATORY COURSE										
22BEBME251	Disaster Management	MC	2	0	0	0	100	0	100	36
SEMESTER TOTAL			20	1	6	22	380	420	800	

SEMESTER-III

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
THEORY										
22BEBME301	Mathematics – III (Linear Algebra and Partial Differential Equations)	BS	3	1	0	4	40	60	100	38
22BEBME302	Human Anatomy and Physiology	PC	3	0	0	3	40	60	100	40
22BEBME303	Analog and Digital Communication	PC	3	0	0	3	40	60	100	42
22BEBME304	Biomaterials	PC	3	0	0	3	40	60	100	44
22BEBME305	Medical Biochemistry	PC	3	0	0	3	40	60	100	46
22BEBME346	C++ and DataStructures	ES	3	0	2	5	40	60	100	48
PRACTICALS										
22BEBME311	Biochemistry & Human Physiology Laboratory	PC	0	0	2	1	40	60	100	51
MANDATORYCOURSE										
22BEBME351	Indian constitution	MC	2	0	0	0	100	0	100	52
SEMESTER TOTAL			20	1	4	22	380	420	800	

SEMESTER-IV

Course Code	Course Title	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
THEORY										
22BEBME401	Probability and Statistics	BS	3	1	0	4	40	60	100	54
22BEBME402	Modeling of Physiology system	PC	3	0	0	3	40	60	100	56
22BEBME403	Biosensors and Transducers	PC	3	0	0	3	40	60	100	58
22BEBME404	Microprocessor and Microcontroller	PC	3	0	0	3	40	60	100	60
22BEBME405	Signals and systems	PC	3	0	0	3	40	60	100	62
22BEBME406	Pathology and Microbiology	PC	3	0	0	3	40	60	100	64
PRACTICALS										
22BEBME411	Biosensors and Transducers Laboratory	PC	0	0	2	1	40	60	100	66
22BEBME412	Microprocessor and Microcontroller Laboratory	PC	0	0	2	1	40	60	100	67
22BEBME413	Pathology and Microbiology Laboratory	PC	0	0	2	1	40	60	100	68
MANDATORY COURSE										
22BEBME451	Soft skills –Verbal &Reasoning	MC	1	0	0	0	100	0	100	69
INTERNSHIP										
22BEBME491	Internship	MC	0	0	0	2	100	0	100	70
SEMESTER TOTAL			19	1	6	24	560	540	1100	

SEMESTER-V

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
THEORY										
22BEBME501	Telemedicine	PC	3	0	0	3	40	60	100	71
22BEBME502	Biomedical Instrumentation	PC	3	0	0	3	40	60	100	73
22BEBME503	Biomedical Signal Processing	PC	3	1	0	4	40	60	100	75
22BEBME504	Professional Ethics and Entrepreneurship development	HS	3	0	0	3	40	60	100	77
21BEBME505	Artificial Intelligence in Medicine	PC	3	0	0	3	40	60	100	79
22BEBME5E--	Professional Elective-I	PE	3	0	0	3	40	60	100	81
PRACTICALS										
22BEBME511	Biomedical Instrumentation Laboratory	PC	0	0	2	1	40	60	100	82
22BEBME512	Biomedical Signal Processing Laboratory	PC	0	0	2	1	40	60	100	83
MANDATORYCOURSE										
22BEBME551	Mini Project	MC	0	0	2	0	100	0	100	84
SEMESTER TOTAL			18	1	6	21	420	480	900	

SEMESTER-VI

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
THEORY										
22BEBME601	Biomedical Image processing	PC	3	0	0	3	40	60	100	85
22BEBME602	Biomechanics	PC	3	1	0	4	40	60	100	87
22BEBME603	Diagnostic and Therapeutic Equipment	PC	3	0	0	3	40	60	100	90
22BEBME604	Hospital Management	PC	3	0	0	3	40	60	100	92
22BEBME605	Rehabilitation Engineering	PC	3	0	0	3	40	60	100	95
22BEBME6E- -	Professional Elective-II	PE	3	0	0	3	40	60	100	98
PRACTICALS										
22BEBME611	Biomedical Image Processing Laboratory	PC	0	0	2	1	40	60	100	99
22BEBME612	Diagnostic and Therapeutic Equipment Laboratory	PC	0	0	2	1	40	60	100	100
INTERNSHIP										
22BEBME691	Internship	MC	0	0	0	2	100	0	100	101
SEMESTER TOTAL			18	1	4	23	420	480	900	

SEMESTER-VII

Course Code	Course Title	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
THEORY										
22BEBME701	Artificial Organs and Implants	PC	3	0	0	3	40	60	100	102
22BEBME702	Medical Regulatory System	PC	3	0	0	3	40	60	100	104
22BEBME7E_—	Professional Elective-III	PE	3	0	0	3	40	60	100	106
22BEBME7E_—	Professional Elective-IV	PE	3	0	0	3	40	60	100	106
22BEBME7OE—	Open Elective-I	OE	3	0	0	3	40	60	100	106
PRACTICALS										
22BEBME711	Bio-Data analytics using R Laboratory	PC	0	0	2	1	40	60	100	107
MANDATORYCOURSE										
22BEBME751	Universal Human Values	MC	1	0	0	0	100	0	100	108
PROJECT WORK										
22BEBME791	Project WorkPhase-I	PW	0	0	8	4	100	0	100	110
SEMESTER TOTAL			16	0	10	20	440	360	800	

SEMESTER-VIII

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
THEORY										
22BEBME8E- -	Professional Elective-V	PE	3	0	0	3	40	60	100	111
22BEBME8OE --	Open Elective-II	OE	3	0	0	3	40	60	100	111
PROJECT										
22BEBME891	Project Work Phase-II andviva-voce	PW	0	0	16	8	120	180	300	112
SEMESTER TOTAL			6	0	16	14	200	300	500	
PROGRAMME TOTAL			149	8	64	171	3040	3280	6400	

LIST OF ELECTIVES
PROFESSIONAL ELECTIVES

Professional Electives are offered, based on three different domains such as Signal & Image processing, IoT & Wearable systems and Assistive & Rehabilitation Technology

S.No	Name of the Courses	Domain
1.	Bioergonomics	Domain –I Signal and Image Processing
2.	Biomedical Transport Phenomena	
3.	Biometric Systems	
4.	Medical Imaging Techniques	
5.	Medical Data Analytics	
6.	Biophotonics	
7.	Medical Informatics	
8.	Electronics for Sensor Design	Domain-II IoT & Wearable Systems
9.	BioMEMS	
10.	Embedded System in Medical devices	
11.	Internet of things in Medicine	
12.	Lasers and Fiber Optics in Medicine	
13.	Robotics and Automation in Medicine	
14.	Wearable Systems	
15.	Biotech Prosthetic Equipment	Domain -III Assistive & Rehabilitation Technology
16.	Design of Medical Implants and devices	
17.	Brain Computer Interface and its applications	
18.	Speech Processing	
19.	Virtual Reality and Augmented Reality	
20.	Nanotechnology in Medicine	

Elective I

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BEBME5E01	Bioergonomics	PE	3	0	0	3	40	60	100	113
22BEBME5E02	Electronics For Sensor Design	PE	3	0	0	3	40	60	100	115
22BEBME5E03	Medical Data Analytics	PE	3	0	0	3	40	60	100	117
22BEBME5E04	Brain Computer Interface and its Applications	PE	3	0	0	3	40	60	100	119

Elective II

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BEBME6E01	Biophotonics	PE	3	0	0	3	40	60	100	121
22BEBME6E02	Medical Informatics	PE	3	0	0	3	40	60	100	123
22BEBME6E03	BIO MEMS	PE	3	0	0	3	40	60	100	125
22BEBME6E04	Wearable Systems	PE	3	0	0	3	40	60	100	127

Elective III

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BEBME7E01	Biometric Systems	PE	3	0	0	3	40	60	100	129
22BEBME7E02	Biomedical Transport Phenomena	PE	3	0	0	3	40	60	100	132
22BEBME7E03	Lasers And Fiber Optics In Medicine	PE	3	0	0	3	40	60	100	134
22BEBME7E04	Bio Tech Prosthetic Equipment	PE	3	0	0	3	40	60	100	136

Elective IV

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BEBME7E05	Speech Processing	PE	3	0	0	3	40	60	100	138
22BEBME7E06	Embedded System In Medical Devices	PE	3	0	0	3	40	60	100	140
22BEBME7E07	Robotics And Automation In Medicine	PE	3	0	0	3	40	60	100	142
22BEBME7E08	Design of Medical Devices and Implants	PE	3	0	0	3	40	60	100	144

Elective V

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
22BEBME8E01	Virtual Reality and Augmented Reality	PE	3	0	0	3	40	60	100	146
22BEBME8E02	Internet of things	PE	3	0	0	3	40	60	100	149
22BEBME8E03	Medical Informatics	PE	3	0	0	3	40	60	100	151
22BEBME8E04	Nanotechnology in Medicine	PE	3	0	0	3	40	60	100	153

OPEN ELECTIVES

Course Code	CourseTitle	Category	Instruction Hours/week			Credit(s)	Maximum Marks			Page No
			L	T	P		CIA	ESE	Total	
ELECTRONICSANDCOMMUNICATIONENGINEERING										
22BEECOE01	Neural Networks and its applications	OE	3	0	0	3	40	60	100	155
22BEECOE02	Principles of Modern Communication System	OE	3	0	0	3	40	60	100	157
COMPUTER SECIENCE AND ENGINEERING										
22BECSOE01	Internet Programming	OE	3	0	0	3	40	60	100	159
22BECSOE02	Machine Learning	OE	3	0	0	3	40	60	100	161
22BECSOE03	Block chain technologies	OE	3	0	0	3	40	60	100	163
BIOTECHNOLOGY										
20BTBTOE01	Bioreactor Design	OE	3	0	0	3	40	60	100	162
20BTBTOE02	Food Processing and Preservation	OE	3	0	0	3	40	60	100	167

FACULTY OF ENGINEERING
DEPARTMENT OF BIOMEDICAL ENGINEERING

S.No.	Course work- subject area	Credits/Semester								Credits Total	No. of courses	Percentage
		I	II	III	IV	V	VI	VII	VIII			
1	Humanities and Social Sciences(HS)	3	6	-	-	3	-	-	-	12	4	7.02
2	Basic Sciences (BS)	14	4	4	4	-	-	-	-	26	6	15.20
3	Engineering Sciences(ES)	8	9	5	-	-	-	-	-	22	6	12.87
4	Professional Core(PC)	-	3	13	18	15	18	7	-	74	30	43.27
5	Professional Electives(PE)	-	-	-	-	3	3	6	3	15	5	8.77
6	Open Electives (OE)	-	-	-	-	-	-	3	3	6	2	3.51
7	Project Work(PW)	-	-	-	-	-	-	4	8	12	2	7.02
8	Mandatory Courses (MC)	-	-	-	2	-	2	-	-	4	6	2.34
9	Total	25	22	22	24	21	23	20	14	171	61	100
TOTAL CREDITS										171		

COURSE OBJECTIVES:

The goal of this course is for students,

- To enable students to attain fluency and accuracy to inculcate proficiency in professional communication to meet the growing demand in the field of Global communication.
- To help students acquire their ability to speak effectively in real life situations.
- To inculcate the habit of reading and to develop their effective reading skills.
- To ensure that students use dictionary to improve their active and passive vocabulary.
- To enable students to improve their lexical, grammatical and communicative competence.
- To enable students to improve their technical writing

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Utilize English language for communication: verbal & non-verbal
- Develop comprehension and acquisition of speaking & writing ability.
- Extend the habit of reading and to develop their effective reading skills.
- Build their listening, speaking, Reading and writing skills.
- Show confidence in using English language in real life situations.
- Make use of word power: lexical, grammatical and communication competence

Unit: I - Basic Writing Skills**9**

Sentence Structures - Use of phrases and clauses in sentences - Importance of proper punctuation - Creating coherence- Organizing principles of paragraphs in documents - Techniques for writing precisely

Unit: II - Vocabulary Building**9**

The concept of word formation - Root words from foreign languages and their use in English - Acquaintance, with prefixes and suffixes from foreign languages in English to form derivatives. - Synonyms, antonyms, and standard abbreviations.

Unit: III - Grammar and Usage**9**

Subject-verb agreement - Noun-pronoun agreement - Misplaced modifiers – Articles – Prepositions – Redundancies – Clichés.

Unit: IV - Listening and Reading Skills**9**

Note taking- viewing model interviews – listening to informal conversations – improving listening / reading comprehension – reading model prose / poems – reading exercise

Unit: V -Writing Practices**9**

Comprehension - Précis writing - essay writing listening comprehension - common everyday situations: conversations and dialogues - communication at workplace – interviews - formal presentations

Total Periods : 45

Note: Students shall have hands on training in improving listening skill in the language laboratory @ 2 periods per each unit.

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Sangeeta Sharma , Meenakshi	Technical Communication: Principles And Practice,	2 nd Edition, OUP	2015
2	Sanjay Kumar and PushpLata	Communication Skills	Oxford University Press	2015
3	Liz Hamp - Lyons and Ben Heasley	Analyze Writing	Cambridge University Press	2006

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	F.T. Wood	Remedial English Grammar	Macmillan.	2007
2	Michael Swan	Practical English Usage	OUP.	2017

Course Objectives

The goal of this course is for the students to:

- Develop the use of matrix algebra techniques that is needed by engineers for practical applications.
- Understand geometrical aspects of curvature and elegant application of differential calculus which are needed in Engineering applications.
- Make the student acquire sound knowledge of techniques in solving ordinary differential equations that model Engineering problems.
- Familiarize the student with functions of several variables which is the foundation for many branches of Engineering.
- Introduce sequence and series which is central to many applications in Engineering.
- Apply differentiation to solve maxima and minima problems which is a foundation course which mainly deals with a single variable.

Course Outcomes

Upon completion of this course the students will be able to:

- Find the rank, Eigen values and eigenvectors, diagonalization of a matrix, Symmetric matrices and the students will be able to use matrix algebra techniques for practical applications.
- Equip the students to have basic knowledge and understanding of differential calculus in the field of Engineering.
- Solve simple standard examples using the ideas of differential equations.
- Apply various techniques to solve Partial Differential Equations
- Develop the tool of power series for learning advanced Engineering Mathematics.
- Apply the knowledge acquired to solve various Engineering problems.

UNIT I - Matrices**12**

Introduction - Characteristic equation – Eigenvalues and Eigenvectors of a real matrix – Properties – Cayley-Hamilton theorem (excluding proof) – Orthogonal transformation of a symmetric matrix to diagonal form – Quadratic forms – Reduction to canonical form through orthogonal reduction. Simple problems using Scilab.

UNIT II – Differential Calculus**12**

Overview of Derivatives - Curvature in Cartesian co-ordinates – Centre and radius of curvature – Circle of curvature – Evolutes – Envelopes- Evolutes as Envelope of normals

UNIT III - Differential Equations**12**

Linear Differential equations of second and higher order with constant coefficients – Homogeneous equation of Euler's and Legendre's type – Method of variation parameters.

UNIT IV – Functions of Several Variables**12**

Partial derivatives- Homogeneous functions and Euler's theorem - Total derivative -Differentiation of implicit functions - Jacobians -Partial differentiation of implicit functions-Taylor's series for functions of two variables- Errors and approximations - Maxima and minima of functions of two variables- Lagrange's method of undetermined multipliers.

UNIT V - Sequences and series**12**

Sequences: Definition and examples – **Series:** Types and Convergence – Series of positive terms – Tests of convergence: Comparison test, Integral test and D'Alembert's ratio test – Alternating series – Leibnitz's test – Series of positive and negative terms – Absolute and conditional convergence.

Total Hours: 60**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Grewal B.S.,	Higher Engineering Mathematics	43rd Edition Khanna Publishers	2014
2	Erwin Kreyszig	Advanced Engineering Mathematics	10th Edition, John Wiley	2016
3	N.P. Bali and Manish Goyal	A text book of Engineering Mathematics, Laxmi Publications	Laxmi Publications	2014

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Veerarajan T	Engineering Mathematics	Tata McGraw-Hill	2017
2	Ramana B.V	Higher Engineering Mathematics	TataMcGraw Hill	2017
3	Hemamalini. P.T	Engineering Mathematics	11th Reprint, Tata McGraw Hill	2014

WEBSITES:

• www.efunda.com • www.mathcentre.ac.uk • www.intmath.com/matrices-determinants • www. Intmath.com/calculus/calculus-intro.php
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Instruction Hours/week: L:3 T:1 P:2**Marks: Internal:40 External:60 Total:100
End Semester Exam:3 Hours****i) Theory****Course Objectives**

The Goal of this course is for students,

- To inculcate the basics of properties of matter and its applications.
- To study the basics of sound and ultrasonics with appropriate applications.
- To disseminate the fundamentals of thermal and quantum physics and their applications.
- To introduce the concepts of light, laser and fiber optics for diverse applications.
- To impart the basic knowledge of crystal and its various crystal structures.
- To analyse the relevant problems in engineering stream.

Course Outcomes

Upon completion of this course, the students will be able to:

- Identify the elastic nature of materials and its thermodynamic properties
- Infer the characteristics of laser and optical fibers for engineering applications.
- Develop the idea of quantum mechanics through applications
- Identify the different atomic arrangements of crystals and its defects.
- Make use of the concepts of sound waves for medical applications
- Illustrate the basic ideas of nuclear reactors for energy resources.

UNIT I – PROPERTIES OF MATTER AND SOUND**9**

Elasticity –stress - strain diagram - factors affecting elastic modulus - tensile strength – Poisson's ratio – Twisting couple - Torsion pendulum- bending of beams – bending moment – young's modulus – uniform and non-uniform bending – I- shaped girders. Basics of sound - Sabine's formula – acoustic quality- Ultrasound– Production, Industrial and medical applications.

UNIT II – LASER AND FIBER OPTICS**9**

Light: Introduction – various phenomena – LASER- characteristics - Einstein's co-efficients derivation. Nd:YAG, CO₂, Semiconductor LASER- Applications of LASER in industry and medicine. Fiber optics - principle– modes of propagation of light in optical fibers – numerical aperture and acceptance angle – Attenuation - types of optical fibers (Material, refractive index and mode) – fiber optical communication system (block diagram) - Fiber optic sensors

UNIT III – THERMAL PHYSICS**9**

Laws of thermodynamics - thermal expansions – bimetallic strips – Mode of heat transfer - thermal conductivity, Forbe's and Lee's disc method: theory and experiment – heat conduction through compound media (series and parallel) – Joule Thomson's effect – porous plug experiment – refrigerating mechanisms – Air conditioning mechanisms - microwave oven and solar water heater.

UNIT IV – QUANTUM PHYSICS**9**

Black body radiation - Photo electric effect – Compton effect – De Broglie hypothesis - uncertainty principle – super position principle - wave function and wave packets – Phase and group velocities - Schrödinger's wave equations – probability of finding a particle in one dimensional box- physical significance of wave function – Expectation values – Degeneracy

UNIT V – CRYSTAL PHYSICS**9**

Crystalline and amorphous solids – crystal structure - unit cell, primitive cell - crystal systems, Bravais lattices, Miller indices – inter-planar distances - Coordination number and packing factor for SC, BCC, FCC, HCP structures– ZnS and diamond structure – quasi crystal and liquid crystal – defects in crystal.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Bhattacharya D.K. & Poonam T.	Engineering Physics	Oxford University Press	2015
2	Gaur R.K. and Gupta S.L	Engineering Physics	Dhanpat Rai Publications	2012
3	Pandey.B.K. & Chaturvedi .S	Engineering Physics	Cengage Learning India	2012

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Halliday.D., Resnick R. & Walker. J	Principles of Physics	Wiley	2015
2	Amitkumar J Patel, and MaheshkumarK.Patel	Engineering Physics	White Falcon Publishing.	2018
3	Steven M. Girvin Yale University and Kun Yang	Modern Condensed matter Physics	Cambridge University Press.	2019

WEBSITES:

1. https://nptel.ac.in/courses/122/103/122103011/
2. https://nptel.ac.in/courses/113/104/113104081/
3. http://hyperphysics.phy-astr.gsu.edu/hbase/optmod/lascon.html

ii) **Laboratory**

Course Objective:

- To develop basic laboratory skills and demonstrating the application of physical principles.
- To prepare for the lab experiment and perform individually a wide spectrum of experiments.
- To present experimental data in various appropriate forms like tabulation, and plots.
- To analyze, Interpret and Summarize experimental results.
- To communicate clearly understanding of various experimental principles, instruments/setup, and procedure.
- To understand basic electric circuits.

Course Outcomes:

- Apply the knowledge on Physics practical experiments and that knowledge will be used by them in different engineering and technology applications.
- Build the lab experiment and perform individually a wide spectrum of experiments.
- Illustrate the experimental data in various appropriate forms like tabulation, and plots.
- Analyze, Interpret and Summarize experimental results
- Communicate clearly understanding of various experimental principles, instruments/setup, and procedure.
- Develop the skills for understanding basic electric circuits

LIST OF EXPERIMENTS – PHYSICS (Any 8 Experiments)

1. Torsional pendulum – Determination of rigidity modulus of wire and moment of inertia of disc
2. Uniform bending (or) Non-uniform Bending – Determination of young's modulus.
3. Viscosity of liquids-Determination of co-efficient of viscosity of a liquid by Poiseuille's flow.
4. Ultrasonic interferometer – determination of the velocity of sound and compressibility of liquids.
5. Laser- Determination of the wave length of the laser using grating, Acceptance angle of optical fiber.
6. Spectrometer- Determination of wavelength using grating.
7. Air wedge – Determination of thickness of a thin sheet/wire.
8. Lee's disc – Determination of thermal conductivity.
9. Determination of Band gap of a semiconductor.
10. Characteristics of photo diode.

i)Theory**COURSE OBJECTIVES:**

The goal of this course is for students :

- To understand the terminologies of atomic and molecular structure
- To study the basics of Periodic properties, Intermolecular forces
- To study about spectroscopic technique
- To understand the thermodynamic functions
- To understand the concept of chemical reactions
- To comprehend the basic organic chemistry and to synthesis simple drug.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Outline the basic principles of chemistry for water treatment
- Examine the electrochemical properties to design non – conventional energy storage devices
- Apply the concepts combustion of different fuels
- Identify the concepts of corrosion and its protection in the engineering field
- Apply the concepts of surface chemistry in the field of engineering
- Integrate the chemical principles in the projects undertaken in field of engineering and technology

UNIT I - Periodic Properties, Intermolecular Forces**9**

Introduction to Periodic Properties- atomic and ionic sizes, ionization energies, electron affinity and electronegativity, effective nuclear charge. Penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations. Polarizability, oxidation states, coordination numbers. Ionic, dipolar and van Der Waals interactions.

UNIT II–Electrochemistry and Storage Device**9**

Thermodynamic functions. Free energy and emf. Cell potentials, the Nernst equation and applications. Types of electrodes Standard Hydrogen Electrode (SHE) & Calomel. Energy storage devices Primary and secondary cells leclanche cell, Lead Acid Battery, Nickel Cadmium Battery, Lithium Battery Charging and discharging reactions.

UNIT III – Corrosion and Its Control**9**

Chemical and Electrochemical corrosion - Galvanic corrosion - Differential aeration corrosion- Corrosion control - Sacrificial anode and Impressed current cathodic methods - Corrosion inhibitors - Protective coatings – Organic coatings-Paints - Constituents and functions –Inorganic coatings- Metallic coatings - Electroplating (Au) and Electro less plating (Ni) - Surface conversion coating - Hot dipping

UNIT IV – Water Technology**9**

Sources-Characteristics – Specification for drinking water, BIS &WHO-Alkalinity – Types of alkalinity and determination – Hardness – Types and estimation by EDTA method - Domestic water treatment – Disinfection methods (Chlorination, Ozonation, UV treatment) – Boiler feed water – Requirements – Disadvantages of using hard water in boilers – Internal conditioning (Phosphate, Calgon and Carbonate conditioning methods) – External conditioning – Demineralization process – Desalination - Reverse osmosis.

UNIT V - Spectroscopic Techniques and Applications**9**

Spectroscopy (Principles and Instrumentation only). Electronic spectroscopy. Vibrational and rotational spectroscopy. Applications. Surface characterization techniques Scanning electron microscope (SEM) and Transmission electron microscopy (TEM). Fluorescence and its applications in medicine

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	B. H. Mahan	University chemistry	, Pearson Education	2010
2	M. J. Sienko and R. A.	Plane Chemistry: Principles and Applications.	Pearson Education	2010
3	C. N. Banwell	Fundamentals of Molecular Spectroscopy	McGraw-Hill,.	2017

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	B. L. Tembe	Engineering Chemistry	NPTEL Web-book	2010
2	P. W. Atkins,	Physical Chemistry	Oxford University Press	2018
3	K. P. C. Volhardt and N. E. Schore,	Organic Chemistry: Structure and Function,	5th Edition W.H. Freeman	2014
4	P C Jain & Monica Jain	Engineering Chemistry	Dhanpat Rai Publishing Company,	2015

ii)Laboratory

Course Objectives

The goal of this course is for students :

- To provide students with practical knowledge of quantitative analysis of materials by classical and instrumental methods for developing experimental skills in building technical competence.
- To estimate the amount of sodium carbonate and sodium hydrogen carbonate, hardness, chloride in water sample
- To make the student acquire practical skills in the determination of conductance of solutions, EMF etc
- To acquaint the students with the determination of rate constant of a reaction
- To estimate acid value of oil
- To carry out different types of titrations for estimation of concerned in materials

Course Outcomes

Upon completion of this course, the student will be able to:

- Analyse water quality parameters and corrosion measurement
- Estimate rate constants of reactions from concentration of reactants/products as a function of time
- Measure molecular/system properties such as surface tension, viscosity, conductance of solutions, redox potentials, chloride content of water, etc
- Determine the partition coefficient of a substance between two immiscible liquids.
- Acquaint the students with the determination of acid value of an oil
- Carry out different types of titrations for estimation of concerned in materials using comparatively more qualities and quantities of materials involved for accurate results

Choice of 10 experiments from the following:

1. Determination of surface tension and viscosity
2. Determination of Sodium Carbonate and Sodium Hydrogen Carbonate in a mixture using volumetric titration
3. Determination of Ca / Mg using complexometric titration
4. Thin layer chromatography
5. Determination of chloride content of water
6. Determination of the rate constant of a reaction
7. Conductometry - Determination of cell constant and conductance of solutions
8. pH Metry – Determination of Acid / Base
9. Potentiometry - determination of redox potentials and emfs
10. Saponification/acid value of an oil
11. Determination of the partition coefficient of a substance between two immiscible liquids
12. Adsorption of acetic acid by charcoal
13. Use of the capillary viscosimeters to demonstrate the isoelectric point as the pH of minimum viscosity for gelatin sols and/or coagulation of the white part of egg

i)Theory**COURSE OBJECTIVES:**

The goal of this course is for students :

- To perceive about core programming basics—including data types, control structures, algorithm development, and program design with functions
- To discuss the fundamental principles of Object-Oriented Programming, as well as in-depth data and information processing techniques
- To solve problems, explore real-world software development challenges, and create practical and contemporary applications
- To develop Python programs with conditionals and loops.
- To define Python functions and use function calls.
- To apply the string handling functions to solve the given problem

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain various operators used in python.
- Elaborate Object oriented concepts with python
- Demonstrate the functions of networking in python
- Develop algorithmic solutions to simple computational problems.
- Read and write data from/to files in Python programs.
- Utilize image processing techniques in python programming to solve a given problem

UNIT I- INTRODUCTION**9**

Installing Python; basic syntax, interactive shell, editing, saving, and running a script variables, assignments; immutable variables; numerical types; arithmetic operators and expressions; comments in the program; understanding error messages.

UNIT II- CONDITIONAL STATEMENT & STRING HANDLING**9**

Conditions, Boolean logic, logical operators; ranges; Control statements: if-else, loops (for, while); short-circuit (lazy) evaluation – Manipulating files and directories, OS and sys modules; text files: reading/writing text and numbers from/to a file; creating and reading a formatted file (csv or tab-separated). String manipulations: subscript operator, indexing, slicing a string; strings and number system: converting strings to numbers and vice versa. Binary, octal, hexadecimal numbers.

UNIT III- OBJECT ORIENTED PROGRAMMING WITH PYTHON**9**

Classes and OOP: classes, objects, attributes and methods; defining classes; design with classes, data modelling; persistent storage of objects – OOP, continued: inheritance, polymorphism,

operator overloading; abstract classes; exception handling, try block

UNIT IV- IMAGE PROCESSING WITH PYTHON

9

Design with functions: hiding redundancy, complexity; arguments and return values; formal vs actual arguments, named arguments. Program structure and design. Recursive functions Simple Graphics and Image Processing: “turtle” module; simple 2d drawing – colors, shapes; digital images, image file formats, image processing Simple image manipulations with ‘image’ module (convert to b/w, rayscale, blur, etc).

UNIT V- NETWORKING WITH PYTHON

9

Multithreading, Networks, and Client/Server Programming; introduction to HTML, interacting with remote HTML server, running html-based queries, downloading pages; CGI programming, programming a simple CGI form.

Total periods : 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Shroff	Learning Python: Powerful Object-Oriented Programming	5th Edition	2013
2	Timothy A	Budd 'Exploring Python'	TATA McGRAW-HILL	2011
3	Vamsi Kurama	Python Programming: A Modern Approach	Pearson Education	2018

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	David M.Baezly	Python Essential Reference	Addison-Wesley Professional; 4 edition	2009
2	David M. Baezly.	Python Cookbook	O'Reilly Media; 3rd edition	2013
3	Guido Van Rossum, Fred . L. Drake	'Introduction to Python'	Network Theory Limited	2011
4	Alex Martelli	'Python in a Nutshell'	O'Reilly - 2nd Edition	2006

WEBSITES:

- <https://www.codecademy.com/learn/python>
- www.learnpython.org/

ii)Laboratory

COURSE OBJECTIVES:

The goal of this course is for students:

- To compile, test, and debug simple Python programs.
- To evaluate Python programs with conditionals and loops.
- To utilize functions for structuring Python programs.
- To outline compound data using Python lists, tuples, and dictionaries.
- To apply data from/to files in Python.
- To do input/output with files in Python.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Compile, test, and debug simple Python programs.
- Illustrate Python programs with conditionals and loops.
- Develop Python programs step-wise by defining functions and calling them.
- Utilize Python lists, tuples, dictionaries for representing compound data.
- Recall and write data from/to files in Python.
- Apply Python features in developing software applications

LIST OF EXPERIMENTS:

1. Compute the GCD of two numbers.
2. Find the square root of a number (Newton's method)
3. Exponentiation (power of a number)
4. Find the maximum of a list of numbers
5. Linear search and Binary search
6. Selection sort, Insertion sort
7. Merge sort
8. First n prime numbers
9. Multiply matrices
10. Programs that take command line arguments (word count)
11. Find the most frequent words in a text read from a file
12. Simulate elliptical orbits in Pygame
13. Simulate bouncing ball in Pygame

i)Theory**Course Objectives**

The goal of this course is for students:

- To impart the basic knowledge about the Electric circuits.
- To understand the concept of Electrical Machines and Transformers.
- To understand the working of Semiconductor devices and Digital Circuits.
- To impart the basic knowledge of Measuring Instruments and Electrical Installation
- Know the fundamentals of Electrical Engineering and practical.
- Practical implementation of fundamental theory concepts.

Course Outcomes

At the end of this course, students will be able to

- Select the electric circuits with DC and AC excitation by applying various circuit la
- Use the Measuring Instruments in Electrical Installation.
- Interpret the various digital circuits in real time applications.
- Develop basic applications using various semiconductor devices.
- Illustrate the working operation of the transformer.
- Apply the fundamental theory concepts in electrical machines.

UNIT I - DC Circuits**(9)**

Electrical circuit elements (R, L and C), voltage and current sources, Kirchoff's current and voltage laws, analysis of simple circuits with dc excitation. Superposition, Thevenin's and Norton Theorems.

UNIT II - AC Circuits**(9)**

Representation of sinusoidal waveforms, peak and rms values, Phasor representation, real power, reactive power, apparent power, power factor. Analysis of single-phase ac circuits consisting of R, L, C, RL, RC, RLC combinations (series and parallel), resonance. Three phase balanced circuits, voltage and current relations in star and delta connections.

UNIT III - Electrical Machines And Transformer**(9)**

Construction and working of a three-phase and Single-phase induction motor. Construction, working and speed control of DC motor. Magnetic materials, BH characteristics, Construction and working principle of ideal and practical transformer.

UNIT IV- Semiconductor Devices And Digital Electronics**(9)**

Bipolar Junction Transistor – Characteristics. Introduction to operational Amplifier –Model– Applications. Number systems – binary codes - logic gates - Boolean algebra, laws & theorems

UNIT V- Measuring Instruments And Electrical Installation**(9)**

Principle, construction, and operation of moving coil and moving iron meters-Measurement of Power. Components of LT Switchgear: Switch Fuse Unit (SFU), MCB, ELCB, RCCB, MCCB. Earthing. Types of Batteries and its application in Electric Vehicle, Important Characteristics for Batteries. Elementary calculations for energy consumption and battery back up.

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	S.K.Bhattacharya	Basic Electrical Engineering	Pearson	2019
2	E. Hughes	Electrical and Electronics Technology	Pearson	2010
3	P. Kothari I. J. Nagrath	Basic Electrical Engineering	Tata McGraw Hill	2017

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	VN Mittle and Arvind Mittal	Basic Electrical Engineering	McGraw Hill	2017
2	A.Sudhakar Shyammohan S Palli	Circuits and Networks	McGraw Hill	2013
3	R.Muthusubramanian S.Salivahanan	Basic Electrical and Electronics Engineering	McGraw Hill	2014

WEBSITES:

- www.nptel.ac.in.
- encyclopedia-magnetica.com/doku.php/coenergy.
- https://en.wikibooks.org/wiki/electronics/measuring_instruments.

ii)Laboratory

Course Objective

The goal of this course is for students:

- To impart the basic knowledge about the AC circuits.
- To impart the basic knowledge about the DC circuits.
- To understand the working of Energy Meter.
- To understand the working of DC Machines.
- To impart the knowledge of Logical digital circuits and their differences.
- Know the fundamentals of Electrical Engineering and practical

Course Outcomes (Cos)

At the end of this course, students will be able

- Summarize the basics of electric circuits.
- Analyze the concept of magnetic circuit.
- Study different meters and instruments for measurement of electrical quantities.
- Interpret the working principles of DC Machines.
- Analyze the working principles of Energy Meter.
- Define the truth table of Logic Gates.

List of Experiments

1. Experimental verification of electrical circuit problems using Ohms law
2. Experimental verification of electrical circuit problems using Kirchoff's Voltage law.
3. Experimental verification of electrical circuit problems using Kirchoff's Current law.
4. Measurement of electrical quantities – voltage, current, power & power factor in R load.
5. Measurement of energy using single phase energy meter.
6. Speed control of DC Shunt Motor.
7. Verification of truth table of Logic Gates.

SEMESTER-II

B.E Biomedical Engineering		2022-2023
22BECC201	COMMUNICATIVE ENGLISH	Semester-II
Instruction Hours/week: L:2 T:0 P:2	Marks: Internal:40 External:60 Total:100	4H-3C
End Semester Exam:3 Hours		

COURSE OBJECTIVES:

Course Objectives

The goal of this course is for students to

- Acquire their ability to speak effectively in real life situations.
- Enable students to communicate in effective way without any barriers.
- Inculcate the habit of listening and to develop their effective listening skills.
- Ensure that students use different aids in order to attain effective communication.
- Realize the barriers of communication and overcome the barriers.
- Enable students to improve their group behavior and presentation skill.

Students undergoing this course will be able to

- Utilize English language for communication: verbal & non-verbal
- Develop comprehension and acquisition of speaking & writing ability.
- Extend the habit of reading and to develop their effective reading skills.
- Build their listening, speaking, Reading and writing skills.
- Show confidence in using English language in real life situations.
- Make use of word power: lexical, grammatical and communication competence

UNIT I - COMMUNICATION SKILLS

9

Communication Skills: Introduction, Definition, The Importance of Communication - The Communication Process – Source, Message, Encoding, Channel, Decoding Receiver, Feedback, Context, Barriers to Communication: Physiological Barriers, Physical Barriers, Cultural Barriers, Language Barriers, Gender Barriers, Interpersonal Barriers, Psychological Barriers, Emotional Barriers Perspectives in Communication: Introduction, Visual Perception, Language, Other factors affecting our perspective-Past Experiences, Prejudices, Feelings, Environment

UNIT II - ELEMENTS OF COMMUNICATION

9

Introduction, Face to Face Communication- Tone of Voice, Body Language (Non-verbal communication), Verbal Communication, Physical Communication.

Communication Styles: Introduction, The Communication Styles Matrix with example for each -Direct Communication Style, Spirited Communication Style, Systematic Communication Style, Considerate Communication Style.

UNIT III - BASIC LISTENING SKILLS**9**

Introduction, Self-Awareness, Active Listening, Becoming an Active Listener, Listening in Difficult Situations. Effective Written Communication: Introduction, When and When Not to Use Written Communication-Complexity of the Topic, Amount of Discussion's Required, Shades of Meaning, Formal Communication.

Writing Effectively: Subject Lines, Put the Main Point First, Know Your Audience Organization of the Message.

UNIT IV - INTERVIEW SKILLS AND GIVING PRESENTATIONS**9**

Purpose of an interview, Do's and Don'ts of an interview- Dealing with Fears, planning your Presentation, Structuring Your Presentation, Delivering Your Presentation, Techniques of Delivery.

UNIT V -WRITING PRACTICES**9**

Group Discussion: Introduction, Communication skills in group discussion, Do's and Don'ts of group discussion

Note: Students shall have hands on training in improving Speaking skill in the language laboratory @ 2 periods per each unit.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Sanjay Kumar and PushpaLata	Communication Skills	Oxford University Press	2011
2	Konarnira	Communication Skills for professionals	New arrivals	2011
3	John Adair	Effective communication,	Cengage Learning India pvt .ltd	2009

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Butter Field,	Soft skill for everyone	Macmillan.	2011
2	Stephen. P. Robbins	Communication skills	Oxford Press	2013

COURSE OBJECTIVES:

The goal of this course is for students to:

- Determine mathematical tools needed in evaluating multiple integrals and their usage.
- Calculate and establish identities connecting the vector quantities, to Evaluate line, surface and volume integrals in simple coordinate systems.
- Utilize Gauss, Stokes and Greens theorems to simplify calculations of integrals and prove simple results.
- Apply the knowledge of Mathematics in various Engineering fields by making them to identify the functions in engineering problems as analytic function and their analyze as a function of a complex variables.
- Develop an understanding of the standard techniques of complex variable theory so as to enable the student to apply them with confidence, to specify some difficult integration that appear in applications can be solved by complex integration in application areas such as fluid dynamics and flow of the electric current.
- Utilize Laplace transforms efficiently for solving the problems that occur in various branches of Engineering disciplines

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Apply integration to compute multiple integrals, area, volume, integrals in polar and Cartesian coordinates, in addition to change of order and vector integration.
- Acquaint the concepts of vector calculus, needed for problems in all engineering disciplines
- Find the Analytic functions using the Cauchy Riemann equations
- Apply the relations between conformal mappings and quadratic differentials and how geometric structures are changing under conformal mappings.
- Evaluate complex integrals using the Cauchy integral formula and the residue theorem and to appreciate how complex methods can be used to prove some important theoretical results.
- Laplace transform and inverse transform of simple functions, properties, various related theorems and application to differential equations with constant coefficients

UNIT-I MULTIPLE INTEGRALS**12**

Double integral – Cartesian coordinates – Polar coordinates – Change of order of integration – Triple integration in Cartesian co-ordinates – Area as double integrals.

UNIT-II VECTOR CALCULUS**12**

Gradient, Divergence and Curl – Directional derivative – Irrotational and Solenoidal vector fields – Vector integration – Green's theorem, Gauss divergence theorem and Stoke's theorems (Statement Only)- Surfaces : hemisphere and rectangular parallelopeds.

UNIT-III ANALYTIC FUNCTIONS**12**

Analytic functions - Cauchy-Riemann equations in Cartesian and polar forms – Sufficient condition for an analytic function (Statement Only) - Properties of analytic functions – Constructions of an analytic function - Conformal mapping: $w = z+a$, az , $1/z$ and bilinear transformation.

UNIT-IV COMPLEX INTEGRATION**12**

Complex Integration - Cauchy's integral theorem and integral formula (Statement Only) – Taylor series and Laurent series - Residues – Cauchy's residue theorem (Statement Only) - Applications of Residue theorem to evaluate real integrals around unit circle and semi circle (excluding poles on the real axis).

UNIT-V LAPLACE TRANSFORM**12**

Transforms of elementary functions – Basic properties – Transforms of derivatives and integrals – Initial and final value theorems. Inverse Laplace transforms – Convolution theorem (statement only) – Solution of Ordinary Differential Equations with constant coefficients using Laplace transforms – Transform of periodic functions.

Total Periods : 60**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Grewal, B.S.	Higher Engineering Mathematics	43 rd Edition Khanna Publishers	2014
2	Kreyszig Erwin	Advanced Engineering Mathematics	John Wiley and Sons, 10 th Edition	2016
3	Kandasamy. P, Thilagavathy. K,	Engineering Mathematics	S Chand & Co. Ltd	2008

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Bali.N., Goyal M. and Watkins C	Advanced Engineering Mathematics,	7 th Edition Firewell Media	2009
2	Venkataraman, M. K	Engineering Mathematics	The National Publishing	2005
3	Narayanan. S, Manicavachagam pillay.T.K and Ramaniah	Advanced Mathematics for Engineering Students	Viswanathan S.(Printers and Publishers)	2002

WEBSITES:

- www.intmath.com
- www.efunda.com
- www.mathcentre.ac.uk

COURSE OBJECTIVES

The Goal of this course is for students:

- To study the complete non-ionizing radiations including light and its effect in human body.
- To demonstrate the principles of ultrasound radiation and its applications in medicine.
- To appraise about radioactive nuclides and also the interactions of radiation with matters and how isotopes are produced.
- To perceive the role of Physics in cardiopulmonary system.
- To analyse the harmful effects of radiation and radiation protection regulations.
- To study the effects of sound and light in human body

COURSE OUTCOMES

Upon completion of this course, the students will be able to

- Analyze the low and high frequency effects of non-ionizing radiation and physics of light.
- Define various clinical applications based on ultrasound wave.
- Explain the process of radioactive nuclide production using different techniques.
- Analyze radiation mechanics involved with various physiological systems.
- Apply the concept of physics in the function of cardiopulmonary system.
- Outline the detrimental effects of radiation and regulations for radiation safety.

UNIT I - ELECTROMAGNETIC SPECTRUM AND ULTRASOUND IN MEDICAL APPLICATION**9**

Physics of light, Reflection, Refraction, absorption and scattering - limits of vision and color vision- Non-ionizing electromagnetic radiation tissue as a leaky dielectric – Relaxation Processes – Overview of non – ionizing radiation effects-Low Frequency Effect-Higher frequency effect-Thermography-Applications, Ultrasound: Ultrasonic Transducers – Interaction of ultrasound with materials – Doppler Effect and its Applications - sonogram.

UNIT II - RADIOACTIVE DECAY AND ITS APPLICATIONS IN MEDICINE**9**

Spontaneous Emission – Isometric Transition – Gamma ray emission, alpha, beta, positron decay, electron capture Principles of Nuclear Physics – Natural radioactivity, Decay series, half life period, types of radiation and their applications. Production of radionuclides - Technetium 99m generator– Cyclotron produced Radionuclide - Reactor produced Radionuclide – Milking Process, Applications of radionuclides in medicine.

UNIT III - INTERACTION OF RADIATION WITH MATTER AND IMAGING PRINCIPLES**9**

Interaction of charged particles with matter – Specific ionization, linear energy Transfer, Bremsstrahlung, Annihilation Interaction of Gamma radiations with matter – Photoelectric effect, Compton Scattering, pair Production, Attenuation of Gamma Radiation, Interaction of neutron with matter. Nuclear magnetic Resonance (NMR) – MRI – SQUID.

UNIT IV - PHYSICS OF CARDIOPULMONARY SYSTEM**9**

The Airways - blood and lung interaction – lung volume – physics of alveoli – breathing mechanism – major components of cardiovascular system – O₂ and CO₂ exchange in the capillary system – Physical activity of heart – transmural pressure – Bernoulli's principle applied to cardiovascular system - Blood flow – laminar and turbulent.

UNIT V - RADIATION MEASUREMENTS AND METHODS**9**

Introduction -exposure-Roentgen - photon fluence and energy fluence - Acute Radiation Effects – KERMA -absorbed dose – CEMA- absorbed dose – stopping power -relationship between the dosimetric quantities - The concept of LD 50 –Radiation syndromes - Stochastic and Deterministic effects – cavity theories – Braggs curve

Total periods: 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	B.H Brown , PV Lawford, R H Small wood , D R Hose , D C Barber	Medical Physics and Biomedical Engineering	CRC Press	1999
2	Gopal B.Saha	Physics and Radiobiology of Nuclear Medicine	Springer, 3 rd ed	2006

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	John R. Cameran and James G. Skofronick	Medical Physics	John–Wiley & Sons	1978
2	R.Hendee and Russell Ritenour	Medical Imaging Physics	Fourth Edition William, Wiley-Liss	2002

WEBSITES:

1. <https://nptel.ac.in/courses/115102025/>
2. <https://nptel.ac.in/courses/113105025/>
3. https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-012-microelectronic-devices-and-circuits-fall-2009/lecture-notes/MIT6_012F09_lec01.pdf

COURSE OBJECTIVES:

The goal of this course is for students to:

- Create the awareness about environmental problems among people.
- Develop an attitude of concern for the environment.
- Motivate public to participate in environment protection and improvement.
- To gain a variety of experiences and acquire a basic understanding of environment and it's associated problems.
- To help the individuals in acquiring skills for identifying and solving environmental problems.
- Relate critically about their roles and identities as citizens, consumers and environmental actors in a complex, interconnected world.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Demonstrate core concepts and methods from ecological and physical sciences and their application in environmental problem solving.
- Identify concepts and methods from economic, political, and social analysis as they pertain to the design and evaluation of environmental policies and institutions.
- Distinguish the ethical, cross-cultural, and historical context of environmental issues and the links between human and natural systems.
- Analyze the transnational character of environmental problems and ways of addressing them, including interactions across local to global scales.
- Outline the social issues and Environment Laws.
- Apply the environmental principles in the projects undertaken in field of engineering and technology.

UNIT I – INTRODUCTION - ENVIRONMENTAL STUDIES& ECOSYSTEMS**9**

Environment Definition, Scope and importance; Ecosystem, Structure and functions of ecosystem. Energy flow, Food chains and food webs, Ecological succession. Classification of ecosystem. Forest ecosystem, Grass and Ecosystem, Desert ecosystem, Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries).

UNIT II - NATURAL RESOURCES - RENEWABLE AND NON RENEWABLE RESOURCES**9**

Natural resources - Renewable and Non – Renewable resources. Land resources and land use change, Land degradation, soil erosion and desertification. Forest resources-Deforestation: Causes and impacts due to mining, dam building on environment, forests, biodiversity and tribal populations. Water resources- Use and over-exploitation of surface and ground water, floods, droughts, conflicts over water. Use of alternate energy sources, growing energy needs, case studies. Role of an individual in conservation of natural resources. Equitable use of resources for sustainable lifestyles.

UNIT III - BIODIVERSITY AND ITS CONSERVATION

9

Levels of biological diversity - genetic, species and ecosystem diversity. Ecosystem and biodiversity services: Ecological, economic, social, ethical, aesthetic and Informational value. Bio- geographical classification of India. Biodiversity patterns (global, National and local levels). Hot- spots of biodiversity. India as a mega-diversity nation. Endangered and endemic species of India. Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts. Conservation of biodiversity: in-situ and ex-situ conservation of biodiversity.

UNIT IV -ENVIRONMENTAL POLLUTION

9

Definition, causes, effects and control measures of Air pollution, Water pollution, Soil pollution, Noise pollution. Nuclear hazards and human health risks. Solid waste management and control measures of urban, industrial and e-wastes. Role of an individual in prevention of pollution. Case studies.

UNIT V - SOCIAL ISSUES AND THE ENVIRONMENT

9

Concept of sustainability and sustainable development. Water conservation -Rain water harvesting, watershed management. Climate change, global warming, ozone layer depletion, acid rain and its impacts on human communities and agriculture. Environment Laws (Environment Protection Act, Air Act, Water Act, Wildlife Protection Act, Forest Conservation Act). International agreements (Montreal and Kyoto protocols). Resettlement and rehabilitation of project affected persons. Disaster management (floods, earthquake, cyclones and landslides). Environmental Movements (Chipko, Silent valley, Bishnois of Rajasthan). Environmental ethics: Role of Indian and other religions and cultures in environmental conservation. Environmental communication and public awareness, case studies (e.g., CNG vehicles in Delhi). Human population growth: Impacts on environment, human health and welfare.

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Anubha Kaushik., and Kaushik, C.P	Perspectives in Environmental Studies	New Age International Pvt.	2004
2	Arvind Kumar.	A Textbook of Environmental Science	APH Publishing Corporation	2004
3	Mishra, D.D	Fundamental Concepts in Environmental Studies	S.Chand & Company Pvt.Ltd	2010
4	Daniel, B. Botkin., and Edward, A. Keller	Environmental Science	John Wiley and Sons, Inc	1995

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Odum,E.P., Odum, H.T. and Andrews, J	Fundamentals of Ecology	Philadelphia: Saunders.	1971
2	Rajagopalan, R	Environmental Studies: From Crisis to Cure	Oxford University Press	2016
3	Sing, J.S., Sing. S.P. and Gupta, S.R.	Ecology, Environmental Science and Conservation	Chand & Publishing Company	2014
4	Singh, M.P., Singh, B.S., and Soma, S. Dey	Conservation of Biodiversity and Natural Resources	Daya Publishing House	2013
5	Tripathy.S.N., and Sunakar Panda	Fundamentals of Environmental Studies	Vrianda Publications Private Ltd	2010
6	Verma, P.S., and Agarwal V.K.	Environmental Biology (Principles of Ecology).	S.Chand and Company	2001
7	Uberoi, N.K.	Environmental Studies.	Excel Books Publications	2010

COURSE OBJECTIVES

The goal of this course is for students,

- To provide basic ideas related to modeling techniques in biological systems
- To familiarize the parameters involved in the biological system
- To model the biological system.
- To know about the biomechanics involved in operation of the human system
- To acquaint the ethical issues in biomedical engineering
- To learn fundamental principles and generalizations of analysis used in Biomedical Engineering

COURSE OUTCOMES

Upon completion of this course, students will be able to:

- Recall the basics and parametric aspects of biological systems.
- Compare the engineering models of biological systems with its biological functions.
- Describe the imaging systems used in biomedical engineering
- Choose the correct material for designing implants
- Implement the ethical issues in biomedical engineering
- Choose the techniques and modern engineering tools for Biomedical applications

UNIT-I BASIC BIOLOGY**9**

Engineering in modern medicine, Physiological Systems-Cell Structure, Structure of nerve cell and functions, Heart & Circulatory system, Respiratory Physiology, Kidney function, Brain & Central Nervous System.

UNIT-II PHYSIOLOGICAL SIGNALS**9**

Signal representation, Signal in time and Frequency domain, Signal Estimation, Bioelectric Signals: ECG & EEG, Vital Signs, Biomagnetic Signals.

UNIT-III MEDICAL IMAGING SYSTEMS**9**

Principles and Applications - X-ray & CT, Ultrasound Imaging, MRI, Nuclear Medicine, Microscopy, Biophotonics.

UNIT-IV BIOMECHANICS & BIOMATERIALS**9**

Mechanical Properties of Tissue - Stress, Strain, Viscosity and Viscoelasticity, Applications of Sports Biomechanics, Biomaterials - Types, Properties, Applications - Artificial heart & Membrane Oxygenators.

UNIT-V SOCIAL AND ETHICAL ISSUES RELATED TO BIOMEDICAL ENGINEERING

9

Principles of clinical research, randomized controlled trials, Technology and community, Environmental aspects of technology related to healthcare delivery, Healthcare economics and health rationing.

Total periods : 45

TEXTBOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	R. S. Khandpur	Biomedical Instrumentation: Technology and Applications	TATA McGraw-Hill	2011
2	Leslie Cromwell	Biomedical Instrumentation and Measurement	Prentice Hall of India	2018

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	John D. Enderle and Susan M. Blanchard	Introduction to Biomedical Engineering	Elsevier International Projects Ltd	2012
2	Laurence J. Street	Introduction to Biomedical Engineering Technology	CRC Press	2008

WEBSITES:

- <https://nptel.ac.in/courses/102101068>
- <https://nptel.ac.in/courses/102105090>

COURSE OBJECTIVES:

The goal of this course is for students :

- To discuss the operational characteristics of a Semiconductor in Equilibrium and Non- Equilibrium conditions.
- To aware with the structure of basic electronic devices.
- To understand the operation and applications of electronic devices.
- To explain the working of PN junction diodes and special purpose diodes.
- To define the basic working of BJT and FET both in ideal and non- ideal conditions.
- To improve knowledge about the working of Rectifiers and Voltage regulators.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain the principles and structure of basic electronic devices
- Discuss the operation and VI Characteristics of electronic devices
- Compare the VI characteristics and operations of BJT and FET.
- Comprehend the specifications of regulators and power supply circuits
- Analyze gain and frequency responses of amplifier.
- Design simple electronic circuits

UNIT I - SEMICONDUCTOR DIODES AND SPECIAL PURPOSE DIODES 9

Semiconductors: Intrinsic semiconductor – extrinsic semiconductor – Fermi level in an intrinsic semiconductor, Semiconductor diodes : Formation of PN junction – working principle – VI characteristics – diode current equation – diode resistance – transition and diffusion capacitance. Special purpose diodes: Tunnel, Varactor, Pin contact, Zener diode, schottky diode- Clippers and clampers

UNIT II - BIPOLAR TRANSISTORS 9

Bipolar Transistors: NPN-PNP- Construction – working – transistor currents –transistor configurations (CB,CE,CC) and input- output characteristics – Early effect (base width modulation) – transistor as an amplifier Transistor as a switch.

UNIT III - FIELD EFFECT TRANSISTORS 9

Field-Effect Transistors: construction, working principle and VI characteristics of JFET – comparison of BJT and JFET – MOSFET : working principle and VI characteristics, enhancement MOSFET, depletion MOSFET - comparison of MOSFET with JFET.

UNIT IV - DC POWER SUPPLIES**9**

Rectifiers – Half wave, full wave and Bridge -Block schematic of a typical DC power supply, single phase HWR, FWR, filters - ripple factor and efficiency analysis , Voltage regulators: voltage regulation, Zener diode shunt regulator, transistor series regulator, transistor shunt regulator, switching regulators. Low dropout Regulators (LDO)

UNIT V - OPERATION AMPLIFIER (OP-AMPS)**9**

Ideal Op-amp - Differential amplifier- differential and common mode operation-common mode rejection ratio (CMRR) - Practical op-amp circuits and its parameter – Compensated and uncompensated Op-amp- inverting amplifier, non -inverting amplifier, weighted summer, integrator, differentiator- Large signal operation of op-amps-Other applications of op-amps.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	S. Salivahanan	Electronic Devices and Circuits:	Tata Mcgraw Hill International	2011
2	G.K.Mithal	Electronic Devices and Circuits	Khanna Publishers	2013

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Thomas L. Floyd	Electron Devices	Charles and Messil Publications	2012
2	Gayakwad A R	Op-amps and Linear Integrated circuits	Pearson Education	2011
3	Millman and Halkias	Electronic devices and Circuits	Tata McGraw Hill International	2010

WEBSITES:

- | |
|--|
| 1. https://nptel.ac.in/courses/108108112
2. https://nptel.ac.in/courses/108108122 |
|--|

COURSE OBJECTIVES

The goal of this course is for students :

- To enhance visualization skills, which will facilitate the understanding of engineering systems.
- To prepare the students to design the components with realistic constraints.
- To make the students to consider economic, environmental, ethical, health and safety when they design.
- To make the students to design the components with considering manufacture ability and sustainability
- To communicate effectively through drawings.
- To make the students to understand to use necessary for engineering practice.

COURSE OUTCOMES

Upon completion of this course, students will be able to:

- Explain the engineering design and its place in society
- Analyse visual aspects of engineering design and engineering graphics standards
- Make use of the computer technologies that impact on graphical communication
- Demonstrate the 2D free hand sketching.
- Infer the basic concept of projection of solids.
- Apply the knowledge of projections of points, lines and plane surfaces.

UNIT I - INTRODUCTION**9**

Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, Bureau of Indian Standards (BIS), Layout of drawing sheets, sizes of drawing sheets, different types of lines used in drawing practice, geometric constructions, principles of dimensioning – linear, angular, aligned system, unidirectional system, parallel dimensioning, chain dimensioning, location dimension and size dimension. Reducing Scale, Enlarging Scale, Plain Scale, Diagonal Scale and Vernier Scale.

UNIT II -FREE HAND SKETCHING**9**

Representation of Three-Dimensional objects – General principles of orthographic projection – Need for importance of multiple views and their placement– First angle projection – layout views – Developing visualization skills through free hand sketching of multiple views from pictorial views of objects.

UNIT III- INTRODUCTION TO COMPUTER GRAPHICS–2D**9**

Overview of Computer Graphics and drafting tools , listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software, Sketching of 2D simple geometries, editing and dimensioning of 2D geometries. Projection of points and straight lines located in the first quadrant inclined to both planes– Determination of true lengths and true inclinations (By using CAD software).

UNIT IV-PROJECTION OF PLANE SURFACES AND SOLIDS**9**

Projection of polygonal surface and circular lamina inclined to both reference planes (By using CAD software). Projection of simple solids like prisms, pyramids, cylinder and cone when the axis is inclined to one reference plane by change of position method.

UNIT V- ISOMETRIC PROJECTIONS

9

Demonstrating knowledge of the theory of CAD software, Introduction to 3D modeling packages. Principles of Isometric projection – Isometric Scale, Isometric Views, Conventions; Isometric Views of lines, Planes, Simple solids, truncated prisms, pyramids, cylinders and cones; Conversion of Isometric Views to Orthographic Views and Vice-versa.

Total periods : 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Venugopal K and Prabhu Raja V	Engineering Graphics,	New Age International Publishers	2021
2	C M Agrawal and Basant Agrawal	Engineering Graphics	Tata McGraw Hill	2019
3	Shah, M.B. & Rana B.C	Engineering Drawing and Computer Graphics	Pearson Education	2017

REFERENCE BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Narayana, K.L. & P Kannaiah, ,	Textbookon Engineering Drawing	ScitechPublis hers	2010
2	Bhatt N.D., Panchal V.M.& Ingle P.R	Engineering Drawing	Charotar Publishing House	2019

WEBSITES:

- <https://nptel.ac.in/courses/112102304>

22BEBME251	DISASTER MANAGEMENT	Semester II 2H-0C
Instruction Hours/week: L:2 T:0 P:0		Marks: Internal:100 External:0 Total:100
End Semester Exam:3 Hours		

Course Objectives

The goal of this course for students is :

- To provide students an exposure to disasters, their significance and types
- to understand a preliminary approaches of Disaster Risk Reduction (DRR)
- To discuss about impact of Development projects
- To analyse Disaster Damage Assessment
- To understand Disaster Risk Management
- To discuss various case studies and field works

Course Outcomes

At the end of the course the students will be able to

- Differentiate the types of disasters, causes and their impact on environment and society.
- Classify vulnerability and various methods of risk reduction measures as well as mitigation
- Discuss inter-relationship between disaster and development
- Analyse Disaster Damage Assessment
- Manage Disaster Risk
- Analyse various case studies

UNIT I- INTRODUCTION TO DISASTERS

6

Definition: Disaster, Hazard, Vulnerability, Resilience, Risks – Disasters: Types of disasters – Earthquake, Landslide, Flood, Drought, Fire etc. – Classification, Causes, Impacts including social, economic, political, environmental, health, psychosocial, etc.- Differential impacts- in terms of caste, class, gender, age, location, disability – Global trends in disasters: urban disasters, pandemics, complex emergencies, Climate change- Dos and Don'ts during various types of Disasters.

UNIT II- APPROACHES TO DISASTER RISK REDUCTION (DRR)

6

Disaster cycle – Phases, Culture of safety, prevention, mitigation and preparedness community based DRR, Structural- non-structural measures, Roles and responsibilities of- community, Panchayati Raj Institutions/Urban Local Bodies (PRIs/ULBs), States, Centre, and other stakeholders- Institutional Processes and Framework at State and Central Level- State Disaster Management Authority(SDMA) – Early Warning System – Advisories from Appropriate Agencies.

UNIT III -INTER-RELATIONSHIP BETWEEN DISASTERS AND DEVELOPMENT

6

Factors affecting Vulnerabilities, differential impacts, impact of Development projects such as dams, embankments, changes in Land-use etc.- Climate Change Adaptation- IPCC Scenario and Scenarios in the context of India – Relevance of indigenous knowledge, appropriate technology and local resources.

UNIT IV- DISASTER RISK MANAGEMENT IN INDIA**6**

Hazard and Vulnerability profile of India, Components of Disaster Relief: Water, Food, Sanitation, Shelter, Health, and Waste Management, Institutional arrangements (Mitigation, Response and Preparedness, Disaster Management Act and Policy – Other related policies, plans, programmes and legislation – Role of GIS and Information Technology Components in Preparedness, Risk Assessment, Response and Recovery Phases of Disaster – Disaster Damage Assessment

UNIT V- DISASTER MANAGEMENT: APPLICATIONS AND CASE STUDIES AND FIELD WORKS**6**

Landslide Hazard Zonation: Case Studies, Earthquake Vulnerability Assessment of Buildings and Infrastructure: Case Studies, Drought Assessment: Case Studies, Coastal Flooding: Storm Surge Assessment, Floods: Fluvial and Pluvial Flooding: Case Studies; Forest Fire: Case Studies, Man Made disasters: Case Studies, Space Based Inputs for Disaster Mitigation and Management and field works related to disaster management.

Total Periods : 30**TEXTBOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Singhal J.P	Disaster Management	Laxmi Publications	2019
2	Tushar Bhattacharya	Disaster Science and Management	McGraw Hill India	2012

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Gupta Anil K, Sreeja S. Nair	Environmental Knowledge for Disaster Risk	NIDM, New Delhi	2011
2	Kapur Anu	Vulnerable India: A Geographical Study of Disasters	IIAS and Sage Publishers	2010

22BEBME301

MATHEMATICS-III

4H-4C

(LINEAR ALGEBRA AND PARTIAL DIFFERENTIAL EQUATIONS)**Instruction Hours/week: L:3 T:1 P:0****Marks: Internal:40 External:60****Total:100****End Semester Exam:3 Hours****Course objectives:**

The goal of this course is for the students,

- To introduce the basic notions of groups, rings, fields which will then be used to solve related problems.
- To understand the concepts of vector space, linear transformations and diagonalization.
- To apply the concept of inner product spaces in orthogonalization.
- To understand the procedure to solve partial differential equations.
- To give an integrated approach to number theory and abstract algebra.
- To provide a firm basis for further reading and study in the subject.

Course outcomes:

Upon Completion of this course the students will be able to:

- Explain the fundamental concepts of advanced algebra and their role in modern mathematics and applied contexts.
- Demonstrate accurate and efficient use of advanced algebraic techniques.
- Demonstrate their mastery by solving non – trivial problems related to the concepts and by proving simple theorems about the statements proven by the text.
- Explain the various types of partial differential equations.
- Solve the various types of partial differential equations.
- Solve Engineering problems using Fourier series.

UNIT I – VECTOR SPACES**12**

Vector spaces- Definition, Axioms, Examples of vector spaces or not vector spaces – Basic Theorems-Subspaces – Linear combinations and linear system of equations – Linear independence and linear dependence of vector spaces-Spanning of a subspace– Bases and dimensions.

UNIT II – LINEAR TRANSFORMATION AND DIAGONALIZATION**12**

Linear transformation on vector space, properties of the function T- Null spaces and ranges, Definition and Examples – Dimension theorem – Matrix representation of a linear transformations – Eigenvalues and eigenvectors – Diagonalizability.

UNIT III – INNER PRODUCT SPACES**12**

Inner product space, Definitions and Examples, norms-Orthogonal and Orthonormal Basis – Gram Schmidt orthogonalization process –Orthonormal complement – Adjoint of linear operations – Least square approximation.

UNIT IV – PARTIAL DIFFERENTIAL EQUATIONS**12**

Formation – Solutions of first order equations – Standard types and equations reducible to standard types – Singular solutions – Lagrange's linear equation – Integral surface passing through a given curve – Classification of partial differential equations – Solution of linear equations of higher order with constant coefficients – Linear non-homogeneous partial differential equations.

UNIT V - FOURIER SERIES SOLUTIONS OF PARTIAL DIFFERENTIAL EQUATIONS 12

Dirichlet's conditions – General Fourier series – Half range sine and cosine series – Method of separation of variables – Solutions of one dimensional wave equation and one-dimensional heat equation.

Total hours: 60

TEXT BOOKS:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Dr. Grewal B.S.	Higher Engineering Mathematics	Khanna Publishers, New Delhi.	2018
2	Burden, R.L. and Faires, J.D	"Numerical Analysis", 9 th Edition	Cengage Learning	2011
3	James, G	Advanced Modern Engineering Mathematics	Pearson Education	2018
4	Kolman, B. Hill, D.R	Introductory Linear Algebra	Pearson Education, New Delhi	2009

REFERENCE BOOKS:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Lay, D.C	Linear Algebra and Applications	Pearson Education	2018
2	O'Neil, P.V	Advanced Engineering Mathematics	Cengage Learning	2017

WEB SITES

www.efunda.com
www.mathcentre.ac.uk
www.intmath.com/matrices-determinants
www.Intmath.com/calculus/calculus-intro.php

COURSE OBJECTIVES

The goal of this course is for students

- To discuss all the organelles of an animal cell and their function.
- To perceive structure and functions of the various types of systems of human body.
- To outline about eye, ear and Endocrine glands of human
- To learn organs and structures involving in system formation and functions.
- To understand all systems in the human body.
- To infer basic understanding of the interconnection of various organ systems in human body

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explain basic structure and functions of cells and its organelles
- Demonstrate about anatomy and physiology of various organ systems
- Compare the functions of eye, ear and Endocrine glands of human
- Explain the interconnect of various organ systems in human body
- Analyse the operation of GI tract
- Interpret organs and structures involving in functions of body

UNIT I – CELL**9**

Structure of Cell– Organelles and description–Function of each component of the cell– Membrane potential– Action Potential–Generation and Conduction –Electrical Stimulation. Blood Cell– Composition –Origin of RBC–Blood Groups–Estimation of RBC, WBC and Platelet–Tissues and its functions.

UNIT II- CARDIAC AND NERVOUS SYSTEM**9**

Heart, Major blood vessels– Cardiac Cycle– ECG–Blood Pressure–Feedback Control for Blood Pressure– Nervous Control of Heart- Cardiac output–Coronary and Peripheral Circulation – Structure and function of Nervous tissue–Neuron –Synapse – Reflexes –Receptors –Brain–Brainstem –Spinal cord– Reflex action– Velocity of Conduction of Nerve Impulses–Electroencephalograph –Autonomic Nervous System.

UNIT III – RESPIRATORY SYSTEM AND MUSCULOSKELETAL SYSTEM**9**

Physiological aspects of respiration –Trachea and lungs –Exchange of gases –Regulation of Respiration –Disturbance of respiration function –Pulmonary function test–Muscles-tissue-types-structure of skeletal muscle-types of muscle and joints.

UNIT IV – DIGESTIVE, EXCRETORY AND LYMPHATIC SYSTEM**9**

Organisation of GI System, Digestion and absorption –Movements of GI tract–Intestine–Liver– Pancreas– Structure of Nephron–Mechanism of Urine formation–Urine Reflex–Skin and Sweat Gland–Temperature regulation, Lymphatic: Parts and Functions of Lymphatic systems – Types of Lymphatic organs and vessels.

UNIT V- EYE, EAR & ENDOCRINE GLANDS**9**

Optics of Eye–Retina –Photochemistry of Vision –Accommodation –Neurophysiology of vision –EOG, Physiology of internal ear–Mechanism of Hearing–Auditory Pathway, Hearing Tests– Endocrine- Pituitary and thyroid glands.

Total periods : 45**TEXTBOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Prabhjot Kaur	Text Book of Anatomy and Physiology	Lotus Publishers	2014
2	Elaine.N. Marieb	Essential of Human Anatomy and Physiology	Pearson Education, New Delhi	2016

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Frederic H. Martini, Judi L. Nath, Edwin F. Bartholomew	Fundamentals of Anatomy and Physiology	Pearson Publishers	2014
2	Gillian Pocock, Christopher D. Richards	The human Body – An introduction for Biomedical and Health	Oxford University Press, USA	2013
3	Eldra Pearl Solomon	Introduction to Human Anatomy and Physiology	W.B. Saunders Company	2015

WEBSITES:

- <https://openstax.org/details/books/anatomy-and-physiology>
- <https://www.visiblebody.com/anatomy-and-physiology-apps/anatomy-and-physiology>

COURSE OBJECTIVES

The goal of this course is for students

- To explain the basic principles and techniques used in analog and digital communications.
- To learn and compare analog and digital communication techniques
- To familiarize with source and error control coding and gain knowledge on multi-user radio communication.
- The course will introduce analog and digital modulation techniques, communication receiver and transmitter design, baseband and band pass communication techniques, line coding techniques, noise analysis, and multiplexing techniques.
- The course also introduces analytical techniques to evaluate the performance of communication systems.
- To appreciate the importance of synchronization in communication systems

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Discuss the concepts of sources noise, analog, digital, data and pulse communication techniques.
- Compare the techniques underlying in analog, digital, data and pulse communication.
- Analyze source and error control coding and apply multi-user radio communication.
- Demonstrate understanding of various analog and digital modulation and demodulation techniques.
- Analyze the performance of modulation and demodulation techniques in various transmission environments
- Explain the importance of synchronization in communication systems

UNIT I- ANALOG COMMUNICATION**9**

Noise: Source of Noise – External Noise- Internal Noise – Noise Calculation. Introduction to Communication Systems: Modulation – Types – Need for Modulation. Theory of Amplitude Modulation- Evolution and Description of SSB Techniques – Theory of Frequency and Phase Modulation –Comparison of various Analog Communication System (AM – FM – PM).

UNIT II- DIGITAL COMMUNICATION**9**

Amplitude Shift Keying (ASK) – Frequency Shift Keying (FSK) Minimum Shift Keying (MSK) – Phase Shift Keying (PSK) – BPSK – QPSK – 8 PSK – 16 PSK – Quadrature Amplitude Modulation (QAM) – 8QAM – 16 QAM – Bandwidth Efficiency– Comparison of various Digital Communication System (ASK– FSK – PSK – QAM).

UNIT III- DATA AND PULSE COMMUNICATION**9**

History of Data Communication – Standards Organizations for Data Communication- Data Communication Circuits – Data Communication Codes – Error Detection and Correction Techniques – Data communication Hardware – serial and parallel interfaces. Pulse Communication: Pulse Amplitude Modulation (PAM) – Pulse Time Modulation (PTM) – Pulse Code Modulation (PCM) – Comparison of various Pulse Communication System (PAM – PTM – PCM)

UNIT IV –SOURCE AND ERROR CONTROL CODING**9**

Entropy Source encoding theorem, Shannon Fano coding, Huffman coding, mutual information, channel capacity, channel coding theorem, Error Control Coding, linear block codes, cyclic codes, convolution codes, Viterbi decoding algorithm.

UNIT V –MULTI-USER RADIO COMMUNICATION**9**

Advanced Mobile Phone System (AMPS) – Global System for Mobile Communications (GSM) – Code Division Multiple Access (CDMA) – Cellular Concept and Frequency Reuse – Channel Assignment and Hand off – Overview of Multiple Access Schemes – Satellite Communication – Bluetooth

Total Periods : 45**TEXT BOOKS:**

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Kennedy G	Kennedy's Electronic Communication Systems	McGraw Hill,	2014
2	Wayne Tomasi	Electronic Communication Systems: Fundamentals through Advanced	Pearson Education	2014

REFERENCE BOOKS:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Rappaport T.S	Wireless Communications: Principles and Practice	Pearson Education	2010
2	B. P.Lathi,	Modern Analog and Digital Communication Systems	Oxford University Press	2011

COURSE OBJECTIVES:

The goal of this course is for students

- To infer characteristics and classification of Biomaterials.
- To understand the response of biomaterials in living system
- To identify different metals, ceramics and its nanomaterial's characteristics as Biomaterials.
- To interpret polymeric materials and its combinations that could be used as a tissue replacement implants.
- To discuss the concepts of Nano Science and Technology.
- To perceive the concept of biocompatibility and the methods for biomaterials testing.

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Compare the different types of Biomaterials and its classification
- Choose materials for design of implants in tissue replacement.
- Assess the response of biomaterials in living system.
- Make use of the concept of compatibility of artificial organs inside the living system.
- Explain the testing standards applied for biomaterials.
- Apply the concept of nano technology towards biomaterials.

UNIT I- INTRODUCTION TO BIO-MATERIALS**9**

Definition and classification of bio-materials, mechanical properties, visco elasticity, biomaterial performance, body response to implants, wound healing, blood compatibility, Nano scale phenomena.

UNIT II- METALLIC AND CERAMIC MATERIALS**9**

Metallic implants – Stainless steels, co-based alloys, Ti-based alloys, shape memory alloy, nanostructured metallic implants, degradation and corrosion, ceramic implant – bio inert, biodegradable or bioresorbable, bioactive ceramics, nanostructured bio ceramics.

UNIT III- POLYMERIC IMPLANT MATERIALS**9**

Polymerization, factors influencing the properties of polymers, polymers as biomaterials, biodegradable polymers, Bio polymers: Collagen, Elastin and chitin. Medical Textiles, Materials for ophthalmology: contact lens, intraocular lens. Membranes for plasma separation and Blood oxygenation, electro spinning: a new approach.

UNIT IV- TISSUE REPLACEMENT IMPLANTS**9**

Small intestinal sub mucosa and other decellularized matrix biomaterials for tissue repair: Extracellular Matrix. Soft tissue replacements, sutures, surgical tapes, adhesive, Percutaneous and skin implants, maxillofacial augmentation, Vascular grafts, hard tissue replacement Implants, joint replacements, tissue scaffolding and engineering using Nano biomaterials.

UNIT V – TESTING OF BIOMATERIALS

Biocompatibility, blood compatibility and tissue compatibility tests, Toxicity tests, sensitization, carcinogenicity, mutagenicity and special tests, Invitro and Invivo testing; Sterilisation of implants and devices: ETO, gamma radiation, autoclaving. Effects of sterilization.

Total periods : 45**TEXT BOOKS:**

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Sujata V. Bhatt	Biomaterials	Narosa Publishing House	2005
2	Sreeram Ramakrishna, Murugan Ramalingam, T. S. Sampath Kumar, and Winston O. Soboyejo	Biomaterials: A Nano Approach	CRC Press	2010

REFERENCE BOOKS:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Myer Kutz	Standard Handbook of Biomedical Engineering & Design	McGraw Hill	2003
2	BD Ratner, AS Hoffmann, FJ Schoen, JE Lemmon	An Introduction to Materials in Medicine	Academic Press	2012

WEBSITES

<https://www.nature.com/subjects/biomaterials>
<https://www.sciencedirect.com/journal/biomaterials>

COURSE OBJECTIVES:

The goal of this course is for students

- To study about the biochemistry of living cells, metabolism of biomolecules and the methods of investigation and diagnostic tools.
- To summarize the role of these biomolecules by providing basic information on specific metabolic diseases and disorders of these biomolecules.
- To analyse the structural and functional properties of carbohydrates, proteins and lipids
- To discuss about functions of each organelles
- To infer about the Transport of substances across biological membranes
- To Know the concepts of biochemistry of living cells

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explain the concepts of biochemistry of living cells
- Compare the biological functions of protein
- Illustrate the structural and functional properties of lipids
- Analyze the etiology and biological parameters in metabolic diseases..
- Discuss the significance of clinical biochemistry
- Identify the biomedical importance of biomolecules in human physiology and pathology

UNIT I- CARBOHYDRATES**9**

Carbohydrates: Definition, classification, biological functions; glycolysis, TCA cycle, glycogenesis, glycogenolysis, Diabetes Mellitus – Blood Sugar analysis and glucose tolerance test.

UNIT II- PROTEINS**9**

Proteins: Definition, classification, architecture, biological functions; Classification of amino acids, Oxidative and nonoxidative deamination, transamination, decarboxylation, urea cycle, Purification of proteins.

UNIT III- LIPIDS**9**

Lipids: Definition, classification, biological functions; biosynthesis of long chain fatty acids, degradation of fatty acids - oxidation of fatty acids.

UNIT IV- METHODS OF INVESTIGATION OF METABOLISM**9**

Liver function tests, Renal function tests, Gastric function tests. Diagnostic tools: Principles and applications of photometry, flurometry, flame photometry, automation in clinical laboratory.

UNIT V- CLINICAL BIOCHEMISTRY**9**

Principles and applications of chromatography, electrophoresis, spectroscopic techniques. Blood gas analysis. Regulation of acid-base balance. Measurement of electrolytes. Uses of radioisotopes in biology and medicine.

Total periods : 45**TEXTBOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of Publication
1	Lehninger.A .L., Nelson D.L., Cox .M.M.,	Principles of Biochemistry	CBS Publications	1993
2	Kumar V., & Gill, K. D.	Basic concepts in clinical biochemistry:	Springer, Singapore	2018

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Keith Wilson and John Walker,	Practical Biochemistry– Principles & Techniques	Oxford University press, 7th Edition	2010
2	Ambiga Shanmugam	Fundamentals of Biochemistry for Medical Students	Karthick Printers, Madras	2016

WEBSITES:

- <https://openstax.org/details/books/anatomy-and-physiology>
- <https://www.sciencedirect.com/journal/biomaterials>

Semester-III

22BEBME346

C++ & DATA STRUCTURES
(Theory & Lab)

3H-5C

Instruction Hours/week: L:3 T:0 P:2

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

i)Theory

COURSE OBJECTIVES:

The goal of this course is for students:

- To analyse Object Oriented Programming concepts and basic characteristics of C++
- To design problem solutions using Object Oriented Techniques.
- To infer the principles of packages, inheritance and interfaces
- To define exceptions and use I/O streams
- To develop a C++ application with threads and generics classes
- To design and build simple Graphical User Interfaces

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Utilize a simple Object Oriented Programming environment for compile programs
- Distinguish the fundamental data types.
- Design classes and organise them into packages.
- Utilize the basic data structures.
- Compare the basic search and sort algorithms.
- Apply appropriate data structure and algorithm to solve a problem.

UNIT I- OBJECTS ORIENTED PROGRAMMING**9**

Object Oriented Programming - Abstraction – objects and classes - Encapsulation- Inheritance - Polymorphism- Characteristics of OOP- Classes-Objects-Data types- Constructors and Destructors- methods -access specifiers - static members –Comments- Control Flow, Arrays- Array Lists –Strings.

UNIT II- INHERITANCE AND INTERFACES**9**

Inheritance – Super classes- sub classes –Protected members – constructors in sub classes- Interfaces – defining an interface, implementing interface, differences between classes and interfaces- Exceptions - exception hierarchy-nput / Output Basics – Streams – Byte streams and Character streams

UNIT III- OPERATOR OVERLOADING**9**

Overloading unary operator – overloading binary operator – data conversion- Differences between multi-threading and multitasking, thread life cycle, creating threads, synchronizing threads, Inter-thread communication, daemon threads, thread groups. Generic Programming – Generic classes – generic methods – Bounded Types – Restrictions and Limitations.

UNIT IV- LINEAR DATA STRUCTURES**9**

Abstract data types- List ADT – array-based implementation – linked list implementation —singly linked lists- circularly linked lists- doubly-linked list Stack ADT – Operations- Evaluating arithmetic expressions- Queue ADT – Operations - Circular Queue – Priority Queue – dequeue

UNIT V - NON LINEAR DATA STRUCTURES AND SEARCHING**9**

Tree- Binary trees-AVL Trees-Heap- Graphs- Types of Graphs- Breadth-first traversal - Depth-first traversal – Topological Sort- Searching- Linear Search - Binary Search. Sorting - Bubble sort - Selection sort - Insertion sort- Hashing.

Total periods : 45**TEXT BOOKS:**

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Herbert Schildt	C++- The Complete Reference	Pearson Education	2017
2	Bhushan Trivedi	Programming with ANSI C++, A Step-By-Step approach	Oxford University Press	2010
3	Mark Allen Weiss	Data Structures and Algorithm Analysis in C++	Addison Wesley	2010

REFERENCES:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Deitel and Deitel	C++, How To Program	Pearson Education	2005
2	Goodrich, Michael T, Roberto Tamassia, David Mount	Data Structures and Algorithms in C++	Wiley	2004

WEBSITES

1. <https://nptel.ac.in/courses/106105151/>
2. <https://nptel.ac.in/courses/106101208/>
3. <https://nptel.ac.in/courses/106102064/>
4. <https://nptel.ac.in/courses/106/106/106106127/>

(ii) Laboratory

COURSE OBJECTIVES:

The goal of this course is for students:

- To analyze Object Oriented Programming concepts and basic characteristics of C++
- To design problem solutions using Object Oriented Techniques.
- To infer the principles of packages, inheritance and interfaces
- To define exceptions and use I/O streams
- To develop a C++ application with threads and generics classes
- To design and build simple Graphical User Interfaces

COURSE OUTCOMES:

Upon completion of this course, the student will be able to

- Utilize a simple Java programming environment, compile programs and interpret compiler errors.
- Distinguish and use the fundamental data types.
- Design classes and organize them into packages.
- Utilize the basic data structures.
- Compare the basic search and sort algorithms.
- Apply appropriate data structure and algorithm to solve a problem.

LIST OF EXPERIMENTS

1. Write a C++ Program to display Names, Roll No., and grades of 3 students who have appeared in the examination. Declare the class of name, Roll No. and grade. Create an array of class objects. Read and display the contents of the array.
2. Write a C++ program to declare a class. Declare pointer to class. Initialize and display the contents of the class member.
3. Given that an EMPLOYEE class contains following members: data members: Employee number, Employee name, Basic, DA, IT, Net Salary and print data members.
4. Write a C++ program that uses functions to perform the following:
 - a. Create a binary search tree of characters.
 - b. Traverse the above Binary search tree recursively in preorder, in order and post order
5. Write a C++ program that uses function templates to perform the following:
 - a. Search for a key element in a list of elements using linear search.
 - b. Search for a key element in a list of sorted elements using binary search.
6. Write a C++ program that implements Insertion sort algorithm to arrange a list of integers in ascending order.
7. Write a template based C++ program that implements selection sort algorithm to arrange a list of elements in descending order.
8. Write a template based C++ program that implements Quick sort algorithm to arrange a list of elements in ascending order.
9. Write a C++ program that implements Heap sort algorithm for sorting a list of integers in ascending order.
10. Write a C++ program that implements Merge sort algorithm for sorting a list of integers in ascending order

22BEBME311 BIOCHEMISTRY AND HUMAN PHYSIOLOGY LABORATORY**2H-1C****Instruction Hours/week: L:0 T:0 P:2****Marks: Internal:40 External:60 Total:100****End Semester Exam:3 Hours****COURSE OBJECTIVE:**

The goal of this course is for students

- To estimate and quantify biomolecules.
- To divide amino acid molecules
- To evaluate and interpret of biochemical parameter
- To understand differential count of WBCs
- To Know the Ishihara chart
- To Comprehend the auditory conduction

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Solve the quantitative test of different biomolecules
- Identify the separation technology of proteins and amino acids.
- Identify the blood group
- Estimate blood glucose
- Estimation of Hemoglobin
- Make use of the Biochemistry laboratory functional parameters

LIST OF EXPERIMENTS:

1. General tests for carbohydrates, proteins and lipids.
2. Preparation of serum and plasma from blood.
3. Estimation of blood glucose.
4. Estimation of serum cholesterol.
5. Estimation of creatinine in urine.
6. Separation of amino acids using thin layer chromatography.
7. Estimation of Hemoglobin
8. Differential count of different WBCs and Blood group identification
9. Ishihara chart for color blindness and Snellen's chart for myopia and hyperopia – by letters reading and ophthalmoscope to view retina.
10. Weber's and Rinnee's test for auditory conduction.

Course Objectives

The goal of this course is for students

- To know about Indian constitution.
- To know about central and state government functionalities in India.
- To know about Indian society.
- To inculcate the students with their Fundamental Duties
- To impart a good knowledge in judicial system in states
- To familiarize the student with constitutional Amendments and Functionaries.

Course Outcomes

Upon completion of the course, students will be able to

- Clarify on functions of the Central government.
- Define functions of the State government
- Elucidate the functions of Constitution
- Explain the rules of the Indian constitution.
- Identify and appreciate different culture among the people.
- Discuss the Indian Society

UNIT I- HISTORY OF MAKING OF THE INDIAN CONSTITUTION

History, Drafting Committee, (Composition & Working), Philosophy of the Indian constitution
Preamble Salient Features

UNIT II - CONTOURS OF CONSTITUTIONAL RIGHTS & DUTIES `

Fundamental Rights, Right to Equality, Right to Freedom, Right against Exploitation, Right to Freedom of Religion, Cultural and Educational Rights, Right to Constitutional Remedies, Directive Principles of State Policy, Fundamental Duties.

UNIT III - ORGANS OF GOVERNANCE

Parliament, Composition, Qualifications and Disqualifications, Powers and Functions, Executive President, Governor, Council of Ministers, Judiciary, Appointment and Transfer of Judges, Qualifications, Powers and Functions

UNIT IV - LOCAL ADMINISTRATION

District's Administration head: Role and Importance, Municipalities: Introduction, Mayor and role of Elected Representative, CEO of Municipal Corporation, Panchayat raj: Introduction, PRI: Zila Panchayat. Elected officials and their roles, CEO Zila Panchayat: Position and role Block level: Organizational Hierarchy (Different departments), Village level: Role of Elected and Appointed

officials, Importance of grass root democracy

UNIT V- ELECTION COMMISSION

Election Commission: Role and Functioning, Chief Election Commissioner and Election Commissioners, State Election Commission: Role and Functioning, Institute and Bodies for the welfare of SC/ST/OBC and women.

TEXTBOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of Publication
1	D.D. Basu	Introduction to the Constitution of India	Lexis Nexis	2015
2	M. P. Jain	Indian Constitution Law	7th Edn., Lexis Nexis	2014

REFERENCES:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Busi S N	Ambedkar B R framing of Indian Constitution	Ava Publishers	2015
2	Bare Act	The Constitution of India	Government Publication,	2021

WEBSITES:

- <https://nptel.ac.in/courses/129106002>

PROBABILITY AND STATISTICS**4H-4C****Instruction Hours/week: L:3 T:1 P:0****Marks: Internal:40 External:60 Total:100****End Semester Exam:3 Hours****COURSE OBJECTIVES:**

The goal of this course is for students

- To provide the required skill to apply the statistical tools in Engineering problems.
- To introduce the basic concepts of probability.
- To introduce the basic concepts of random variables.
- To acquaint the knowledge of testing of hypothesis for small and large samples which plays an important role in real life problems.
- To understand the notion of sampling distributions and statistical techniques
- To introduce the basic concepts of classifications of design of experiments which plays very important roles in the field of agriculture and statistical quality control.

COURSE OUTCOMES:

Upon successful completion of the course, students will be able to:

- Explain the fundamental concepts of probability and standard distributions which can describe real life phenomenon.
- Explain the basic concepts of one and two dimensional random variables and their applications in engineering.
- Apply the concept of testing of hypothesis for small and large samples in real life problems.
- Apply the basic concepts of classifications of design of experiments in the field of agriculture and statistical quality control.
- Discuss the notion of sampling distributions and statistical techniques used in engineering and management problems.
- Discuss about the techniques in quality control that model engineering problems.

UNIT I- PROBABILITY AND RANDOM VARIABLES**12**

Probability – The axioms of probability – Conditional probability – Baye's theorem - Discrete and continuous random variables – Moments – Moment generating functions – Binomial, Poisson, Geometric, Uniform, Exponential and Normal distributions.

UNIT II- TWO - DIMENSIONAL RANDOM VARIABLES**12**

Joint distributions – Marginal and conditional distributions – Covariance – Correlation and linear regression using SPSS tool– Transformation of random variables – Central limit theorem (for independent and identically distributed random variables).

UNIT III- TESTING OF HYPOTHESIS**12**

Sampling distributions - Estimation of parameters - Statistical hypothesis - Large sample tests based on Normal distribution for single mean and difference of means -Tests based on t, Chisquare and F distributions for mean, variance and proportion - Contingency table (test for independent) - Goodness of fit.

UNIT IV- DESIGN OF EXPERIMENTS**12**

One way and Two way classifications - Completely randomized design – Randomized block design – Latin square design – 2^2 factorial design using SPSS tool.

UNIT V- STATISTICAL QUALITY CONTROL**12**

Control charts for measurements (X and R charts) – Control charts for attributes (p, c and np charts) – Tolerance limits - Acceptance sampling.

Total periods : 60**TEXT BOOKS:**

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Johnson, R.A., Miller, I and Freund J	Miller and Freund's Probability and Statistics for Engineers	Pearson Education, Asia	8th Edition, 2015
2	Milton. J. S. and Arnold. J.C.	Introduction to Probability and Statistics	Tata McGraw Hill	4th Edition, 2007
3	Devore. J.L.	Probability and Statistics for Engineering and the Sciences	Cengage Learning, New Delhi	8th Edition, 2014

REFERENCE BOOKS:

S. NO.	AUTHOR(S) NAME	TITLE OF THE BOOK	PUBLISHER	YEAR OF PUBLICATION
1	Papoulis, A. and Unnikrishna pillai,S.	Probability, Random Variables and Stochastic Processes	McGraw Hill Education India	4th Edition, 2010.
2	Ross, S.M.	Introduction to Probability and Statistics for Engineers and Scientists	Elsevier	3rd Edition, 2004.

WEBSITES:

1. www.cut-theknot.org/probability.shtml
2. www.mathworld. Wolfram.com
3. www.mathcentre.ac.uk

COURSE OBJECTIVES:

The goal of this course is for students:

- To provide insights of modeling techniques of physiological systems
- To discuss the concept of different systems of body in mathematical model.
- To demonstrate the working of physiological system in terms of equations.
- To define the parameters involved in thermal regulatory system
- To apply mathematical modeling principles in understanding the various fundamental biological systems
- To study the system concept of biological control

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Develop the application of basic physiological system
- Examine the concept of circulatory system
- Analyse the process of regulation
- Describe the parameters involved in thermal regulatory system
- Discuss the process of ultra-filtration system
- Explain the mechanism of respiration

UNIT I - BASICS OF PHYSIOLOGICAL SYSTEMS**9**

Systems Analysis, examples of physiological control systems, differences between engineering and physiological control systems. Generalized system properties, mathematical approximate electrical analogy, linear models, lung mechanics, muscle mechanics, distributed parameter versus lumped parameter models, static analysis, regulation of cardiac output, blood glucose regulation, chemical regulation of ventilation, electrical model of neural control mechanism

UNIT II - CIRCULATORY SYSTEM**9**

Physical, chemical and rheological properties of blood, problems associated with extracorporeal blood flow, dynamics of circulatory system.

UNIT III - THERMAL REGULATORY SYSTEM**9**

Parameters involved, Control system model etc. Biochemistry of digestion, types of heat from body, models of heat transfer between subsystem of human body like skin core, etc systems like within body, body- environment, etc.

UNIT IV - ULTRA FILTRATION SYSTEM**9**

Transport through cells and tubules, diffusion, facilitated diffusion and active transport, method waste removal, counter current model of urine formation in nephron, Modeling Henle's loop

UNIT V- RESPIRATORY SYSTEM**9**

Modelling oxygen uptake by RBC and pulmonary capillaries, Mass balancing by lungs, Gas transport mechanisms of lungs, oxygen and carbon di oxide transport in blood and tissues

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	David O. Cooney	Biomedical Engineering Principles	Marcel Decker Pub.Co	2000
2	Michael C.K.Kho	Physiological Control Systems	Prentice Hall of India	2018

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	John Enderly Susan Blanchard, Joseph Bronzino	Introduction to Biomedical Engineering	Second Edition, Academic Press Series in	2012
2	Claudio Cobelli, Ewart Carson	Introduction to Modeling in Physiology and Medicine	Academic Press	2019

WEBSITES:

- <https://nptel.ac.in/courses/107106081>
- <https://ocw.mit.edu/courses/hst-542j-quantitative-physiology-organ-transport-systems-spring-2004/pages/readings/>

COURSE OBJECTIVES:

The goal of this course is for students

- To define the basic concepts and characteristics of transducers and sensors
- To comprehend the principle of sensors
- To learn the principle of transduction, classifications and the characteristics of different transducers
- To discuss the working transducers and sensors.
- To infer the different display and recording devices.
- To build the biomedical applications using the transducers and sensors.

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

- Compare the general properties and characteristics of transducers and sensors.
- Demonstrate the various transducers and sensors in the course.
- Describe the purpose and methods of measurements.
- Explain the principle of different sensors and its applications
- Make use of the transducers and sensors learnt in the course in suitable medical contexts.
- Implement the working knowledge of some of the transducers and sensors

UNIT I -INTRODUCTION TO TRANSDUCERS AND ITS CHARACTERISTICS 9

Introduction: Generalized Instrumentation System, General Properties of Input Transducer Static Characteristics: Accuracy, Precision, Resolution, Reproducibility, Sensitivity, Drift, Hysteresis, Linearity, Input Impedance and Output Impedance. Dynamic Characteristics: First Order and Second Order Characteristics, Time Delay, Error Free Instrument, Transfer Functions. Design Criteria, Generalized Instrument Specifications.

UNIT II- MEASUREMENTS 9

Displacement, motion and Pressure Measurement: (with applications) Resistive: Potentiometers, Strain Gauges and Bridge Circuits. Inductive: Variable Inductance and LVDT Capacitive type, Piezoelectric Transducers. Types of Diaphragms, Bellows, Bourdon Tubes.

UNIT III- THERMAL MEASUREMENTS 9

Temperature Measurement: Thermistor, Thermocouple, Resistive Temperature Detector, IC based Temperature Measurement, Radiation Sensors and Applications .

UNIT IV - SENSOR FABRICATION**9**

Sensor Fabrications – an overview, Fabrication Technique – Pressure sensor – Piezoresistive pressure sensor – capacitive pressure sensor – Micro sensor – Biomedical Application.

UNIT V- BIOSENSORS**9**

Chemical Sensors: Blood gas and Acid- Base Physiology Potentiometric Sensors, Ion Selective Electrodes, ISFETS. Amperometric Sensors, Clark Electrode with examples - pH, pO₂, pCO₂ Electrodes, Transcutaneous Arterial Oxygen Tension, Carbon Dioxide measurements: capnostat. Biosensor: Classifications: Biological phenomenon, Transduction Phenomenon i.e. Enzyme Sensor and Electrode based: Affinity Sensors (Catalytic Biosensors), Two examples of each Biosensors and Immunosensors.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Hermann K P. Neubert	Instrument Transducer– An Intro to their performance and design	Hermann K P. Neubert	2000
2	Harry N, Norton.	Biomedical sensors – fundamentals and application	Harry N, Norton.	2001
3	Nandini K	Electronics in Medicine and Biomedical Instrumentation	Jog PHI Second Edition	2013

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	LaGeddes and L.E. Baker	Principles of applied Biomedical Instrumentation	LaGeddes and L.E. Baker	1997
2	Leslie Cromwell, Fred. J. Weibell and Pfeiffer	Biomedical instrumentation and measurement	Leslie Cromwell, Fred. J. Weibell and Pfeiffer	2002
3	Jacob Fraden	Handbook of Modern Sensors – Physics, Design and Application	AIP press	2000

WEBSITES:

- <https://nptel.ac.in/courses/1081081471>.

COURSE OBJECTIVES:

The goal of this course is for students :

- To infer processor architecture and its programming
- To apply the interfacing concepts
- To analyse advanced processor architecture
- To utilize the concepts of Interfacing with Peripherals for building applications
- To demonstrate the concepts Reduced Instruction Set Computer (RISC) architecture and Advanced RISC Machine (ARM) architecture.
- To develop the skill to explore system design technique

COURSE OUTCOMES:

At the end of this course students will demonstrate the ability to

- Interpret the architecture, instruction set, memory organization and addressing modes
- Apply the knowledge of microprocessor, microcontroller and ARM for implementing assembly language programming.
- Match the hardware interfacing peripheral devices.
- Develop applications using microprocessor, microcontrollers and ARM Processor
- Compare the architectures of Reduced Instruction Set Computer (RISC) and Advanced RISC Machine (ARM) processors
- Build the Interfacing peripherals in real time scenario.

UNIT I- 8086 MICROPROCESSOR**9**

Introduction to 8086 – Microprocessor architecture – Addressing modes - Instruction set and assembler directives – Assembly language programming – Modular Programming - Linking and Relocation – Stacks - Procedures – Macros – Interrupts and interrupt service routines – Byte and String Manipulation.

UNIT II - 8051 MICROCONTROLLER**9**

Architecture of 8051 – Special Function Registers (SFRs) - I/O Pins Ports and Circuits - Programming 8051 Timers - Serial Port Programming - Interrupts Programming - Instruction set - Addressing modes - Assembly language programming.

UNIT III- I/O INTERFACING**9**

LCD & Keyboard Interfacing - ADC, DAC & Sensor Interfacing - Stepper Motor and Waveform generation, Serial communication interface, Traffic Light control.

UNIT IV- HIGH PERFORMANCE RISC ARCHITECTURE – ARM**9**

Organization of CPU – Bus architecture –Memory management unit - ARM instruction set- Thumb Instruction set- addressing modes – Programming the ARM processor.

UNIT V - ARM APPLICATION DEVELOPMENT**9**

Peripheral Interface – Application of ARM Processor - Caches – Memory protection Units – Memory Management units – Future ARM Technologies

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	R. S. Gaonkar	Microprocessor Architecture: Programming and Applications with the 8085	Penram International Publishing	2000
2	D A Patterson and J H Hennessy	Computer Organization and Design The hardware and software	Morgan Kaufman Publishers	2018
3	Douglas Hall	Microprocessors Interfacing	Tata McGraw Hill	2017

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Jonathan W Valvano	Introduction to Arm(r) Cortex-M Microcontrollers	Clifton Park, NY : Thomson Delmar Learning	2005.
2	Sergio Franco	Design with operational amplifier and analog integrated circuits	Create space Independent Publisher	2016

WEBSITE:

<https://nptel.ac.in/courses/117104072>
<https://nptel.ac.in/courses/106108100>
<https://nptel.ac.in/courses/108105102>

COURSE OBJECTIVES**COURSE OBJECTIVES**

The goal of this course is for students :

- To explain the basic properties of signal and systems
- To understand the signal types, properties and analysis
- To analyze continuous time signals
- To analyze discrete time signals
- To find the concepts of Fourier Transform in signal analysis.
- To demonstrate the DTFT and Z –transforms for discrete signals

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Apply Laplace and Fourier transform for continuous signals
- Analyze continuous time LTI systems using Fourier and Laplace Transforms
- Apply DTFT and Z transforms for discrete signals
- Apply Fourier series and Transforms on signals
- Classify the continuous and discrete systems in time and frequency domain
- Analyze discrete time LTI systems using Z transform and DTFT

UNIT I- CLASSIFICATION OF SIGNALS AND SYSTEMS**9**

Continuous time signals (CT signals) - Discrete time signals (DT signals) - Step, Ramp, Pulse, Impulse, Sinusoidal, Exponential, Classification of CT and DT signals - Periodic & Aperiodic signals, Deterministic & Random signals, Energy & Power signals - CT systems and DT systems- Classification of systems – Static & Dynamic, Linear & Nonlinear, Time-variant & Time-invariant, Causal & Noncausal, Stable & Unstable.

UNIT II -ANALYSIS OF CONTINUOUS TIME SIGNALS**9**

Fourier series analysis-spectrum of Continuous Time (CT) signals- Fourier and Laplace Transforms in CT Signal Analysis - Properties.

UNIT III- LINEAR TIME INVARIANT- CONTINUOUS TIME SYSTEMS**9**

Differential Equation-Block diagram representation-impulse response, convolution integrals- Fourier and Laplace transforms in Analysis of CT systems.

UNIT IV- ANALYSIS OF DISCRETE TIME SIGNALS**9**

Baseband Sampling - DTFT – Properties of DTFT - Z Transform – Properties of Z Transform

UNIT V - LINEAR TIME INVARIANT-DISCRETE TIME SYSTEMS**9**

Difference Equations-Block diagram representation-Impulse response - Convolution sum-Discrete. Fourier and Z Transform Analysis of Recursive & Non-Recursive systems.

Total periods : 45**TEXTBOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of Publication
1	S Salivahanan	Signals and Systems	McGraw Hill Education	2018
2	B. P. Lathi	Principles of Linear Systems and Signals	Second Edition Oxford	2009

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Allan V. Oppenheim, S. Wilsky and S. H. Nawab	Signals and Systems	Pearson	2007
2	R. E. Zeimer, W. H. Tranter and R. D. Fannin	Signals & Systems Continuous and Discrete	Pearson	2007
3	John Alan Stuller	An Introduction to Signals and Systems	Thomson	2007
4	M. J. Roberts	Signals & Systems Analysis using Transform Methods & MatLab	Tata McGraw Hill	2003

WEBSITES:

1. https://nptel.ac.in/courses/117104074
2. https://nptel.ac.in/courses/117101055

COURSE OBJECTIVES:

The goal of this course is for students:

- To apply the concept of cell biology.
- To explain the structural and functional aspects of living organisms.
- To build the etiology and remedy in treating the pathological diseases.
- To define the importance of public health.
- To explain the structural and functional aspects of living organisms.
- To know the etiology and remedy in treating the pathological diseases.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Define about the structural and functional aspects of living organisms.
- Demonstrate about the function of microscope
- Elaborate about the importance of public health.
- Explain about the methods involved in treating the pathological diseases
- Build knowledge on Disease caused by bacteria, fungi and protozoa
- Compare natural and artificial immunity

UNIT I- CELL DEGENERATION, REPAIR AND NEOPLASIA**9**

Cell injury - Reversible cell injury and Irreversible cell injury and Necrosis, Apoptosis, Intracellular accumulations, Pathological calcification- Dystrophic and Metastatic. cellular adaptations of growth and differentiation, Inflammation and Repair including fracture healing, Neoplasia, Classification, Benign and Malignant tumours, carcinogenesis, spread of tumours Autopsy and biopsy.

UNIT II- FLUID AND HEMODYNAMIC DERANGEMENTS**9**

Edema, Hyperemia /Ischemia, normal hemostasis, thrombosis, disseminated intravascular coagulation, embolism, infarction, shock, Chronic venous congestion. Hematological disorders Bleeding disorders, Leukaemias, Lymphomas Haemorrhage.

UNIT III - MICROBIOLOGY**9**

Structure of Bacteria and Virus. Routes of infection and spread; endogenous and exogenous infections, Morphological features and structural organization of bacteria and virus, growth curve, identification of bacteria , culture media and its types , culture techniques and observation of culture. Disease caused by bacteria, fungi, protozoal, virus and helminthes.

UNIT IV- MICROSCOPES**9**

Light microscope – bright field, dark field, phase contrast, fluorescence, Electron microscope & SEM). Preparation of samples for electron microscope. Staining methods – simple, gramstaining and AFB staining.

UNIT V- IMMUNOPATHOLOGY**9**

Natural and artificial immunity, types of Hypersensitivity, antibody and cell mediated tissue injury: opsonization, phagocytosis, inflammation, Secondary immunodeficiency including HIV infection. Auto-immune disorders: Basic concepts and classification, SLE. Antibodies and its types, antigen and antibody reactions, Sterilization: heat, filtration, chemicals and radiation, immunological techniques: immune diffusion, immuno electrophoresis, RIA and ELISA, monoclonal antibodies.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Ramzi S Cotran Vinay Kumar & Stanley L Robbins	Pathologic Basis of Diseases	7th edition, WB Saunders Co.	2005
2	Ananthanarayanan & Panicker	Microbiology	10 th edition Orient black swan	2017

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Underwood	JCE: General and Systematic Pathology	Churchill Livingstone, 3rd edition,	2000
2	Dube RC and Maheswari DK	A Text Book of Microbiology	Chand & Com	2007

WEBSITES:

1. <https://nptel.ac.in/courses/102103015>
2. <https://ocw.mit.edu/courses/hst-035-principle-and-practice-of-human-pathology-spring-2003/>
3. <https://ocw.mit.edu/courses/20-106j-systems-microbiology-fall-2006/pages/lecture-notes/>

COURSE OBJECTIVES:

The goal of this course is for students,

- To learn about data acquisition of physiological signal
- To learn about various biosensors
- To distinguish contact and non-contact method of skin temperature measurement
- To outline the working principle of Transducers
- To experiment with basic functions of Biosensors.
- To identify different physiological signals.

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Interpret the characteristics of various biosensors
- Illustrate the importance of the sensors and transducers for medical applications.
- Analyse the characteristics of physiological signals
- Identify skin temperature
- Solve data acquisition of physiological signals
- Choose the biosensors for relevant application

LIST OF EXPERIMENTS

1. Study of Biological Sensors
2. Displacement measurement using LVDT
3. Characteristics of temperature sensors – thermistor and RTD.
4. Characteristics of thermocouple
5. Characteristics of Flow Transducer
6. Characteristics of photo diodes, phototransistor
7. Characteristics of Piezoelectric Transducer.
8. Data acquisition of physiological signals
9. Measurement of skin temperature by both contact and non-contact method.
10. Study of the characteristics of capacitor level sensor for saline level measurement in a I-V set.

**22BEBME412 MICROPROCESSOR AND MICROCONTROLLER LABORATORY
2H-1C****Instruction Hours/week: L: 0 T: 0 P:2****Marks: Internal:40 External:60 Total:100****End Semester Exam:3 Hours****COURSE OBJECTIVES:**

The goal of this course is for students,

- To outline ALP concepts and features
- To construct programs on 8085 microprocessor
- To build ALP for arithmetic and logical operations in 8086 and 8051
- To classify Serial and Parallel Interface
- To relate different I/O's with Microprocessors
- To experiment with MASM

COURSE OUTCOME:

Upon completion of this course, students will be able to:

- Build ALP Program for fixed and Floating Point and Arithmetic operations
- Relate different I/O's with processor
- Develop waveforms using Microprocessors
- Examine Programs in 8051
- Apply Serial and Parallel Interface
- Demonstrate A/D and D/A conversion

LIST OF EXPERIMENTS

1. Programs for 8/16 bit Arithmetic operations (Using 8085 and 8086).
2. Programs for Sorting and Searching (Using 8086).
3. Programs for String manipulation operations (Using 8086).
4. Programs on Subroutines (Using 8051).
5. Interfacing ADC and DAC (Using MSP 430 Controllers/8051).
6. Interfacing with 8255.
7. Transfer data serially between two kits (8253/8251).
8. Interfacing with 8279.
9. Traffic Control Using MSP 430 controller/8051.
10. Interfacing with 8259 Programmable Interrupt Controller.
11. Interfacing and Programming of Stepper Motor and (8051).
12. Programming using Arithmetic, Logical & Bit Manipulation instructions of 8051 microcontroller.

COURSE OBJECTIVES

The goal of this course is for students :

- To utilize the Compound microscope
- To examine on chemical examinations,
- To examine Cryo processing,
- To demonstrate the histopathological examinations
- To compare anemia and leukemia
- To classify bone marrow charts

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Plan practical experiments on tissue processing using Compound microscope.
- Analyse the concept of cryoprocessing.
- Analyse the staining Processes
- Compare anemia and leukemia
- Build bone marrow charts
- Explain about Antigen-Antibody reaction

LIST OF EXPERIMENTS:

1. Urine physical and chemical examination (protein, reducing substances, ketones, bilirubin and blood)
2. Study of parts of compound microscope
3. Histopathological slides of benign and malignant tumours.
4. Manual paraffin tissue processing and section cutting (demonstration)
5. Cryo processing of tissue and cryosectioning (demonstration)
6. Basic staining – Hematoxylin and eosin staining.
7. Special stains – cresyl fast Blue (CFV)- Trichrome – oil red O – PAS
8. Capsule stain
9. Simple stain.
10. Gram stain.
11. AFB stain.
12. Antigen-Antibody reaction Immuno electrophoresis
13. Slides of malarial parasites, micro filaria and leishmania donovani.
14. Haematology slides of anemia and leukemia.
15. Study of bone marrow charts.

COURSE OBJECTIVE:

The goal of this course is for students:

- To elevate the students into productivity powerhouses who can employ life skills to better their performances.
- To examine the interpersonal skills.
- To take part in building interpersonal skills.
- To outline the ability to work with others.
- To identify students to right attitudinal and behavioural aspects
- To build the right attitudinal through activities

INTENDED OUTCOMES

Upon completion of this course, the student will be able to:

- Plan and communicate smartly and effectively with co-workers, relationship enhancement
- Examine time management and organizational skill.
- Development of leadership teamwork, creativity, efficiency & productivity
- Development of presentation skills
- Demonstrate stress symptom & develop stress deflecting strategies
- Summarize Brain storming & problem solving strategies to increase creativity and collaborative outcomes

UNIT I**(4)**

Overview to communication, self-Introduction, Presentation on their own topic, Extempore, Group Activity (Verbal Reasoning – Proposition)

UNIT II**(3)**

Group Discussion, Do's and Don'ts of Group Discussion, Body language, Grooming and Resume, Resume correction, Activity (Verbal Reasoning - Premise)

UNIT III**(4)**

Introduction to HRM – Questions - Do's and Don'ts - Interview - Mock GD – Stress Management, Activity (Verbal Reasoning – Syllogism)

UNIT IV**(4)**

Personality Development - Presentation skills, Interpersonal skills, Critical thinking, Confidence building and Stress management, Activity (Verbal Reasoning – Verbal Analogies)

Total periods : 15

TEXTBOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of Publication
1	Barun K Mitra	Personality Development and SoftSkills	Oxford University Press-New Delhi	2012
2	Rajiv K Mishra	Personality Development	Rupa & Co	2012

B.E Biomedical Engineering**2022-2023****Semester-IV****22BEBME491****INTERNSHIP****2C****Instruction Hours/week: L:0 T:0 P:0****Marks: Internal:100 External:0 Total:100****End Semester Exam:3 Hours**

SEMESTER V

B.E Biomedical Engineering

2022-2023

Semester-V

22BEBME501

TELEMEDICINE

3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To infer the key principles for telemedicine and health.
- To define telemedical technology.
- To build telemedicine standards, mobile telemedicine and its applications.
- To explain the principles of clinical telehealth
- To interpret the scope, benefits and limitations of Telemedicine and security in telemedicine applications
- To classify e-health and cyber medicine

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain the basic principle and technology underlying in the telemedicine.
- Explain Protocols behind encryption techniques for secure transmission of data.
- Apply multimedia technologies in telemedicine.
- Outline the basic concepts involved in telemetry based transmission and reception
- Compare the communication devices and Networks of telemedicine
- Apply telehealth technologies in various fields of healthcare with its ethical and legal aspects.

UNIT I - FUNDAMENTALS OF TELEMEDICINE

9

History of telemedicine, definition of telemedicine, tele-health, tele-care, scope, Telemedicine Systems, benefits & limitations of telemedicine.

UNIT II - TYPE OF INFORMATION & COMMUNICATION INFRASTRUCTURE FOR TELEMEDICINE

9

Audio, video, still images, text and data, fax-type of communications and network: PSTN, POTS, ANT, ISDN, internet, air/ wireless communications, GSM satellite, micro wave, Mobile health and ubiquitous healthcare.

UNIT III - ETHICAL AND LEGAL ASPECTS OF TELEMEDICINE

9

Confidentiality, patient rights and consent: confidentiality and the law, the patient doctor relationship, access to medical records, consent treatment - data protection & security, jurisdictional issues, intellectual property rights.

UNIT IV - PICTURE ARCHIVING AND COMMUNICATION SYSTEM

9

Introduction to radiology information system and ACS, DICOM, PACS strategic plan and needs assessment, technical Issues, PACS architecture.

UNIT V - APPLICATIONS OF TELEMEDICINE**9**

Teleradiology, Telepathology, Telecardiology, Teleoncology, Teledermatology, Telesurgery, e Health and Cyber Medicine.

Total periods: 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Khandpur R S	TELEMEDICINE– Technology and Applications	PHI Learning Pvt Ltd.	2017
2	Norris A C	Essentials of Telemedicine and Telecare	John Wiley, New York	2002

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	H K Huang	PACS and Imaging Informatics: Basic Principles and Applications	Wiley, New Jersey	2010
2	Richard Wootton, John Craig, Victor Patterson	Introduction to Telemedicine	Taylor & Francis	2017

WEBSITES:

- <https://ocw.mit.edu/courses/hst-s14-health-information-systems-to-improve-quality-of-care-in-resource-poor-settings-spring-2012/pages/lectures-and-videos/>
- <https://www.coursera.org/lecture/healthcare-it/module-4-telehealth-IKo2k>

COURSE OBJECTIVES:

The goal of this course is for students:

- To illustrate the origin of bio potentials and its propagations
- To apply the basic idea of Bio potential Electrodes and Bio potential measurement.
- To analyze the different types of electrodes and its placement for various recordings
- To build bio amplifier for various physiological recordings
- To explain the various bio chemical measurements.
- To analyze the different measurement techniques for non-physiological parameters.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Compare bio potentials and its propagations.
- Outline the electrode behaviour and circuit models
- Illustrate different electrode placement for various physiological recordings
- Build bio amplifier for various physiological recordings
- Explain various technique for non-electrical physiological measurements
- Classify various biochemical parameters.

UNIT I- BIOPOTENTIAL ELECTRODES**9**

Origin of bio potential and its propagation. Electrode-electrolyte interface, electrode-skin interface, half-cell potential, Contact impedance, polarization effects of electrode – non polarizable electrodes. Types of electrodes - surface, needle and micro electrodes and their equivalent circuits. Biochemical transducers – transducers for biomedical applications Recording problems - motion artifacts, measurement with two electrodes.

UNIT II- BIOPOTENTIAL MEASUREMENTS**9**

Bio signals characteristics – frequency and amplitude ranges. ECG – Einthoven's triangle, standard 12 lead system, Principles of vector cardiograph. EEG – 10-20 electrode system, unipolar, bipolar and average mode. EMG– unipolar and bipolar mode. Recording of ERG, EOG and EGG

UNIT III- SIGNAL CONDITIONING CIRCUITS**9**

Need for bio-amplifier - single ended bio-amplifier, differential bio-amplifier, Impedance matching circuit, isolation amplifiers – transformer and optical isolation - isolated DC amplifier and AC carrier amplifier., Power line interference, Right leg driven ECG amplifier, Band pass filtering

UNIT IV- MEASUREMENT OF NON-ELECTRICAL PARAMETERS**9**

Blood Pressure: indirect methods - Auscultatory method, oscillometric method, direct methods: electronic manometer, Pressure amplifiers, Systolic, diastolic, mean detector circuit. Blood flow and cardiac output measurement: Indicator dilution, thermal dilution and dye dilution method, Electromagnetic and ultrasound blood flow measurement. Measurement of heart sounds – phonocardiography. Cardiac pacemakers – internal and external pacemakers, defibrillators. Plethysmography –photo electric and impedance plethysmographs

UNIT V- BIOCHEMICAL MEASUREMENT AND BIOSENSORS**9**

Biochemical sensors - pH, pO₂ and pCO₂, Ion selective Field effect Transistor(ISFET), Immunologically sensitive FET (IMFET), Blood glucose sensors, Blood gas analyzers - colorimeter, Sodium Potassium Analyser, spectrophotometer, blood cell counter, auto analyser (simplified schematic description) – Bio Sensors – Principles – amperometric and voltometric techniques.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Khandpur R.S	Handbook of Biomedical Instrumentation	Tata McGraw-Hill, New Delhi	2014
2	John G. Webster	Medical Instrumentation Application and Design	John Wiley and sons	2020

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Leslie Cromwell	Biomedical Instrumentation and	Prentice hall of India, New Delhi	2007
2	Myer Kutz	Standard Handbook of Biomedical Engineering & Design	McGraw-Hill	2003

WEBSITES:

- <https://ocw.mit.edu/courses/20-309-biological-engineering-ii-instrumentation-and-measurement-fall-2006/pages/syllabus/>

COURSE OBJECTIVES

The goal of this course is for students:

- To find the characteristics of different bio signals.
- To find the linear and non-linear filtering techniques to extract desired information
- To analyse the characteristics of some of the most commonly used biomedical signals, including ECG, EEG, EOG, and EMG.
- To match the choice of filters to remove noise and artefacts from biomedical signals.
- To apply engineering methods to analyse ECG signal problems.
- To interpret the engineering methods and analyze the neurological signals.

COURSE OUTCOMES

Upon completion of this course, the student will be able to:

- Compare different types of biomedical signals and identify their spectral components.
- Identify different filters on biomedical signals and judge filter performance.
- Identify physiological interferences and artifacts affecting ECG signal.
- Compare power and correlation spectra of EEG signal.
- Analyse bio signals in time and frequency domains.
- Outline the classification and recognition Bio signals

UNIT I- INTRODUCTION TO BIOMEDICAL SIGNALS**9**

Biosignal Characteristics of Electro Cardiogram (ECG), Electroencephalogram (EEG), Electromyogram (EMG), Electrooculogram (EOG), Electroretinogram (ERG), Electrogastrogram (EGG), Electroneurogram (ENG), Event related potentials (ERPs), Phonocardiogram (PCG), Speech signal, Objectives of Biomedical signal analysis, Difficulties in Biomedical signal analysis, Computer-aided diagnosis.

UNIT II- FILTERING FOR REMOVAL OF ARTIFACTS**9**

Time-domain Filters - synchronized averaging, Moving Average Filters, Derivative-based operators to remove low-frequency artifacts. Frequency-domain filters - Removal of High Frequency noise, Removal of low frequency noise, Removal of periodic artifacts, optimal filter-Wiener filter, Adaptive filters for removal of interference.

UNIT III- CARDIOVASCULAR APPLICATIONS**9**

Noise & Artifacts, ECG Signal Processing: Baseline Wandering, Power line interference, Muscle noise filtering – QRS detection, Adaptive noise cancelling in ECG, improved adaptive filtering in FEKG, Wavelet detection in ECG – structural features, matched filtering, adaptive wavelet detection, detection of overlapping wavelets. Wavelet transforms - mother wavelet, Computation of diagnostic signal parameters of ECG like Heart rate and QRS detection using Multivariate analysis (PCA and ICA). Segmentation of PCG, intensity patterns, Spectral modelling and analysis of PCG signals.

UNIT IV- NEUROLOGICAL APPLICATIONS**9**

EEG rhythms & waveforms, EEG applications- Epilepsy, sleep disorders, brain computer interface. Modelling EEG- linear, stochastic models - Nonlinear modelling of EEG - artifacts in EEG & their characteristics and processing – Nonparametric spectral analysis, Model based spectral analysis - EEG segmentation - Joint Time-Frequency analysis - correlation analysis of EEG channels - coherence analysis of EEG channels. Evoked potentials- noise characteristics, Noise reduction by linear filtering.

UNIT V- ANALYSIS ON WAVESHAPE, SIGNAL CLASSIFICATION AND RECOGNITION**9**

Modelling intramuscular EMG-Intramuscular signal decomposition-Fractal analysis of EMG signals. Statistical analysis of VAG signals. Analysis on amplitude and latency of MEG signals. Analysis of ERP effect. Signal classification and recognition – Statistical signal classification, linear discriminant function, direct feature selection and ordering, Back propagation neural network based classification. Analysis of EEG using Empirical mode decomposition (EMD)

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	John G. Proakis and Dimitris G.Manolakis	Digital Signal Processing, Algorithms and Applications	PHI of India Ltd., New Delhi, 3rd Edition	2021
2	Rangaraj.M.Rangayyan	Biomedical signal Analysis	Wiley-IEEE Press	2015

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Sanjit K.Mitra	Digital Signal Processing', A Computer Based Approach	Tata McGraw-Hill, New Delhi	2001
2	D.C.Reddy	Biomedical Signal Processing, Principles and Technique	TMH, New Delhi	2005

WEBSITES:

tel.ac.in/courses/108105101 w.mit.edu/courses/res-6-008-digital-signal-processing-spring-2011/
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COURSE OBJECTIVE

The goal of this course is for students:

- To develop managerial and entrepreneurial skills our culture and ethics
- To compare Direct and Preventive Control
- To apply knowledge on the principles of management is essential for all kinds of people in all kinds of organisations
- To simplify the understanding of the managerial functions like planning, organizing, staffing, leading and controlling
- To classify the awareness and practice through engineering ethics and human values
- To outline how business ethics works

COURSE OUTCOMES

Upon completion of this course, the student will be able to:

- Apply philosophical knowledge of the profession of recreation and leisure
- Build of trends and issues as related to current professional practice
- Evaluate organizational theories and human resource management principles
- Analyse the information competency
- Classify Ethical practice and ethical management
- Analyze the mmodels of Professional Roles

UNIT I - HISTORICAL DEVELOPMENT, PLANNING, ORGANISING 9

Definition of Management – Management and Administration – Development of Management Thought – Contribution of Taylor and Fayol – Levels and Functions of Management – Steps involved in Planning – Objectives – Setting Objectives – Process of Managing by Objectives – Strategies, Policies and Planning Premises– Forecasting – Decision-making – Formal and informal organization – Organization Chart –.

UNIT II - DIRECTING AND CONTROLLING 9

Human Factors – Creativity and Innovation – Harmonizing Objectives – Leadership – Types of Leadership Motivation – Hierarchy of needs – Motivation theories – Motivational Techniques – Job Enrichment –Process of Communication – System and process of Controlling – Requirements for effective control – Control of Overall Performance – Direct and Preventive Control – Reporting

UNIT III - ENGINEERING ETHICS 9

Senses of 'Engineering Ethics' – variety of moral issued – types of inquiry – moral dilemmas – moral autonomy – Kohlberg's theory – Gilligan's theory – consensus and controversy – Models of Professional Roles – theories about right action – Self-interest – customs and religion – uses of ethical theories, case studies. Engineers as Managers – Consulting Engineers – Engineers as Expert Witnesses and Advisors -Moral Leadership -Code of Conduct – Corporate Social Responsibility.

UNIT IV- FACTORS OF CHANGES**9**

Forces that shape culture, social control – Meaning, Agencies, Institution, Customs, Values, Folkways, Norms and Laws. Social changes – Meaning and nature – theories, Gender Sensitivity.

UNIT V - ENTREPRENEURSHIP AND MOTIVATION**9**

Entrepreneur – Types of Entrepreneurs – Difference between Entrepreneur and Intrapreneur – Entrepreneurship in Economic Growth– Major Motives Influencing an Entrepreneur – Achievement Motivation Training, self-rating, Business Game, Thematic Apperception Test – Stress management, Entrepreneurship Development Programs – Need, Objectives, Startup – History of the startup terminology, 5 Steps from Concept to Startup, Special Considerations – Business Incubators: Meaning, Definition, Services, Development and Types.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Harold Kooritz and Heinz Weihrich	Essentials of Management	Tata McGraw Hill, New Delhi	2010
2	Khanka S.S	Entrepreneurial Development	S.Chand and Co. Ltd., New Delhi	2006
3	Mike Martin and Roland Schinzinger	Ethics in Engineering	McGraw–Hill, New York	2005

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Tripathy P.C and Reddy P.N	Principles of Management	Tata McGraw Hill, New Delhi	2008
2	Rabindra N Kanungo	Entrepreneurship and innovation	Sage Publications, New Delhi	1998
3	Charles E Harris, and Michael J Rabins	Engineering Ethics – Concepts and Cases	Wadsworth Thompson Learning, New	2013

WEBSITES:

- http://www.managementstudyguide.com/taylor_fayol.htm
- http://tutor2u.net/business/gcse/people_motivation_theories.htm
- <http://lfkbb.tripod.com/eng24/gilliganstheory.html>
- <http://www.developingeyes.com/five-types-of-entrepreneurs/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To discuss problems that are amenable to solution by AI methods.
- To analyze given problem in the language/framework of different AI methods.
- To summarize appropriate AI methods to solve a given problem.
- To solve problems that are amenable to solution by AI methods drugs
- To analyse empirical evaluation of different algorithms
- To define knowledge inference

COURSE OUTCOMES:

Upon completion of this course, the students will be able to:

- Identify problems that are amenable to solution by AI methods.
- Identify appropriate AI methods to solve a given problem.
- Interpret a given problem in the language/framework of different AI methods.
- Implement basic AI algorithms.
- Build an empirical evaluation of different algorithms on a problem formulation, and state the conclusions that the evaluation supports.
- Explain about expert systems

UNIT I- INTRODUCTION TO AI AND PRODUCTION SYSTEMS**9**

Introduction to AI-Problem formulation, Problem Definition -Production systems, Control strategies, Search strategies. Problem characteristics, Production system characteristics - Specialized production system- Problem solving methods – Problem graphs, Matching, Indexing and Heuristic functions -Hill Climbing-Depth first and Breath first, Constraints satisfaction – Related algorithms, Measure of performance and analysis of search algorithms.

UNIT II- REPRESENTATION OF KNOWLEDGE**9**

Game playing – Knowledge representation, Knowledge representation using Predicate logic, Introduction to predicate calculus, Resolution, Use of predicate calculus, Knowledge representation using other logic-Structured representation of knowledge. Control Systems - Artificial Intelligence based optimal control - Reinforcement learning

UNIT III- KNOWLEDGE INFERENCE**9**

Knowledge representation -Production based system, Frame based system. Inference – Backward chaining, Forward chaining, Rule value approach, Fuzzy reasoning – Certainty factors, Bayesian Theory-Bayesian Network-Dempster – Shafer theory.

UNIT IV- PLANNING AND MACHINE LEARNING**9**

Basic plan generation systems – Strips -Advanced plan generation systems – K strips -Strategic explanations -Why, Why not and how explanations. Learning- Machine learning, adaptive Learning.

UNIT V- EXPERT SYSTEMS**9**

Expert systems – Architecture of expert systems, Roles of expert systems – Knowledge Acquisition – Meta knowledge, Heuristics. Typical expert systems – MYCIN, DART, XOON, Expert system shells.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Peter Jackson	Introduction to Expert Systems	3rd Edition, Pearson Education,	2007
2	Stuart Russel and Peter Norvig	“AI – A Modern Approach”	2nd Edition Pearson Education 2007.	2020

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Deepak Khemani	Artificial Intelligence	Tata McGraw Hill Education	2013.
2	N.P.Padhy	Artificial Intelligence and Intelligent Systems	Oxford University Press	2006

WEBSITES:

<https://nptel.ac.in/courses/106105077>

<https://ocw.mit.edu/courses/6-034-artificial-intelligence-spring-2005/> www.nptel.com

B.E Biomedical Engineering

2022-2023

Semester-V

22BEBME5E--

PROFESSIONAL ELECTIVE I

3H-3C

Instruction Hours/week: L:3 T: 0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To discuss the working principle of Biomedical Instrumentation systems
- To infer the basic acquisition techniques of bioelectric signals.
- To apply different Detection, correlation and averaging of various biomedical signals.
- To construct PCB layout design
- To examine blood pressure measurement
- To examine pH measurement

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Develop preamplifiers and amplifiers for various bio signal recordings.
- Build various non-electrical parameters using suitable sensors/transducers
- Explain about Isolation amplifier
- Demonstrate blood pressure measurement using sphygmomanometer
- Explain about Photo transducer
- Construct PCB layout for any bio amplifier

LABORATORY EXPERIMENTS:

1. Design of ECG Amplifiers with appropriate filter to remove power line and other artifacts.
2. Design of EMG amplifier
3. Design of EOG amplifier to detect eye blink
4. Design of EEG amplifier.
5. Design and study the characteristics of optical Isolation amplifier
6. Measurement of pulse-rate using Photo transducer.
7. Measurement of pH and conductivity.
8. Measurement of blood pressure using sphygmomanometer.
9. Measurement and recording of peripheral blood flow
10. Design a PCB layout for any bio amplifier using suitable software tool.

COURSE OBJECTIVES:

The goal of this course is for students:

- To experiment with basic signal processing operations
- To infer about Discrete time signals
- To demonstrate the filtering operation
- To interpret the architecture of DSP
- To analyse FIR and IIR filters and DSP Processor
- To develop the architecture of DSP processor

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Apply coding for different convolution and correlation techniques.
- Build random noise
- Compare between FIR and IIR filters.
- Analyse the various filtering operation
- Implement sampling operation in DSP Processor
- Experiment with various biomedical signals using Digital Signal Processor.

LABORATORY EXPERIMENTS:

1. Linear and Circular convolutions.
2. Generation of elementary Discrete-Time sequences
3. Auto correlation and Cross Correlation
4. Frequency Analysis using DFT
5. Design of FIR filters (LPF/HPF/BPF/BSF) and demonstrates the filtering operation .
6. Study of architecture of Digital Signal Processor
7. Generation of various signals and random noise.
8. Perform MAC operation using various addressing modes
9. Design of Butterworth and Chebyshev IIR filters (LPF/HPF/BPF/BSF) and demonstrate the filtering operations.
10. Implement an Up-sampling and Down-sampling operation in DSP Processor

COURSE OBJECTIVES:

The goal of this course is for students:

- To experiment with self education and understand the value of achieving perfection in project implementation & completion
- To apply the theoretical concepts to solve industrial problems
- To implement project planning with teamwork and multidisciplinary approach
- To demonstrate professionalism with ethics
- To explain the concepts of Project and financial Management
- To classify the skills and relate engineering issues to broader societal context

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Demonstrate the knowledge, skills and attitudes of a professional engineer
- Identify the problem, formulation and solution.
- Apply societal and cultural issues in professional engineering practice
- Demonstrate the skills to use modern engineering tools and software to solve complex engineering problems.
- Function as a member of multidisciplinary team and Communicate with engineers and the community
- Build on project management and finance

SEMESTER VI

B.E Biomedical Engineering

2022-2023

Semester-VI

22BEBME601

BIOMEDICAL IMAGE PROCESSING

3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To study the image fundamentals and transforms necessary for image processing.
- To represent image in frequency domain Fourier transform
- To learn various image enhancement techniques used in biomedical image Processing
- To study techniques for improving quality of information in corrupted images
- To introduce schemes for compressing images to save storage space
- To gain knowledge about the development of feature extraction algorithms.

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Review the fundamental concepts of an image processing system
- Analyze images in the frequency domain using various transforms
- Evaluate the techniques for image enhancement and image restoration.
- Categorize various compression techniques.
- Interpret Image compression standards.
- Interpret image segmentation and representation techniques.

UNIT I- DIGITAL IMAGE FUNDAMENTAL

9

Elements of digital image processing systems, Elements of Visual perception, structure of human eye and image formation, Image sampling and quantization, Some Basic relationships between pixels, Matrix and Singular Value representation of discrete images.

UNIT II - IMAGE TRANSFORMS

9

2D DFT, Cosine, Sine Hadamard, Haar, Slant, KL transform , Fourier transform and their properties.

UNIT III- IMAGE ENHANCEMENT**9**

Histogram – Modification and specification techniques, Enhancement by point processing
Image smoothening, Image sharpening, generation of spatial masks from frequency domain
specification, Homomorphic filtering, and color image processing, color model – RGB Colormodel and
HIS color model.

UNIT IV- IMAGE RESTORATION AND RECONSTRUCTION OF MEDICAL IMAGE**9**

Image degradation models, Unconstrained and Constrained restoration, mean filters, inverse
filtering, Band pass filter, Band reject filter, Least mean square filter, Image reconstruction
from projections–Radon transforms, Filter back projection algorithm, Fourier slice theorem,
Fourier reconstruction of MRI Images.

UNIT V- MEDICAL IMAGE COMPRESSION TECHNIQUES**9**

Run length, Huffman coding, arithmetic coding, Golomb coding, Symbol based coding, Pixel
coding, transform coding, JPEG Standard, predictive techniques, Wavelet coding. Application
of image processing techniques in thermography, SPECT, PET images.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Rafael C., Gonzalez and Richard E. Woods	Digital Image Processing	Pearson Education Asia	2017
2	Anil K. Jain	Fundamentals of Digital Image Processing	Prentice Hall of India	2015

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	William K. Pratt	Digital Image Processing	John Wiley	2010
2	S.Sridhar	Digital Image processing	Oxford University press	2016

WEBSITES:

- <https://nptel.ac.in/courses/102105090>
- <https://nptel.ac.in/courses/108105091>
- <https://ocw.mit.edu/courses/hst-582j-biomedical-signal-and-image-processing-spring-2007/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To acquire knowledge on fundamental concepts of biomechanics
- To learn in deriving the mathematical models related to blood vessels
- To understand about musculoskeletal mechanics
- To learn about the mechanics of joints.
- To represent mathematical models in biomechanical systems
- To analyze mechanics of physiological systems and related applications

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Illustrate the mechanics of physiological systems
- Explain the use of mechanics in medicine
- Develop various bio mechanic models
- Distinguish the reason for cardiac and respiratory abnormal patterns
- Analyze the biomechanical systems using mathematical models
- Design and develop the models specific to ergonomics.

UNIT I- INTRODUCTION TO MECHANICS**12**

Introduction – Scalars and vectors, Statics – Force types, Resolution and composition of forces, Moments of force and couple, Resultant force determination, parallel forces in space, equilibrium coplanar forces, Dynamics, Basic principles – Linear motion, Newton's laws of motion, Impulse and Momentum, Work and Energy Kinetics – Velocity and acceleration, Kinematics – Link segment models, Force transducers, Force plates, Introduction to Constitutive equations –Constitutive equations of Non viscous fluid, Newtonian Viscous fluid and Hookean Elastic solid

UNIT II- MUSCULOSKELETAL MECHANICS**12**

Bone structure and composition, Mechanical properties of bone: Electrical properties of bone, fracture mechanism and crack propagation in bones, fracture fixators, repairing of bones. Muscle structure, Muscle Actions - Mechanical Methods of Muscle Action Analysis, Tissue loads, Response of tissue to forces, Biomechanics of passive Muscle - Tendon units, ligament, Mechanical characteristics of muscles - Stretch shortening cycles, Force -Time Principle, Gait Analysis.

UNIT III- CARDIOVASCULAR AND RESPIRATORY MECHANICS**12**

Hook's law, Newtonian Fluid, Non-Newtonian fluid-. Blood flow: Laminar and Turbulent - Haematology and Blood Rheology - Relationship between diameter, velocity and pressure of blood flow - Resistance against flow. Mechanical properties of blood vessels - Arteries, arterioles, capillaries, veins, vascular graft-Prosthetic heart valves- TAH - heart valve dynamics. Interaction of blood and lung-Alveoli mechanics, P-V curve of lung - Airway resistance - Physics of lung diseases

UNIT IV- BIOMECHANICS OF JOINTS**12**

Skeletal joints, forces and stresses in human joints, Analysis of rigid bodies in equilibrium, Freebody diagrams, Structure of joints, Types of joints, Biomechanical analysis of elbow, shoulder, spinal column, hip, knee and ankle, Lubrication of synovial joints, Gait analysis, Motion analysis using video.

UNIT V- MODELING AND ERGONOMICS**12**

Introduction to Finite Element Analysis, finite element analysis of lumbar spine; Ergonomics –Musculoskeletal disorders, Ergonomic principles contributing to good workplace design , Process of seeing – visual capabilities-factors affecting visual acuity and contrast sensitivity – human factor aspects of hard copy text and computer screen text Whole body vibrations, Hand transmitted vibrations.

Total periods : 60**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Y.C. Fung	Bio-Mechanics-Mechanical Properties of Tissues	Springer-Verlag	1998
2	Duane Knudson	Fundamentals of Biomechanics, Second Edition	Springer	2007

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Sheraz S. Malik and Shahbaz S. Malik	Orthopaedic Biomechanics Made Easy	Cambridge University Press	2015
2	Jay D. Humphrey, Sherry De Lange	An Introduction to Biomechanics: Solids and Fluids, Analysis and Design	Springer Science Business Media	2004
3	Shrawan Kumar	Biomechanics in Ergonomics, Second Edition	CRC Press	2007
4	Neil J. Mansfield	Human Response to Vibration	CRC Press	2005

WEBSITES:

- <https://nptel.ac.in/courses/112105305>
- <https://nptel.ac.in/courses/112106248>
- <https://ocw.mit.edu/courses/hst-021-musculoskeletal-pathophysiology-january-iap-2006/pages/lecture-notes/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To understand the medical equipment used in the measurement of cardiology and neurology
- To learn some of the cardiac assist devices.
- To understand the principle of biotelemetry
- To gain knowledge about the diathermy
- To study special equipment diagnostics
- To learn about patient safety design metrics

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Apply medical devices applied in measurement of parameters related to cardiology and neurology
- Analyze cardiac assist devices, its continuous monitoring and transmission
- Measure and analyze signals generated by muscles
- Identify the electrical hazards and Implement methods of patient safety
- Perform continuous monitoring and transmission of vital parameters.
- Comprehend the need for special diagnostic and therapeutic devices

UNIT I- CARDIAC EQUIPMENT**9**

Electrocardiograph, Normal and Abnormal Waves, Heart rate monitor, Holter Monitor, Phonocardiography, Plethysmography. Cardiac Pacemaker- Internal and External Pacemaker– Batteries, AC and DC Defibrillator- Internal and External

UNIT II- NEUROLOGICAL EQUIPMENT**9**

Clinical significance of EEG, Multi channel EEG recording system, Sleep patterns, Epilepsy, Evoked Potential–Visual, Auditory and Somatosensory, MEG (Magneto Encephalo Graph). EEG Bio Feedback Instrumentation, Psychophysiological Measurements for testing sensory Responses.

UNIT III- SKELETAL MUSCULAR EQUIPMENT**9**

Generation of EMG, recording and analysis of EMG waveforms, fatigue characteristics, Muscle stimulators, nerve stimulators, Nerve conduction velocity measurement, EMG Bio Feedback Instrumentation.

UNIT IV- DIATHERMY**9**

Short wave diathermy, ultrasonic diathermy, Microwave diathermy, Electro surgery machine
Current waveforms, Tissue Responses, Electro surgical current level, Hazards and safety
procedures.

UNIT V- SPECIAL DIAGNOSTIC TECHNIQUES & PATIENT SAFETY**9**

Lithotripsy, Principles of Cryogenic technique and application, Need for heart lung machine,
Endoscopy, Laproscopy., ophthalmic instruments. Physiological effects of electricity –
important susceptibility parameters – Macro shock – Micro shock hazards – Patient's
electrical environment –Electrical safety codes and standards.

Total periods : 45**TEXT BOOK:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Khandpur R.S	Handbook of Biomedical Instrumentation	Tata McGraw Hill	2003
2	John G. Webster	Medical Instrumentation Application and Design	John Wiley and Sons	2006

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Myer Kutz	Standard Handbook of Biomedical Engineering & Design	Mc Graw Hill	2003
2	Leslie Cromwell	Biomedical Instrumentation and Measurement	Pearson Education	2007
3	Antony Y.K.Chan	Biomedical Device Technology, Principles and design	Charles Thomas Publisher Ltd	2008
4	Joseph J. Carr and John M. Brown	Introduction to Biomedical Equipment Technology	Pearson education	2004

WEBSITES:

- <https://nptel.ac.in/courses/127106136>
- <https://ocw.mit.edu/courses/7-346-rnai-a-revolution-in-biology-and-therapeutics-spring-2010/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To Introduce the practices followed in hospital administration
- To gain knowledge about Human resource management
- To understand national and international policies related to hospital management
- To understand about working of hospital information systems
- To learn the quality and safety aspects in hospital.
- To learn about health insurance schemes

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain the principles, practices and areas of application in Hospital Management.
- Demonstrate about human resource management in hospital
- Interpret market related research processes in healthcare and hospital sectors.
- Discuss the role of hospital system and hospital information system.
- Illustrate the quality and safety aspects in hospital.
- Explain the importance of supportive services

UNIT I- OVERVIEW OF HOSPITAL ADMINISTRATION**9**

Distinction between Hospital and Industry, Challenges in Hospital Administration – Hospital Planning- Equipment Planning – AMC-Functional Planning - Current Issues in Hospital Management–Telemedicine - Bio-Medical Waste Management.

UNIT II- HUMAN RESOURCE MANAGEMENT IN HOSPITAL**9**

Principles of HRM – Functions of HRM – Profile of HRD Manager – Tools of HRD –Human Resource Inventory – Manpower Planning. Different Departments of Hospital, Recruitment, Selection, Training Guidelines –Methods of Training – Evaluation of Training – Leadership grooming and Training, Promotion – Transfer, Communication – nature, scope, barriers, styles and modes of communication.

UNIT III- MARKETING RESEARCH PROCESS**9**

Marketing information systems - assessing information needs, developing & disseminating information - Market Research process - Other market research considerations – Consumer Markets & Consumer Buyer Behavior - Model of consumer behavior - The buyer decision process Model of business buyer behavior – Major types of buying situations - WTO and its implications.

UNIT IV- HOSPITAL INFORMATION SYSTEMS & SUPPORTIVE SERVICES 9

Management Decisions and Related Information Requirement - Clinical Information Systems - Administrative Information Systems - Support Service Technical Information Systems –Medical Transcription, Medical Records Department – Central Sterilization and Supply Department– Pharmacy– Food Services - Laundry Services.

UNIT V- QUALITY AND SAFETY ASPECTS IN HOSPITAL 9

Quality system – Elements, implementation of quality system, Documentation, Quality auditing, International Standards ISO 9000 – 9004 – Features of ISO 9001 – ISO 14000 – ISO 13485- Environment Management Systems. NABH, JCI, NABL, NABA. Security – Loss Prevention – Fire Safety – Alarm System – Safety Rules. Health Insurance & Managing Health Care – Medical Audit – Hazard and Safety in a hospital Setup.

Total periods : 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	R.C.Goyal,	Hospital Administration and Human Resource Managementl	PHI – Fourth Edition	2006
2	G.D.Kunders	Hospitals – Facilities Planning and Management – TMH, New Delhi	Fifth Reprint	2007

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Cesar A.Caceres and Albert Zara	The Practice of Clinical Engineering	Academic Press, NewYork	1977
2	Norman Metzger	Handbook of Health Care Human Resources Management	2nd edition Aspen Publication Inc. Rockville, Maryland, USA	1990
3	Peter Berman	Health Sector Reform in Developing Countries	Harvard University Press	1995
4	William A. Reinke	Health Planning For Effective Management	Oxford University Press	1988
5	Blane, David, Brunner	Health and SOCIAL Organization: Towards a Health Policy for the 21 st Century	Eric Calrendon Press	2002

WEBSITES:

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| <ul style="list-style-type: none">• https://nptel.ac.in/courses/110104095• https://ocw.mit.edu |
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COURSE OBJECTIVES:

The goal of this course is for students:

- To explain the need for medical aids.
- To understand the sensory rehabilitation systems.
- To learn the use of the orthopaedic prosthetics and orthotics in rehabilitation.
- To have an understanding of rehabilitation medicine and advocacy
- To gain knowledge about the prosthetic devices

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain the terminologies used by the rehabilitation team
- Demonstrate Engineering Concepts in Sensory & Motor rehabilitation.
- Design and develop different sensory assist devices, orthotics and prosthetics for rehabilitation applications.
- Design specific Hearing aids and visual aids and assistive devices
- Understand the need of virtual reality based rehabilitation
- Appreciate the legal aspects for building rehabilitation aids for the needed people.

UNIT I- INTRODUCTION TO REHABILITATION**9**

What is Rehabilitation, Epidemiology of Rehabilitation, Health, Levels of Prevention, Preventive Rehabilitation, Diagnosis of Disability, Functional Diagnosis, Importance of Psychiatry in Functional diagnosis, Impairment disability handicap, Primary & secondary Disabilities, Rehabilitation team Classification of members, Need for proper delivery of rehabilitation care, Community based rehabilitation and its aspects.

UNIT II- PRINCIPLES OF REHABILITATION**9**

Introduction, The Human Component, Principles of Assistive Technology Assessment, Principles of Rehabilitation Engineering- Key Engineering Principles, Key Ergonomic Principles Practice of Rehabilitation and Assistive Technology.

UNIT III- THERAPEUTIC EXERCISE TECHNIQUE**9**

Co-ordination exercises, Frenkels exercises, Gait analyses-Pathological Gaits, Gait Training, Relaxation exercises-Methods for training Relaxation, Strengthening exercises-Strength training, Types of Contraction, Mobilization exercises, Endurance exercises.

UNIT IV- MANAGEMENT OF COMMUNICATION & VIRTUAL REALITY**9**

Impairment-introduction to communication, Aphasia, Types of aphasia, Treatment of aphasic patient, Augmentative communication-general form of communication, types of visual aids, Hearing aids, Types of conventional hearing aid, Writing aids. Introduction to virtual reality, Virtual reality based rehabilitation, Hand motor recovery systems with Phantom haptics, Robotics and Virtual Reality Applications in Mobility Rehabilitation.

UNIT V- ORTHOTIC, PROSTHETIC DEVICES & RESTORATION TECHNIQUES**9**

General orthotics, Classification of orthotics-functional & regional, General principles of Orthosis, Calipers- FO, AFO, KAFO, HKAFO. Prosthetic devices: Hand and arm replacement, Body powered prosthetics, Myoelectric controlled prosthetics and Externally powered limb prosthetics. Functional Electrical Stimulation systems-Restoration of hand function, restoration of standing and Walking, Hybrid Assistive Systems (HAS).

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Sunder	Textbook of Rehabilitation	Jaypee Brothers Medical Publishers Pvt. Ltd, New Delhi	2007
2	Joseph D. Bronzino	Biomedical Engineering Fundamentals Medical Devices and Systems Tissue Engineering and Artificial Organs	Taylor & Francis	2000
3	Joseph D. Bronzino	The Biomedical Engineering Handbook	CRC press 2 nd edition Volume 1	2000

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Warren E. Finn, Peter G. Lo Presti	Hand book of Neuroprosthetic Methods	Plenum press, New York	2003
2	Rory A Cooper (Editor), Hisaichi Ohnabe (Editor), Douglas A. Hobson (Editor),	An Introduction to Rehabilitation Engineering Series in Medical Physics and Biomedical Engineering	CRC Press	2006

WEBSITES:

- | |
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| <ol style="list-style-type: none">1. https://nptel.ac.in2. https://ocw.mit.edu/ |
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COURSE OBJECTIVES:

The goal of this course is for students :

- To implement fundamental image processing techniques in Biomedical Images.
- To realize enhancement and Transformation of Medical Images.
- To construct and reconstruct images
- To understand the CT images
- To analyse the images
- To implementation using scilab

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- comprehend and appreciate the significance of image
- Illustrate the image enhancement techniques.
- Demonstrate their abilities using filtering techniques
- Implement the image restoration procedures.
- Carry out simulation of image processing system
- Analyse the normal and abnormal patterns
- .

LIST OF EXPERIMENTS:

1. Digital image Fundamentals.
2. Image Enhancement and Transformation.
3. Edge detection and boundary tracing techniques.
4. Removal of noise in medical images.
5. Image compressions.
6. Restoration of CT images.
7. Reconstruction of images.
8. Image Analysis.
9. MATLAB implementation.

22BEBME612 DIAGNOSTIC AND THERAPEUTIC EQUIPMENT LABORATORY 2H-1C**Instruction Hours/week: L:0 T:0 P:2****Marks: Internal:40 External:60 Total:100****End Semester Exam:3 Hours****COURSE OBJECTIVES:**

The goal of this course is for students:

- To understand the concept the ultrasonic diathermy
- To study the control function in diagnostic and therapeutic equipment
- To provide practice on recording and analysis of different Bio potentials
- To understand patient safety procedures
- To learn different non-electrical parameters using various methods.
- To provide practice on circuit debugging

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Measure different bioelectrical signals using various methods
- Assess different non-electrical parameters using various methodologies
- Illustrate various diagnostic and therapeutic techniques.
- Handle therapeutic equipment
- Design ECG amplifier
- Design and simulate by using Lab View

LIST OF EXPERIMENTS:

1. Simulation of ECG – detection of QRS complex and heart rate
2. Study of shortwave and ultrasonic diathermy
3. Study of biotelemetry
4. Electrical safety measurements.
5. Measurement of Respiratory parameters using spirometry.
6. Study of medical stimulator.
7. Study of ESU – cutting and coagulation modes
8. Recording of Audiogram
9. Design of ECG amplifier, recording and analysis using Lab View

SEMESTER VII

B.E Biomedical Engineering

2022-2023

Semester-VII

22BEBME701 ARTIFICIAL ORGANS AND IMPLANTS

3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To discuss the overview of artificial organs & transplants
- To extend the principles of implant design with a case study
- To explain the implant design parameters and solution in use
- To simplify about various blood interfacing implants
- To know the biocompatibility of artificial organs
- To learn about the implantable medical devices

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explain the implant design parameters and solution in use
- Analyze about various blood interfacing implants
- Evaluate response of biomaterials in living system
- Discuss the scope of artificial organs & transplants
- Demonstrate different types of soft tissue replacement and hard tissue replacement
- Assess biocompatibility of artificial organs

UNIT I- ARTIFICIAL ORGANS

9

Artificial blood, Artificial skin, Artificial Heart, Prosthetic Cardiac Valves, Artificial lung (oxygenator), Artificial Kidney (Dialyzer membrane), Dental Implants

UNIT II- IMPLANT DESIGN & MATERIALS

9

Principles of implant design, Clinical problems requiring implants for solution. Metallic implant materials, stainless steels, co-based alloys, Ti-based alloys, ceramic implant materials, aluminum oxides, hydroxyapatite, glass ceramics, carbons, medical applications

UNIT III- IMPLANT DESIGN PARAMETERS AND ITS SOLUTION

9

Biocompatibility, local and systemic effects of implants, Design specifications for tissue bonding and modulus matching, Degradation of devices, natural and synthetic polymers, corrosion, wear and tear, Implants for Bone, Devices for nerve regeneration.

UNIT IV- BLOOD INTERFACING IMPLANTS

9

Neural and neuromuscular implants, heart valve implants, heart and lung assist devices, artificial heart, cardiac pacemakers, artificial kidney- dialysis membrane and artificial blood.

UNIT V- IMPLANTABLE MEDICAL DEVICES AND ORGANS**9**

Gastrointestinal system, Dentistry, Maxillofacial and craniofacial replacement, Soft tissue repair, replacement and augmentation, recent advancement and future directions.

Total periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Kopff W.J	Artificial Organs	John Wiley and sons, New York, 1st edition	1976
2	Park J.B.,	Biomaterials Science and Engineering	Plenum Press	1984

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	J D Bronzino	Biomedical Engineering handbook Volume II	CRC Press / IEEE Press	2000
2	R S Khandpur	Handbook of Biomedical Instrumentation	Tata McGraw Hill	2003
3	Joon B Park	Biomaterials – An Introduction	Plenum press, New York	1992
4	Yannas, I. V	Tissue and Organ Regeneration in Adults	New York, NY: Springer	2001
5	Yadin David, Wolf W. von Maltzahn, Michael R. Neuman, Joseph.D, Bronzino	Clinical Engineering	CRC Press, 1st edition	2010
6	Myer Kutz	Standard Handbook of Biomedical Engineering & Design	McGraw- Hill	2003

WEBSITE:

- <https://nptel.ac.in/courses/113108071>
- <https://ocw.mit.edu/courses/2-782j-design-of-medical-devices-and-implants-spring->

COURSE OBJECTIVES:

The goal of this course is for students:

- To understand the medical device classes and regulatory efforts
- To familiarize the international medical device regulations and standards.
- To know about the patents and intellectual property rights
- To understand Indian medical device regulations
- To explain the global policies on medical device regulations.
- To inspect and investigate national and international device regulation standards

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explain the basic concepts of medical device regulations
- Explore the purpose and types of testing.
- Discuss the FDA regulations and standards
- Apply global policies on medical device regulations.
- Inspect and investigate national and international device regulation standards
- Analyze the way design concepts are imbibed in practical scenarios

UNIT I - DEFINITION OF TESTING**9**

Parsing test requirements, test protocol, test methodology, purpose of the test, failure definition, determining sample size and test length, types of testing. Analysis of test data- failure rate, mean time between failure, reliability, confidence level, confidence limits, minimum life, graphical analysis. Reliability and liability- negligence, strict liability, breach of warranty, defects, plaintiff's conduct, defendant's conduct, defendant related issues, manufacturers and physicians responsibilities, accident reconstruction and forensics.

UNIT II - FOOD AND DRUG ADMINISTRATION**9**

History of device regulation, device classification, registration and listing, 510(k) process, declaration of conformance to a recognized standard, PMA application, investigational device exemptions, good laboratory practices, good manufacturing practices, human factors, design control, FDA and software classification, FDA inspection, advice on dealing with the FDA regulations and standards- definition of medical device, MDD, United States Domestic Standards, rest of the world standards.

UNIT III- INDIAN MEDICAL DEVICE RULES AND REGULATIONS 9

Indian medical device rules and regulations-2017, licensing patents, copyrights and trade secrets, trademarks. Manufacturing and quality control- GMP regulations, design for manufacturability, design for assembly, manufacturing process.

UNIT IV - MISCELLANEOUS ISSUES

9

Learning from failure, design for failure, design for convenience, universal design, design for assembly, prevention through design, design for the environment, pokayoke, product life issues, product testing issues. Product safety and legal issues, accident reconstruction and forensics, biomechanics and traffic-accident investigations. professional issues, BME – related professional societies, standards setting groups, professional engineering licensure, rules of professional conduct, codes of ethics, forensics and consulting, continuing education.

UNIT V- DESIGN OF CASE STUDIES

9

Multidetector brain scanning system development, testing of anesthetists, apnea detection system, cancer clinic charting, EKG analysis techniques & module.

Total Periods: 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Seeram Ramakrishna	Medical Devices Regulations, Standards and Practices	Wood Head Publishing series in Biomaterials, UK	2015

REFERENCE:

S.NO.	Author(s) Name	Title of the book	Publisher	year of publication
1	Val Theisz	Medical Device Regulatory Practices, An International Perspective	CRC Press	2016

WEBSITE:

- <https://nptel.ac.in/courses/127106136>
- <https://nptel.ac.in/courses/127106010>

B.E Biomedical Engineering	2022-2023
	Semester-VII
22BEBME7E--	PROFESSIONAL ELECTIVE-III
Instruction Hours/week: L:3 T:0 P:0	Marks: Internal:40 External:60 Total:100
	End Semester Exam:3 Hours

B.E Biomedical Engineering	2022-2023
	Semester-VII
22BEBME7E--	PROFESSIONAL ELECTIVE-IV
	3H-3C
Instruction Hours/week: L:3 T:0 P:0	Marks: Internal:40 External:60 Total:100
	End Semester Exam:3 Hours

B.E Biomedical Engineering	2022-2023
	Semester-VII
22BEECOE--	OPEN ELECTIVE-1
22BEC SOE--	3H-3C
22BEEEOE--	
Instruction Hours/week: L:3 T:0 P:0	Marks: Internal:40 External:60 Total:100
	End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To Understand concepts around Intelligence and Analytics
- To understand the concept of collaborative filtering
- To understand the concept of R algorithm
- To provide practice on real-life project
- To learn different supervised and unsupervised machine learning techniques.
- To explain analysis of variance.

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explore Recommendation Systems with various functions
- Apply various supervised machine learning techniques
- Perform Analysis of Variance (ANOVA)
- Choose appropriate algorithms for various analysis
- Analyse supervised and unsupervised machine learning techniques
- Distinguish user-based collaborative filtering and Item-based collaborative filtering

LIST OF EXPERIMENTS:

1. Introduction to Data Analytics.
2. Introduction to R Programming.
3. Data Manipulation in R.
4. Data Import Techniques in R.
5. Exploratory Data Analysis.
6. Data Visualization in R.
7. Data Mining: Clustering Techniques.
8. Data Mining: Association Rule Mining & Collaborative filtering.

COURSE OBJECTIVES:

The goal of this course is for students:

- To see the need for developing a holistic perspective of life
- To facilitate the development of a Holistic perspective among students towards life and profession
- To highlight plausible implications of such a Holistic understanding in terms of ethical human conduct
- To sensitise the student about the scope of life – individual, family (inter-personal relationship), society and nature
- To Strengthen self-reflection
- To develop more confidence and commitment to understand, learn and act accordingly

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Build comfortable relationship with each other
- Recall technical education without study of human values can generate more problems than solutions
- Infer that lack of understanding of human values is the root cause of most of the present day problems
- Demonstrate that the natural acceptance (intention) is always for living in harmony
- Differentiate between the characteristics and activities of different orders
- Explain the concept of whole existence confidently

UNIT I- ASPIRATIONS AND CONCERNS

Individual academic – career - Expectations of family – peers - society - nation - Fixing one's goals

UNIT II- SELF MANAGEMENT

Self confidence- peer pressure- time management- anger- stress- Personality development- self improvement

UNIT III- HARMONY OF THE SELF AND BODY

Understanding Human being as the Co-existence of the Self and the Body - Distinguishing between the Needs of the Self and the Body-Mental and physical health -Health issues- healthy diet- healthy lifestyle -Hostel life

UNIT IV- HARMONY IN RELATIONSHIP

Harmony in the Family – the Basic Unit of Human Interaction - Values in Human-to-Human Relationship- gratitude towards parents, teachers and others - Ragging and interaction- Competition and cooperation -Peer pressure

UNIT V- HARMONY IN THE SOCIETY

Understanding Harmony in the Society- Participation in society- Participation in nature/ existence.

Total periods : 15

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	R R Gaur, R Asthana, G P Bagaria,	A Foundation Course in Human Values and Professional Ethics	2nd Revised Edition, Excel Books, New Delhi	2019
2	A.N. Tripathi	Human Values	New Age Intl. Publishers, New Delhi	2004

WEBSITE:

- <http://uhv.ac.in>
- <http://www.uptu.ac.in>
- <http://www.storyofstuff.com>

B.E Biomedical Engineering		2022-2023
		Semester-VII
22BEBME791	PROJECT WORK PHASE I	8H-4C
Instruction Hours/week: L:0 T:0 P:8		Marks: Internal:100 External:0 Total:100
End Semester Exam:3 Hours		

COURSE OBJECTIVES:

The goal of this course is for students:

- To get capable of self education and understand the value of achieving perfection in project implementation & completion
- To apply the theoretical concepts to solve industrial problems
- To enable the students to implement project planning with teamwork and multidisciplinary approach
- To demonstrate professionalism with ethics
- To Understand concepts of Project and financial Management
- To present effective communication skills and relate engineering issues to broader societal context

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Demonstrate the knowledge, skills and attitudes of a professional engineer
- Undertake problem identification, formulation and solution.
- Apply societal and cultural issues in professional engineering practice
- Demonstrate the skills to use modern engineering tools and software to solve complex engineering problems.
- Function as a member of multidisciplinary team and Communicate with engineers and the community
- comprehend the project management and finance

SEMESTER VIII

B.E Biomedical Engineering		2022-2023
		Semester-VIII
22BEBME8E--	PROFESSIONAL ELECTIVE-V	3H-3C
Instruction Hours/week: L:3 T:0 P:0		Marks: Internal:40 External:60 Total:100
End Semester Exam:3 Hours		

B.E Biomedical Engineering		2022-2023
		Semester-VIII
22BEECOE--		
22BEC SOE—	OPEN ELECTIVE-II	3H-3C
22BEBTOE--		
Instruction Hours/week: L:3 T:0 P:0		Marks: Internal:40 External:60 Total:100
End Semester Exam:3 Hours		

COURSE OBJECTIVES:

The goal of this course is for students:

- To get capable of self education and understand the value of achieving perfection in project implementation & completion
- To apply the theoretical concepts to solve industrial problems
- To enable the students to implement project planning with teamwork and multidisciplinary approach
- To demonstrate professionalism with ethics
- To Understand concepts of Project and financial Management
- To present effective communication skills and relate engineering issues to broader societal context

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Demonstrate the knowledge, skills and attitudes of a professional engineer
- Undertake problem identification, formulation and solution.
- Apply societal and cultural issues in professional engineering practice
- Demonstrate the skills to use modern engineering tools and software to solve complex engineering problems.
- Function as a member of multidisciplinary team and Communicate with engineers and the community
- comprehend the project management and finance

PROFESSIONAL ELECTIVES FOR SEMESTER V

PROFESSIONAL ELECTIVE I

B.E Biomedical Engineering

2022-2023

Semester-V

22BEBME5E01

BIOERGONOMICS

3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To be exposed to principles of ergonomics.
- To develop the mechanics of muscle physiology concepts.
- To understand the anthropometric design principles
- To learn the process of hearing
- To infer the factors in design of work space surfaces
- To familiar with the mathematical models, analysis and design of biomedical devices using case studies.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain the principles of ergonomics.
- Analyse the factors in design of work space surfaces
- Elaborate with the mathematical models, analysis and design of biomedical devices using case studies.
- Apply the principles of good ergonomic design in anthropometry
- Design biomedical devices by applying ergonomics.
- Optimize the design of Medical Equipment.

UNIT I - VISUAL AND AUDITORY ERGONOMICS

9

Process of seeing – visual capabilities-factors affecting visual acuity and contrast sensitivity – human factor aspects of hard copy text and computer screen text, factors in selecting graphic representations symbols, qualitative visual display-process of hearing-principles of auditory display.

UNIT II - MUSCLE PHYSIOLOGY

9

Muscle physiology -muscle metabolism-respiratory response-joint motion study- measure of physiological in- efficiency and energy consumption-work rest cycles-aspects of manual and posture study, material handling (MMH) Bio-mechanical recommended limits of MMH.

UNIT III- CONTROLS AND DISPLAYS

9

Spatial compatibility physical arrangement of displays and controls- movement capability- rotary controls and rotar displays movement of displays orientation of the operator and movement relationships control orders and control responses- human limitations in tracking task.

UNIT IV- ANTHROPOMETRY

9

Anthropometry- anthropometric design principles –work space envelope- factors in design of work space surfaces- principles of seat design –principles of control panel. Organization classification of human errors theories of accident causation-reducing accidents by altering behaviour.

UNIT V- CASE STUDIES

9

Case Study 1: computer design, control panel design of an electronic instrument, computer key board, hand drill etc. Case Study 2: Biomedical Application, Design optimization of Medical Equipments.

Total Periods : 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Pascale Carayon	Handbook of Human Factors and Engineering, Second Edition	CRC Press	2016
2	Robert.N. Bailey	Human Performance Engineering, Third Edition	Prentice Hall	1996

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Shrawan Kumar	Biomechanics in Ergonomics, Second Edition	CRC Press	2007
2	Stephen Pheasant, Christine M. Haslegrave	Body space: Anthropometry Ergonomics and the Design of Work	CRC Press	2016

WEBSITES:

1. <https://nptel.ac.in/>
2. <https://ocw.mit.edu/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To define the mechanisms working of electronic components.
- To demonstrate the design of transmitters with industrial standard.
- To apply concepts of electronics in the various sensor designs.
- To study system concept of data acquisition.
- To familiarize with the electronic conditioning circuits for transducers.
- To perform data acquisition and artificial intelligence techniques for enhancing sensor characteristics.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Describe the different working of electronic components.
- Discuss the concepts of design of transmitters with industrial standard.
- Explain the system concept of electronics in the various sensor designs.
- Design signal conditioning circuits for resistive and capacitive transducers
- Explain the procedure to design conditioning circuits for temperature measuring transducers and design the transmitters for a given measurement application
- Design data acquisition systems and use fuzzy logic and neural networks for sensor linearization.

UNIT I - ANALOG ELECTRONIC CIRCUITS FOR R & C TRANSDUCERS 9

Design of Basic and Advanced Analog Electronic Circuits for Resistive and Capacitive Transducers. Conditioning Circuits for Linearization – Sensitivity Improvement – Offset and Span Adjustments.

UNIT II- ANALOG ELECTRONIC CONDITIONING CIRCUITS 9

Review of Thermocouple, RTD and thermistor characteristics – Review of basic analog electronic conditioning circuits. Importance of linearity – Hardware and software methods for linearization, Importance of sensitivity – design aspects for sensitivity improvement. EMI and EMC design aspects.

UNIT III - REVIEW AND DESIGN OF TRANSMITTERS 9

Review of transmitters – design of two wire and four wire transmitters using analog electronic circuits and IC's.

UNIT IV- DATA ACQUISITION SYSTEM**9**

Introduction to data acquisition system, issues related to interfacing of static and dynamic sensors. Design of data acquisition for a given measurement application.

UNIT V- FUZZY LOGIC AND NEURAL NETWORKS FOR SENSOR LINEARIZATION 9

Introduction to Fuzzy logic and neural networks. Use of Fuzzy logic and neural networks for sensor linearization and improvement of other characteristics.

Total periods: 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	William C. Dunn	Introduction to Instrumentation, Sensors, and Process Control	Artech House	2005
2	Jacob Fraden	Handbook of Modern Sensors: Physics, Designs, and Applications	Springer	1993

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	H.R. Taylor	Data Acquisition for Sensor Systems	Springer	2010
2	Manabendra Bhuyan	Intelligent Instrumentation: Principles and Applications	CRC Press Taylor & Francis Group	2010

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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COURSE OBJECTIVE

The goal of this course is for students:

- To learn about data analytics and different statistical tools related to medical data applications.
- To understand and apply the algorithms in health care applications.
- To learn the principles of statistical learning.
- To have a clear understanding of the algorithms for regression.
- To use new user friendly computational tools.
- To outline algorithms for classification.

COURSE OUTCOMES

Upon completion of this course, the student will be able to:

- Outline the basics of health care data, statistical methods, computational tools, data modelling, supervised and unsupervised learning.
- Apply and compare the regression and classification algorithms.
- Use various data computational tools in health care.
- Synthesis of trends and issues as related to current professional practice in data management.
- Evaluate organizational data management systems.
- Analyse the information competency needed for different applications

UNIT I - INTRODUCTION TO HEALTHCARE DATA**9**

Typical problems of data analysis, descriptive statistics, predictive modelling, health surveys - measure of disease risk and association, standardization of data - Introduction to spreadsheet - importing, coding and manipulating data, writing formulas and linking tables, pivot tables, what-if analysis and data displays.

UNIT II - INTRODUCTION TO STATISTICAL LEARNING**9**

Probability - Bayes Rule - types of distributions - estimation of unknown function, prediction accuracy and model interpretability, supervised versus unsupervised learning, regression versus classification, modeling data, principles of guesstimation.

UNIT III - ALGORITHMS FOR REGRESSION**9**

Linear regression, ridge regression, the lasso, logistic regression, linear discriminant analysis- case studies.

UNIT IV - ALGORITHMS FOR CLASSIFICATION**9**

K-nearest neighbours - splines - generalized additive models, tree-based methods - support vector machines, Random forests, case studies.

UNIT V - COMPUTATIONAL TOOLS**9**

Introduction to R Package: Basic commands, Graphics, indexing data, loading data, time series analysis, graphical multivariate analysis - analytics in clinical trials- predicting the medical devices life cycle- predicting deterioration of patient's condition using EMR.

Total periods: 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Gareth James, Daniela Witten, Trevor Hastie and Robert Tibshirani	An Introduction to Statistical Learning	Springer	2010
2	Trevor Strome	Healthcare Analytics for Quality and Performance Improvement	Hoboken: John Wiley & Sons Inc	2013

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Victor A. Bloomfield	Using R for Numerical Analysis in Science and Engineering	Chapman & Hall	2014
2	Sarah Stowell	Instant R: An Introduction to R for Statistical Analysis	Jotun heim Publishing	2012

WEBSITES:

1. https://nptel.ac.in/ 2. https://ocw.mit.edu/
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COURSE OBJECTIVE

The goal of this course is for students

- To Understand the basic concepts of brain computer interface
- To Study the various signal acquisition methods
- To Learn about the signal processing methods used in BCI
- To Learn the various applications of BCI
- To analyse feature extraction methods
- To explain machine learning methods for BCI

Course Outcomes

Upon completion of this course, students will be able to:

- Evaluate concept of BCI.
- Assign functions appropriately to the human and to the machine.
- Select appropriate feature extraction methods
- Use machine learning algorithms for translation.
- Design classifier for a BCI system.
- Implement BCI for various applications

UNIT I- INTRODUCTION TO BCI**9**

Introduction - Brain structure and function, Brain Computer Interface Types - Synchronous and Asynchronous - Invasive BCI -Partially Invasive BCI - Non Invasive BCI, Structure of BCI System, BCI Monitoring Hardware, EEG, ECoG, MEG, fMRI.

UNIT II- BRAIN ACTIVATION**9**

Brain activation patterns - Spikes, Oscillatory potential and ERD, Slow cortical potentials, Movement related potentials-Mu rhythms, motor imagery, Stimulus related potentials - Visual Evoked Potentials – P300 and Auditory Evoked Potentials, Potentials related to cognitive tasks.

UNIT III- FEATURE EXTRACTION METHODS**9**

Data Processing – Spike sorting, Frequency domain analysis, Wavelet analysis, Time domain analysis, Spatial filtering -Principal Component Analysis (PCA), Independent Component Analysis (ICA), Artefacts reduction, Feature Extraction - Phase synchronization and coherence

UNIT IV- MACHINE LEARNING METHODS FOR BCI**9**

Classification techniques –Binary classification, Ensemble classification, Multiclass Classification, Evaluation of classification performance, Regression - Linear, Polynomial, RBF's, Perceptron's, Multilayer neural networks, Support vector machine, Graph theoretical functional connectivity analysis

UNIT V- APPLICATIONS OF BCI**9**

Case Studies - Invasive BCIs: decoding and tracking arm (hand) position, controlling prosthetic devices such as orthotic hands, Cursor and robotic control using multi electrode array implant, Cortical control of muscles via functional electrical stimulation. Noninvasive BCIs:P300 Mind Speller, Visual cognitive BCI, Emotion detection. Ethics of Brain Computer Interfacing.

Total Periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Rajesh.P.N.Rao,	Brain-Computer Interfacing: An Introduction	Cambridge University Press, First edition	2013
2	Jonathan Wolpaw, Elizabeth Winter, Wolpaw	Brain Computer Interfaces: Principles and practice	Oxford University Press, USA, Edition 1,	2012

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Ella Hassianien, A& Azar.A.T	Brain-Computer Interfaces Current Trends and Applications	Springer	2015
2	Andrew Webb	Statistical Pattern Recognition	Wiley International	2002.

WEBSITE:

1. <https://nptel.ac.in/>
2. <https://ocw.mit.edu/>

PROFESSIONAL ELECTIVE II

B.E Biomedical Engineering

2022-2023

Semester-VI

22BEBME6E01

BIOPHOTONICS

3H-3C

Instruction Hours/week: L:3 T:1 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To provide insights of development of optical devices
- To discuss the concept of different light-tissue interactions.
- To demonstrate the mechanisms of light-tissue interactions.
- To define the different therapeutic applications of light.
- To apply concepts of light in the various fields of diagnostics.
- To study system concept of fiber optic sensors.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Recall the principles of optical devices development.
- Reproduce the concept of different light-tissue interactions.
- Analyze the process and mechanisms of light-tissue interactions
- Describe the different therapeutic applications of light.
- Discuss the concepts of light in the various fields of diagnostics.
- Explain the system concept of fiberoptic sensors.

UNIT I - INTRODUCTION

12

Development of Optical Devices-Historical Events in Therapeutic and Diagnostic Use of Light - Definitions in Optics-Electromagnetic Wave Theory-Electromagnetic Wave Spectrum-Light Sources - Applications of Various Lasers.

UNIT II - LIGHT-TISSUE INTERACTIONS

12

Laser Laser Power- Light Delivery Protocol- Power Density Profile of the Beam- Optical Properties of the Tissue- Thermodynamic Tissue Properties- Light Transportation Theory-Temperature Effects of Light-Tissue Interaction.

UNIT III - LIGHT-TISSUE INTERACTION MECHANISMS AND APPLICATIONS

12

Photo ablation- Photo acoustics- Spectroscopy- Endoscopy Medical Applications Using Endoscopy- Photosynthesis- Photodynamic Therapy- Wound Healing- Photo biological Bio stimulation Photo biological Effects- Optic Nerve-Cell Depolarization under the Influence of Light.

UNIT IV THERAPEUTIC APPLICATIONS OF LIGHT**12**

Pulsed Laser Use in Cardiology- Optical Tweezers- Vascular Welding- Inflammatory Disease Lesion Treatment- Laser Photocoagulation- Lasers in the Treatment of Benign Prostatic Hyperplasia.

UNIT V DIAGNOSTIC APPLICATIONS OF LIGHT& FIBEROPTIC SENSORS**12**

Optical Microscopy-Near-Field Scanning Optical Microscope-Lab-on-a-CHIP Holographic Imaging-Transillumination Imaging-Optical Coherence Tomography-Medical Thermography. Biosensors-Fiberoptic Fluorescence Sensors-Fiber optic Sensors in Gastrointestinal Applications Fiberoptic Immunosensors-Medical Immunosensors

Total periods: 60**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Robert Splinter, Brett A. Hooper	An Introduction to Biomedical Optics	Taylor & Francis, CRC Press, New York	2006
2	Paras N. Prasad	Introduction to Biophotonics	A. John Wiley and sons, Inc. Publications	2003

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Tuan Vo Dinh	Biomedical photonics – Handbook	CRC Press LLC	2003
2	Brian Wilson, Valeriï Viktorovich Tuchin, S. Tanev	Advances in Biophotonics	IOS Press	2005

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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COURSE OBJECTIVES:

The goal of this course is for students:

- To understand the theories and practices adopted in Hospital Information systems.
- To gain knowledge on Health informatics and recent Trends in Medical informatics.
- To illustrate in the importance of medical standards, medical data formats in bioinformatics.
- To understand the basic concepts of medical data storage and automation.
- To appraise the knowledge about health informatics.
- To USE techniques to design modern informatics system.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Recall about hospital information system.
- State medical standards in informatics.
- Explain about storage of medical data, databases used in
- Demonstrate various levels of health informatics.
- Enumerate the functions and importance of health informatics.
- Describe the recent trends in medical informatics.

UNIT I- MEDICAL INFORMATICS**9**

Introduction, Medical Informatics, Bioinformatics, Health Informatics, Structure of Medical Informatics Functional capabilities of Hospital Information System, On-line services and off, line services, History taking by computer, Dialogue with the computer.

UNIT II- MEDICAL STANDARDS**9**

Evolution of Medical Standards, IEEE 11073, HL7, DICOM, IRMA, LOINC, HIPPA, Electronics Patient Records, Healthcare Standard Organizations, JCAHO (Join Commission on Accreditation of Healthcare Organization), JCIA (Joint Commission International Accreditation), Evidence Based Medicine, Bioethics.

UNIT III- MEDICAL DATA STORAGE AND AUTOMATION**9**

Plug in Data Acquisition and Control Boards, Data Acquisition using Serial Interface, Medical Data formats, Signal, Image and Video Formats, Medical Databases, Automation in clinical laboratories, Intelligent Laboratory Information System, PACS, Data mining.

UNIT IV- HEALTH INFORMATICS**9**

Bioinformatics Databases, Bio-information technologies, Semantic web and Bioinformatics, Genome projects, Clinical informatics, Nursing informatics, Public health informatics, Education and Training.

Medical Expert Systems, Virtual reality applications in medicine, Virtual Environment, Surgical simulation, Radiation therapy and planning, Telemedicine, virtual Hospitals, Smart Medical Homes, Personalized e-health services, Biometrics, GRID and Cloud Computing in Medicine.

Total periods: 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	R.D.Lele	Computers in medicine progress in medical informatics	Tata McGraw Hill Publishing computers Ltd, New Delhi,	2005
2	Mohan Bansal	Medical informatics	Tata McGraw Hill Publishing Computers Ltd, NewDelhi	2003

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	N.Mathivanan	PC-Based Instrumentation	Prentice Hall of India Pvt Ltd, New Delhi	2007
2	Orpita Bosu and Simminder Kaur Thukral	Bioinformatics Databases, Tools and Algorithms	Oxford University press, New Delhi	2007

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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COURSE OBJECTIVES

The goal of this course is for students

- To gain knowledge on the principles and application of BioMEMS
- To learn various MEMS fabrication techniques.
- To explain different types of sensors and actuators and their principles of operation at the micro scale level.
- To Know the application of MEMS in different field of medicine
- To comprehend the characteristics of fluid flow and actuation through micro channels.
- To Explain the need and use of CAD for MEMS design

COURSE OUTCOMES

Upon completion of this course, students will be able to:

- Discuss various MEMS fabrication techniques.
- Explain different types of sensors and actuators and their principles of operation at the micro Scale level.
- Analyse the Properties of piezoelectric materials
- Apply MEMS in different field of medicine.
- Acquainted with micro fluidic systems
- Design MEMS devices for different medical applications.

UNIT I - MEMS MATERIALS AND FABRICATION**9**

Typical MEMs and Microsystems, materials for MEMS - active substrate materials- Silicon and its compounds, Silicon piezoresistors, Gallium Arsenide, quartz, polymers. Micromachining photolithography, thin film deposition, doping, etching, bulk machining, wafer bonding, LIGA

UNIT II - . MECHANICAL AND THERMAL SENSORS AND ACTUATORS**9**

Mechanics for MEMs design- static bending of thin plates, mechanical vibration, thermomechanics, fracture and thin film mechanics. Mechanical sensors and actuators – beam and cantilever – microplates, strain, pressure and flow measurements, Thermal sensors and actuators- actuator based on thermal expansion, thermal couples, thermal resistor, Shape memory alloys- Inertia sensor, flow sensor

UNIT III - ELECTROSTATIC AND PIEZOELECTRIC SENSORS AND ACTUATORS**9**

Parallel plate capacitor, pull in effect, Electrostatic sensors and actuators- Inertia sensor, Pressure sensor, flow sensor, tactile sensor, comb drive. Properties of piezoelectric materials, Piezoelectric sensor and actuator – inchworm motor, inertia sensor, flow sensor.

UNIT IV - MICROFLUIDIC SYSTEMS**9**

Fluid dynamics, continuity equation, momentum equation, equation of motion, laminar flow in circular conduits, fluid flow in micro conduits, in sub micro meter and nano scale. Micro scale fluid, expression for liquid flow in a channel, fluid actuation methods, dielectrophoresis, micro fluid dispenser, micro needle, micro pumps-continuous flow system, micro mixers

UNIT V- APPLICATIONS OF BIOMEMS**9**

CAD for MEMs, Drug delivery, micro total analysis systems (MicroTAS) detection and measurement methods, microsystem approaches to polymerase chain reaction (PCR), DNA sensor, MEMS based drug delivery

Total Periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Tai Ran Hsu	MEMS and Microsystems Design and Manufacture	Tata McGraw Hill Publishing Company, New Delhi	2002
2	WanJun Wang, Stephen A.Soper	BioMEMs: Technologies and Applications	CRC Press, New York	2007

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Marc J. Madou	Fundamentals of Microfabrication: the Science of Miniaturization	CRC Press	2002
2	Nadim Maluf, Kirt Williams	An introduction to Microelectro Mechanical Systems Engineering	Second Edition, Artech House Inc, MA,	2004
3	Chang Liu,	Foundations of MEMS	Pearson Education International, New Jersey, USA	2006
4	Nitaigour Premchand Mahalik	MEMS	Tata McGraw Hill Publishing Company, New Delhi	2007

WEBSITES:

1. <https://nptel.ac.in/>
2. <https://ocw.mit.edu/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To relate the the recent trends and need for wearable systems.
- To acquire knowledge on choosing appropriate sensors and signal processing techniques for wearable systems.
- To appraise the knowledge about energy requirement for a wearable system.
- To identify techniques to design modern wearable system.
- To Compare about sensors and signal processing
- To acquire knowledge on wearable devices and its application

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Describe the recent trends and need for wearable systems.
- Choose appropriate sensors and signal processing techniques for wearable systems.
- Infer the energy requirement for a wearable system.
- Analyse and experiment energy harvesting techniques for wearable systems.
- Examine the need for BAN and the challenges involved in the design of BAN.
- Design basic wearable systems for medical applications.

UNIT I - SENSORS**9**

Need for wearable systems, Sensors for wearable systems-Inertia movement sensors, Respiration activity sensor, Inductive plethysmography, Impedance plethysmography, pneumography, Wearable ground reaction force sensor, GSR, Radiant thermal sensor, Wearable motion sensors, CMOS, Based Biosensors, E-Textiles, Bio compatibility.

UNIT II- SIGNAL PROCESSING**9**

Wear ability issues-physical shape and placement of sensor, Technical challenges-sensor design, signal acquisition, Constraint on sampling frequency for reduced energy consumption, light weight signal processing, Rejection of irrelevant information, Data mining.

UNIT III- ENERGY HARVESTING FOR WEARABLE DEVICES**9**

Solar cell, Vibration based, Thermal based, Human body as a heat source for power generation, Hybrid thermoelectric photovoltaic energy harvests, Thermopiles.

UNIT IV - WIRELESS HEALTH SYSTEMS**9**

Need for wireless monitoring, Definition of Body area network, BAN and Healthcare, Technical Challenges- System security and reliability, BAN Architecture, Introduction, Wireless communication techniques.

UNIT V- APPLICATIONS OF WEARABLE SYSTEMS**9**

Medical Diagnostics, Medical Monitoring- Patients with chronic disease, Hospital patients, Elderly patients, Multi parameter monitoring, Neural recording, Gait analysis, Sports Medicine, Smart Fabrics.

Total periods: 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Annalisa Bonfiglio, Danilo De Rossi	Wearable Monitoring Systems	Springer	2014
2	Sandeep K.S. Gupta, Tridib Mukherjee, Krishna Kumar Venkatasubramanian	Body Area Networks Safety, Security, and Sustainability	Cambridge University Press	2013

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Hang, Yuan-Ting	Wearable medical sensors and systems	Springer	2013
2	Mehmet R. Yuce, Jamil Y.Khan	Wireless Body Area Networks Technology, Implementation and Applications	Pan Stanford Publishing Pvt. Ltd	2012

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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PROFESSIONAL ELECTIVES FOR SEMESTER VII

PROFESSIONAL ELECTIVE III

B.E Biomedical Engineering		2022-2023
		Semester-VII
20BEBME7E01	BIOMETRIC SYSTEMS	3H-3C
Instruction Hours/week: L:3 T:0 P:0		Marks: Internal:40 External:60 Total:100
		End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for students:

- To rephrase the relevance of this course to the existing technology
- To infer the technologies of finger print, iris, face and speech recognition.
- To develop the general principles of design of biometric systems and the underlying trade-offs.
- To experiment with personal privacy and security implications of biometrics-based identification technology.
- To inspect issues in the realistic evaluation of biometrics-based systems.
- To explain biometric authentication

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Demonstrate the technologies of finger print, iris, face and speech recognition
- Analyse the general principles of design of biometric systems and the underlying trade-offs.
- Identify personal privacy and security implications of biometrics-based identification technology.
- Develop fingerprint recognition technique.
- Analyse Feature Extraction and Pattern Classification
- Explain the issues in the realistic evaluation of biometrics-based systems

UNIT I - INTRODUCTION TO BIOMETRICS

9

Introduction and background–biometric technologies–passive biometrics–active biometrics– Biometric systems–Enrolment–templates–algorithm–verification–Biometric applications– biometric characteristics–Authentication technologies–Need for strong authentication–Protecting privacy and biometrics and policy–Biometric applications–biometric characteristics

UNIT II - FINGERPRINT TECHNOLOGY

9

History of fingerprint pattern recognition- General description of finger prints- Fingerprint feature processing techniques-fingerprint sensors using RF imaging

techniques–fingerprint quality assessment–computer enhancement and modelling of fingerprint images–fingerprint enhancement –Feature extraction–fingerprint classification–fingerprint matching

UNIT III -FACE RECOGNITION AND HAND GEOMETRY

9

Introduction to face recognition, Neural networks for face recognition–face recognition from correspondence maps–Hand geometry–scanning–Feature Extraction–Adaptive Classifiers- Visual-Based Feature Extraction and Pattern Classification–feature extraction–types of algorithm–Biometric fusion.

UNIT IV- MULTIMODAL BIOMETRICS AND PERFORMANCE EVALUATION

9

Voice Scan–physiological biometrics–Behavioral Biometrics-Introduction to multimodal biometric system–Integration strategies–Architecture–level of fusion–combination strategy–training and adaptability– examples of multimodal biometric systems–Performance evaluation-Statistical Measures of Biometrics–FAR–FRR–FTE– EER–Memory requirement and allocation.

UNIT V- BIOMETRIC AUTHENTICATION

9

Introduction-Biometric Authentication Methods-Biometric Authentication Systems–Biometric authentication by fingerprint-Biometric Authentication by Face Recognition.-Expectation- Maximization theory-Support Vector Machines. Biometric authentication by fingerprint–biometric authentication by hand geometry-Securing and trusting a biometric transaction–matching location–local host-authentication server–match on card (MOC)–Multibiometrics and Two-Factor Authentication

Total Periods : 45

TEXTBOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	JamesWayman, AnilJain, DavideMaltoni	Biometric Systems, Technology Design and Performance Evaluation	Springer	2005
2	S.Y.Kung,S.H.Lin,M. W.Mak	Biometric Authentication: A Machine Learning Approach	Prentice Hall	2005

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Paul Reid	Biometrics for Network Security	Pearson Education	2004
2	Nalini K Ratha, Ruud Bolle	Automatic fingerprint Recognition System	Springer	2003
4	John Chirillo, Scott Blaul	Implementing Biometric Security	John Wiley	2003
5	Arun A. Ross, Karthik Nanda Kumar, Anil K. Jain	Handbook of Multi biometrics	Springer	2006

WEBSITES:

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| <ol style="list-style-type: none">1. https://nptel.ac.in/2. https://ocw.mit.edu/ |
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COURSE OBJECTIVES:

The goal of this course is for students:

- To develop the concepts and law of thermodynamics.
- To analyze the physical properties of the body fluids and cell membrane.
- To apply solute transport in biological systems.
- To infer knowledge about oxygen transport in biological systems and bio artificial organs.
- To explain the Modelling approaches for drug distribution
- To define the factors affecting drug distribution

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Demonstrate the concepts and law of thermodynamics.
- Analyse the physical properties of the body fluids and cell membrane.
- Examine the solute transport in biological systems.
- Extend the knowledge about oxygen transport in biological systems and bio artificial organs.
- Analyse the factors affecting drug distribution
- Develop a model for drug distribution

UNIT I- THERMODYNAMIC CONCEPTS**9**

First law and second law of thermodynamics, properties, fundamental property relations.

UNIT II - PHYSICAL PROPERTIES OF THE BODY FLUIDS AND CELL MEMBRANE**9**

Body fluids and compositions, capillary plasma retention, osmotic pressure, formation of interstitial fluid, net capillary filtration rate, lymphatic system, solute transport across the capillary endothelium, cell membrane and ion pumps. Blood transport: Momentum transport fundamentals, marginal zone theory for blood flow and rheology. Mechanical energy balance (Bernoulli's equation.).

UNIT III- SOLUTE TRANSPORT IN BIOLOGICAL SYSTEMS**9**

Capillary properties, capillary flow rates, solute diffusion-Fick's law, solute transport by capillary function, diffusion with in heterogeneous media, solute permeability, solute transport across the capillary wall, shell balances, Krogh tissue cylinder. Active transport of biological solutes.

UNIT IV - OXYGEN TRANSPORT IN BIOLOGICAL SYSTEMS AND BIOARTIFICIAL ORGANS **9**

Diffusion of oxygen in multicellular systems, haemoglobin, hill equation, tissue oxygenation, oxygen transport in bioartificial organs, oxygen transport in krogh tissue cylinder.

UNIT V- PHARMACOKINETIC ANALYSIS **9**

Modelling approaches, factors affecting drug distribution, drug clearance, model for intravenous injection of drug, accumulation of drug in urine, constant infusion of drug, first order drug absorption and elimination, two compartment model. Extracorporeal devices: haemodialysis and blood oxygenators.

Total Periods : 45

TEXTBOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Fournier R	Basic Transport Phenomena in Biomedical Engineering	3rd Edition, CRC press, Taylor and Francis Group.	2011

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Truskey G.A, Yuan F, Katz D.F	Transport phenomena in biological systems	Pearson Prentice Hall	2009
2	Cooney D.O	Biomedical engineering principles: An Introduction To Fluid, Heat, and Mass Transport Processes	Marcel Dekker, Inc	2000

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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Instruction Hours/week: L:3 T:0 P:0**Marks: Internal:40 External:60 Total:100****End Semester Exam:3 Hours****Course Objectives**

The goal of this course is for students

- To define Biomedical Laser principles and applications.
- To make use of optical properties of tissues
- To infer the knowledge of photonics
- To be take part in Optical Holography
- To explain the various applications of Laser
- To recall photodynamic therapy.

Course Outcomes

Upon completion of this course, students will be able to:

- Analyse the optical properties of tissues
- Utilize the Photonics instrumentation
- Apply lasers in different areas of medicine.
- Choose the lasers in ophthalmology
- Inspect about optical hologram
- Explain the special techniques of Lasers.

UNIT I - OPTICAL PROPERTIES OF THE TISSUES**9**

Refraction, scattering, absorption, light transport inside the tissue, tissue properties, Light interaction with tissues, optothermal interaction, fluorescence, speckles.

UNIT II- INSTRUMENTATION IN PHOTONICS**9**

Instrumentation for absorption, scattering and emission measurements, excitation light sources – high pressure arc lamp, solid state LEDs, optical filters, polarisers, time resolved and phase resolved detectors.

UNIT III- APPLICATIONS OF LASERS**9**

Laser in tissue welding, lasers in dermatology, lasers in ophthalmology, otolaryngology, urology.

UNIT IV - OPTICAL HOLOGRAPHY**9**

Wave fronts, interference patterns, principle of hologram, optical hologram, applications.

UNIT V- SPECIAL TECHNIQUES**9**

Near field imaging of biological structures, in-vitro clinical diagnostic, fluorescent spectroscopy, photodynamic therapy.

Total Periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Leon Goldman, M.D., & R.James Rockwell, Jr.,	Lasers in Medicine	Gordon and Breach, Science Publishers Inc.,	1975
2	Abraham Katzir	Lasers and Optical Fibers in Medicine	Academic Press Edition	1998

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Tuan Vo Dirh	Biomedical Photonics – Handbook	CRC Press, Bocaraton	2003
2	Glasser, O.,	Medical Physics -- Vol 1, 2, 3	Adam Hilgar, Brustol Inc	1987
3	G.David Baxter	Therapeutic Lasers – Theory and practice	Churchill Livingstone Publications	2001

WEBSITES:

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| 1 https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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COURSE OBJECTIVES:

The goal of this course is for students:

- To model heart lung machine and artificial heart
- To analyze some of the cardiac assist devices.
- To explain the need of artificial kidney
- To distinguish about the prosthetic and orthodic devices.
- To summarize the need and use of some respiratory and hearing aids
- To explain Materials for Prosthetic and orthodic devices.

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Demonstrate about heart lung machine and artificial heart
- Explain about cardiac assist devices, its continuous monitoring and transmission
- Explain about prosthetic and orthodic devices
- Interpret the need and use of the extracorporeal devices.
- Compare the types of deafness
- Analyse various materials for Prosthetic and Orthodic devices

UNIT I - HEART LUNG MACHINE AND ARTIFICIAL HEART 9

Condition to be satisfied by the H/L system. Different types of oxygenators, pumps, pulse and continuous types, monitoring process, shunting, the indication for cardiac transplant, driving mechanism, blood handling system, functioning and different types of Artificial heart, mock test setup for assessing its functions.

UNIT II - CARDIAC ASSIST DEVICES 9

Synchronous counter pulsation, assisted through respiration right ventricular by-pass pump, left ventricular bypass pump, open chest and closed chest type, Principle and problems --Intra Aortic balloon pumping, Veno Arterial Pumping, Prosthetic Cardio Valves, Biomaterials for purposes, its characteristics and testing.

UNIT III - ARTIFICIAL KIDNEY 9

Indication and principle of Hemodialysis, Membrane, Dialysate, different types of Hemodialysers, monitoring systems, wearable artificial kidney, implanting type.

UNIT IV - PROSTHETIC AND ORTHODIC DEVICES**9**

Hand and Arm replacement – Different Types of Models Externally Powered Limb Prosthesis Feedback in Orthodic System, functional Electrical Stimulation, Sensory Assist Devices, Materials for Prosthetic and Orthodic devices.

UNIT V - RESPIRATORY AND HEARING AIDS**9**

Intermittent positive pressure, breathing apparatus operating sequence, electronic IPPB unit with monitoring for all respiratory parameters, audiograms , types of deafness, conductive and nervous, hearing aids, constructional and functional characteristics.

Total Periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Levine S.N.(Ed.)	Advances in Biomedical Engineering and Medical Physics	Inter University Publications, New York	1968
2	Kolff W.J.	Artificial Organs	John Wiley and Sons, New York	1979

REFERENCES:

1	Andreas.F. Von racum	“Hand book of Bio material Evaluation,”	Mc.Millan Publishers	1980
2	Albert M.Cook and WebsterJ.G.,	Therapeutic Medical Devices	Prentice Hall Inc., New Jersey	1992
3	R.S.Khandpur	Handbook of Biomedical Instrumentation	Tata McGraw Hill, 2nd Edition	2003

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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PROFESSIONAL ELECTIVES FOR SEMESTER VII

PROFESSIONAL ELECTIVE IV

B.E Biomedical Engineering		2022-2023
		Semester-V
22BEBME7E05	SPEECH PROCESSING	3H-3C
Instruction Hours/week: L: 3 T: 0 P: 0		Marks: Internal:40 External:60 Total:100
End Semester Exam:3 Hours		

COURSE OBJECTIVES

The goal of this course is for students:

- To relate knowledge on speech analysis and different speech modeling procedures.
- To distinguish speech Recognition and Speech Synthesis Methods and its application.
- To analyze the techniques of speech signal processing.
- To infer choice of filters to remove noise and artifacts from biomedical signals.
- To apply established engineering methods to analyse ECG signal problems.
- To extend established engineering methods to analyse neurological signals.

COURSE OUTCOMES

Upon completion of this course, the student will be able to:

- Describe the concept of speech production mechanism and
- Design different types of speech Recognition and Speech Synthesis Methods and its application.
- Utilize different speech signal processing techniques.
- Design algorithms for extracting parameters from the speech signal and implement speech modeling techniques.
- Discuss simple pattern recognition applications of speech processing.
- Assess reproduce the concepts of speech synthesis and its applications.

UNIT I- BASIC CONCEPTS

9

Speech Fundamentals: Articulatory Phonetics, Production and Classification of Speech Sounds; Acoustic Phonetics, Acoustics of speech production; Review of Digital Signal Processing concepts; Short-Time Fourier Transform, Filter-Bank and LPC Methods.

UNIT II- SPEECH ANALYSIS

9

Features, Feature Extraction and Pattern Comparison Techniques: Speech distortion measures, mathematical and perceptual, Log, Spectral Distance, Cepstral Distances, Weighted Cepstral Distances and Filtering, Likelihood Distortions, Spectral Distortion using a Warped Frequency Scale, LPC, PLP and MFCC Coefficients, Time Alignment and Normalization, Dynamic Time Warping, Multiple Time, Alignment Paths.

UNIT III- SPEECH MODELING**9**

Hidden Markov Models: Markov Processes, HMMs, Evaluation, Optimal State Sequence, Viterbi Search, Baum-Welch Parameter Re-estimation, Implementation issues

UNIT IV - SPEECH RECOGNITION**9**

Large Vocabulary Continuous Speech Recognition: Architecture of a large vocabulary continuous speech recognition system, acoustics and language models, n-grams, context dependent sub-word units; Applications and present status.

UNIT V - SPEECH SYNTHESIS**9**

Text-to-Speech Synthesis: Concatenative and waveform synthesis methods, sub-word units for TTS, intelligibility and naturalness, role of prosody, Applications and present status.

Total periods: 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Lawrence Rabiner and Biing-Hwang Juang	Fundamentals of Speech Recognition	Pearson Education	2003
2	Daniel Jurafsky and James H Martin	Speech and Language Processing, An Introduction to Natural Language Processing Computational Linguistics, and Speech Recognition	Pearson Education	2019

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Thomas F Quatieri	Discrete-Time Speech Signal Processing, Principles and Practice	Pearson Education	2004
2	Ben Gold and Nelson Morgan	Speech and audio signal processing, Processing and Perception of Speech and Music	Wiley- India Edition	2006

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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Course Objectives

The goal of this course is for students

- To examine the basics of Embedded System.
- To apply the basics of Application Specific Instruction-Set Processors.
- To compare various Communication Interfaces
- To summarize about Real Time Operating System
- To recall the Need for communication interfaces
- To utilize Real time application

Course Outcomes

Upon completion of this course, students will be able to:

- Analyse various kinds Application Specific Instruction-Set Processors
- Analyse communication interfaces
- Apply the concept of Real Time Operating System in engineering applications
- Analyse various Digital Signal Processors
- Explain the Need for communication interfaces
- Develop Real time application

UNIT I- INTRODUCTION TO EMBEDDED SYSTEM**9**

Introduction, design challenge, processor technology, IC technology, Design technology, Trade-offs, Single purpose processors, custom single purpose processor design and optimization techniques.

UNIT II- GENERAL PURPOSE PROCESSORS**9**

State Machine and Concurrent Process Models: Basic architecture, operation, Pipelining, Programmer's view, development environment, Application Specific Instruction-Set Processors (ASIPs)

UNIT III- MICRO CONTROLLERS AND DIGITAL SIGNAL PROCESSORS**9**

models Vs. languages, finite state machines with data path model (FSMD), using state machines, program state machine model (PSM), concurrent process model, concurrent processes, communication among processes, synchronization among processes, implementation, data flow model, real-time systems

UNIT IV- COMMUNICATION INTERFACES**9**

Need for communication interfaces, RS232 / UART, RS422 / RS485, USB, Infrared, IEEE 1394 Firmwire, Ethernet, IEEE 802.11, Blue tooth.

UNIT V - BASIC CONCEPTS OF REAL TIME OPERATING SYSTEM**9**

ARCHITECTURE OF the Kernel, Tasks and Task scheduler, Interrupt service routines, Semaphores, Mailboxes , Message Queues, Event Registers, Pipes, Signals, Timers, Memory Management, Priority inversion problem.

Total periods : 45**TEXT BOOKS**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Frank Vahid, Tony D. Givargis	Embedded System Design – A Unified Hardware/Software Introduction	John Wiley	2002
2	K.V.K.K. Prasad	Embedded / Real Time Systems	Dreamtech Press	2005

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Sri Ram V Iyer, Pankaj Gupta	Embedded / Real Time Systems	Tata McGraw Hill	2005
2	Steve Heath	Embedded System Design	Elsevier, Second Ed.,	2004

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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Course Objectives

The goal of this course is for students

- To analyse the basic mechanism of robotic subsystems.
- To utilize knowledge on design, analysis and working principle of robotics in medical field.
- To infer knowledge on the principles of management is essential for all kinds of people in all kinds of organizations
- To recall a clear understanding of the managerial functions like planning, organizing, staffing, leading and controlling
- To build an awareness and practice through engineering ethics and human values
- To outline how business ethics works

Course Outcomes

Upon completion of this course, students will be able to:

- Outline the basic concept of robotics and discuss the sensors, actuators, manipulators.
- Explain the working principle underlying in sensors, actuators, manipulators with power source in robotics.
- Distinguish the applications of robotic systems in medical field.
- Utilize philosophical knowledge of the profession of recreation and leisure
- Summarize trends and issues as related to current professional practice
- Illustrate organizational theories and human resource management principles

UNIT I - INTRODUCTION OF ROBOTICS**9**

Introduction to Robotics and its history, Overview of robot subsystems, Degrees of freedom, configurations and concept of workspace, Automation, Mechanisms and movements, Dynamic stabilization- Applications of robotics in medicine

UNIT II - ACTUATORS AND GRIPPERS**9**

Pneumatic and hydraulic actuators, Stepper motor control circuits, End effectors, Various types of Grippers, Design consideration in vacuum and other methods of gripping, PD and PID feedback actuator models

UNIT III- MANIPULATORS & BASIC KINEMATICS**9**

Construction of Manipulators, Manipulator Dynamic and Force Control, Electronic and pneumatic manipulator, Forward Kinematic Problems, Inverse Kinematic Problems, Solutions of Inverse Kinematic problems

UNIT IV- POWER SOURCES AND SENSORS**9**

Sensors and controllers, Internal and external sensors, position, velocity and acceleration sensors,

Proximity sensors, force sensors, laser range finder, variable speed arrangements, Path determination - Machinery vision, Ranging, Laser- Acoustic, Magnetic fiber optic and Tactile sensor

UNIT V- ROBOTICS IN MEDICINE

9

Da Vinci Surgical System, Image guided robotic systems for focal ultrasound based surgical applications, System concept for robotic Tele-surgical system for off-pump CABG surgery, Urologic applications, Cardiac surgery, Neuro-surgery, Pediatric-, and General- Surgery, Gynecologic Surgery, General Surgery and Nano robotics.

Total periods: 45

TEXT BOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Nagrath and Mittal	Robotics and Control	Tata McGraw-Hill	2018
2	Reza N.Jazar	Theory of Applied Robotics Kinematics, Dynamics and Control	Springer, First Indian Reprint	2016

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Spong and Vidhyasagar	Robot Dynamics and Control	John Wiley and Sons, First edition	2015
2	Jacob Rosen, Blake Hannaford & Richard M Satava	Surgical Robotics: System Applications & Visions	Springer	2011

WEBSITES:

1. <https://nptel.ac.in/>
2. <https://ocw.mit.edu/>

B.E Biomedical Engineering		2022-2023
22BEBME7E08	DESIGN OF MEDICAL DEVICES AND IMPLANTS	Semester-VII 3H-3C
Instruction Hours/week: L:3 T:0 P:0		Marks: Internal:40 External:60 Total:100
		End Semester Exam:3 Hours

Course Objectives

The goal of this course is for students

- To construct the Principles of Implant Design.
- To define design parameters
- To inspect the concept of Biocompatibility.
- To classify Dental and Otologic Implants.
- To list the properties of implant materials
- To explain surgical Tools and Implants.

Course Outcomes

Upon completion of this course, students will be able to:

- Analyse design parameters
- Explain the design of Clinical Trials.
- Utilize various Implants.
- Choose materials for design of implants in tissue replacement.
- Compare the Soft Tissue and Hard Tissue Replacement Implants
- Extend knowledge on Surgical Tools and Implants

UNIT I- CLINICAL PROBLEMS REQUIRING IMPLANTS FOR SOLUTION 9

Principles of Implant Design / Design Parameters: Permanent versus Absorbable Devices; The Missing Organ and its Replacement

UNIT II- BIOCOMPATIBILITY 9

Local and Systemic Effects; Design Specifications: Tissue Bonding and Modulus Matching; Degradation of Devices: Natural and Synthetic Polymers; Biocompatibility: Scar Formation and Contraction; Degradation of Devices: Corrosion and Wear; Federal Regulation of Devices I; Oral Presentations of Proposals for Design II; Federal Regulation of Devices II

UNIT III - SCAFFOLDS FOR CARTILAGE REPAIR**9**

Implants for Bone; Implants for Plastic Surgery; Cardiovascular Prostheses: Heart Valves and Blood Vessels; Devices for Nerve Regeneration; Musculoskeletal Soft Tissues: Meniscus, Intervertebral Disk; Dental and Otologic Implants; Other Devices: Spinal Cord, Heart Lung; Final Oral Presentation of Designs (Mock FDA Panel)

UNIT IV- CRITERIA FOR MATERIALS SELECTION**9**

Tissue Engineering I: Scaffolds; Tissue Engineering II: Cells and Regulators Medical Device Classification, Bioethics and Privacy, Biocompatibility and Sterilization Techniques, Design of Clinical Trials, Design Control & Regulatory Requirements.

UNIT V- INTRODUCTION TO SPECIFIC MEDICAL TECHNOLOGIES**9**

Biopotentials measurement (EMG, EOG, ECG, EEG), Medical Diagnostics (In-vitro diagnostics), Medical Diagnostics (Imaging), Minimally Invasive Devices, Surgical Tools and Implants.

Total Periods : 45**TEXT BOOK:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Kopff W.J	Artificial Organs	John Wiley	1976
2	Park J.B.	Biomaterials Science and Engineering	Plenum Press	1984

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	J D Bronzino,	Biomedical Engineering handbook	CRC Press	2000
2	Yadin David, Wolf W. von Maltzahn, Michael R. Neuman,	Clinical Engineering	CRC Press	2010
3	C. Ross Ethier, Craig A Simmons	Introduction to Biomechanics From Cells to Organisms	Cambridge Texts in Biomedical Engineering	2007

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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PROFESSIONAL ELECTIVES FOR SEMESTER VIII

PROFESSIONAL ELECTIVE V

B.E Biomedical Engineering

2022-2023

22BEBME8E01 VIRTUAL REALITY AND AUGMENTED REALITY 3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

Course Objectives

The goal of this course is for students

- To outline the fundamental aspects, principles of virtual reality technology.
- To infer knowledge about applications of virtual reality.
- To show the relevance of this course to the existing technology through demonstrations and applications.
- To relate virtual reality, augmented reality and using them to build Biomedical engineering applications
- To find the intricacies of these platform to develop PDA applications with better optimality

Course Outcomes

Upon completion of this course, students will be able to:

- Understand the basic concepts of Virtual reality.
- Infer the importance of virtual reality.
- Illustrate the significance Virtual reality in present scenario.
- Analyse VR on the mobile and VR on the web.
- Rephrase various modeling concepts.
- Develop the Virtual Reality applications in different areas.

UNIT I - INTRODUCTION

9

The three I's of virtual reality-commercial VR technology and the five classic components of a VR system - Input Devices: (Trackers, Navigation, and Gesture Interfaces): Three-dimensional position trackers, navigation and manipulation-interfaces and gesture interfaces-Output Devices: Graphics displays-sound displays & haptic feedback..

UNIT II - VR DEVELOPMENT PROCESS

9

Geometric modeling - kinematics modeling- physical modeling - behaviour modeling - model Management.

UNIT III - CONTENT CREATION CONSIDERATIONS FOR VR**9**

Methodology and terminology-user performance studies-VR health and safety issues-Usability of virtual reality system- cyber sickness -side effects of exposures to virtual reality environment

UNIT IV - VR ON THE WEB & VR ON THE MOBILE**9**

JS-pros and cons-building blocks (WebVR, WebGL, Three.js, device orientation events)-frameworks (A-frame, React VR)-Google VR for Android-Scripts, mobile device configuration, building to android-cameras and interaction-teleporting-spatial audio-Assessing human parameters-device development and drivers-Design Haptics

UNIT V - APPLICATIONS**9**

Medical applications-military applications-robotics applications- Advanced Real time Tracking other applications- games, movies, simulations, therapy.

Total Periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	C. Burdea & Philippe Coiffet	Virtual Reality Technology	Second Edition, Gregory, John	2008
2	Jason Jerald	. The VR Book: Human-Centred Design for Virtual Reality. Association for Computing Machinery and Morgan & Claypool	New York, NY, US	-

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Dieter Schmalstieg & Tobias Hollerer	Augmented Reality: Principles and Practice (Usability)	Pearson Education (US), Addison-Wesley Educational Publishers Inc, New Jersey, United States	2016
2	Steve Aukstakalnis,	Practical Augmented Reality: A Guide to the Technologies, Applications, and Human Factors for AR and VR (Usability)	Addison-Wesley Professional 1 edition,	2016

3	Robert Scoble & Shel Israel	The Fourth Transformation: How Augmented Reality & Artificial Intelligence Will Change Everything	Patrick Brewster, Press	2016
4	Tony Parisi,	Learning Virtual Reality: Developing Immersive Experiences and Applications for Desktop, Web, and Mobile	O'Reilly Media; 1 edition	2015
5	Tony Parisi	Programming 3D Applications with HTML5 and WebGL: 3D Animation	O'Reilly Media; 1 edition	2014
6	Jos Dirksen	Learning Three.js: The JavaScript 3D Library for WebGL	Packt Publishing - ebooks Account; 2nd Revised ed. Edition	2015

WEBSITE:

1. <https://nptel.ac.in/>
2. <https://ocw.mit.edu/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To infer Smart Objects and IoT Architectures.
- To demonstrate about various IOT-related protocols.
- To summarize about design and development of IoT systems.
- To build simple IoT Systems using Arduino and Raspberry Pi.
- To define data analytics and cloud in the context of IoT
- To develop IoT infrastructure for popular applications

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Explain the concept of IoT.
- Analyse various protocols for IoT.
- Build simple IoT Systems using Arduino and Raspberry Pi.
- Apply data analytics and use cloud offerings related to IoT.
- Analyse applications of IoT in real time scenario.
- Utilize concepts of design and development of IoT systems.

UNIT I - FUNDAMENTALS OF IoT**9**

Evolution of Internet of Things - Enabling Technologies – IoT Architectures: oneM2M, IoT World Forum (IoT WF) and Alternative IoT models – Simplified IoT Architecture and Core IoT Functional Stack – Fog, Edge and Cloud in IoT – Functional blocks of an IoT ecosystem – Sensors, Actuators, Smart Objects and Connecting Smart Objects

UNIT II- IoT PROTOCOLS**9**

IoT Access Technologies: Physical and MAC layers, topology and Security of IEEE 802.15.4, 802.15.4g, 802.15.4e, 1901.2a, 802.11ah and LoRa WAN – Network Layer: IP versions, Constrained Nodes and Constrained Networks – Optimizing IP for IoT: From 6LoWPAN to 6Lo, Routing over Low Power and Lossy Networks – Application Transport Methods: Supervisory Control and Data Acquisition – Application Layer Protocols: CoAP and MQTT

UNIT III - DESIGN AND DEVELOPMENT**9**

Design Methodology - Embedded computing logic - Microcontroller, System on Chips - IoT system building blocks - Arduino - Board details, IDE programming - Raspberry Pi - Interfaces and Raspberry Pi with Python Programming.

UNIT IV- DATA ANALYTICS AND SUPPORTING SERVICES**9**

Structured Vs Unstructured Data and Data in Motion Vs Data in Rest – Role of Machine Learning – No SQL Databases – Hadoop Ecosystem – Apache Kafka, Apache Spark – Edge Streaming Analytics and Network Analytics – Xively Cloud for IoT, Python Web Application Framework –Django – AWS for IoT – System Management with NETCONF-YANG

UNIT V- CASE STUDIES/INDUSTRIAL APPLICATIONS**9**

Cisco IoT system - IBM Watson IoT platform – Manufacturing - Converged Plant wide Ethernet Model (CPwE) – Power Utility Industry – Grid Blocks Reference Model - Smart and Connected Cities: Layered architecture, Smart Lighting, Smart Parking Architecture and Smart Traffic Control

Total Periods : 45**TEXTBOOK:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	David Hanes, Gonzalo Salgueiro, Patrick Grossetete, Rob Barton and Jerome Henry	IoT Fundamentals: Networking Technologies, Protocols and Use Cases for Internet of Things	Cisco Press	2017

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Arshdeep Bahga, Vijay Madiseti	Internet of Things – A hands-on approach	Universities Press	2015
2	Olivier Hersent, David Boswarthick, Omar Elloumi	The Internet The Internet of Things – Key applications and Protocols	Wiley	2012(for Unit 2).
3	Vlasios Tsiatsis, Catherine ulligan, Stamatis, Karnousko Stefan Avesand, David Boyle	From Machine-to-Machine to the Internet of Things - Introduction to a New Age of Intelligence		2014
4	Dieter Uckelmann, Mark Harrison, Michahelles, Florian (Eds),	Architecting the Internet of Things	Springer	2011
5	Michael Margolis	Cookbook, Recipes to Begin, Expand, and Enhance Your Projects-2nd Edition	O'Reilly Media	2011

WEBSITES:

1. <https://nptel.ac.in/>
2. <https://ocw.mit.edu/>

COURSE OBJECTIVES:

The goal of this course is for students:

- To apply the theories and practices adopted in Hospital Information systems.
- To build knowledge on Health informatics and recent Trends in Medical informatics.
- To illustrate the importance of medical standards, medical data formats in bioinformatics.
- To recall the basic concepts of medical data storage and automation.
- To examine the knowledge about health informatics.
- To make use of techniques to design modern informatics system.

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Recall about hospital information system.
- Outline medical standards in informatics.
- Explain about storage of medical data, databases used in hospital sectors.
- Compare various levels of health informatics.
- Summarize the functions and importance of health informatics.
- Examine the recent trends in medical informatics.

UNIT I- INTRODUCTION**9**

Introduction – Invention and Creativity – Intellectual Property (IP) – Importance – Protection of IPR – Basic types of property (i) Movable Property (ii) Immovable Property and (iii) Intellectual Property.

UNIT II - PATENTS, COPYRIGHTS AND TRADEMARKS**9**

IP – Patents – Copyrights and related rights – Trade Marks and rights arising from Trademark registration – Definitions – Industrial Designs and Integrated circuits – Protection of Geographical Indications at national and International levels – Application Procedures.

UNIT III- INTERNATIONAL STANDARDISATION**9**

International convention relating to Intellectual Property – Establishment of WIPO – Mission and Activities – History – General Agreement on Trade and Tariff (GATT).

UNIT IV - INDIAN STRATEGIES**9**

Indian Position Vs WTO and Strategies – Indian IPR legislations – commitments to WTO-Patent Ordinance and the Bill – Draft of a national Intellectual Property Policy – Present against unfair competition.

UNIT V - CASE STUDIES**9**

Case Studies on – Patents (Basmati rice, turmeric, Neem, etc.) – Copyright and related rights – Trade Marks – Industrial design and Integrated circuits – Geographic indications – Protection against unfair competition.

Total Periods : 45**TEXT BOOKS:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Subbaram N.R	Handbook of Indian Patent Law and Practice	S. Viswanathan, Printers and Publishers Pvt. Ltd	1998

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Eli Whitney	United States Patent Number: 72X	Cotton Gin	March 14, 1794
2	Derwent IP Matters	Using the Internet for non-patent prior art searches	-	July 2000.

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| <ol style="list-style-type: none">1. https://nptel.ac.in/2. https://ocw.mit.edu/ |
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COURSE OBJECTIVES:

The goal of this course is for students:

- To illustrate the basic science behind the properties of materials at nanoscale
- To infer knowledge of Nanotechnology in the field of medicine.
- To inspect the Importance of nanomaterials in drug delivery.
- To explain nanomaterials and Nano systems in Medical Diagnostics and Therapeutics
- To infer nanotechnology in health monitoring systems.
- To explain nanotechnology in health monitoring systems

COURSE OUTCOMES:

Upon completion of this course, the student will be able to:

- Plan the basic science behind the properties of materials at nanoscale
- Utilize of nanotechnology health care and medicine.
- Interpret nanomaterials in drug delivery.
- Describe Biosensors based nano system
- Apply nanomaterials and Nano systems in Medical Diagnostics and Therapeutics.
- Make use of nanotechnology in health monitoring systems.

UNIT I -INTRODUCTION**9**

Cellular Nano machines and the Building Blocks of Life, A New Generation of Nano tools, Importance of various nano materials in health and medicine.

UNIT II- NANOPARTICLES FOR DIAGNOSTICS**9**

Nanoparticles in Medical Diagnostics and Therapeutics, Targeted drug delivery, Magnetic Nanoparticles as Contrast Agents for Medical Diagnosis, Liposome based delivery, Bio Inspired Nano materials for a New Generation of Medicine.

UNIT III- THERAPEUTIC NANO DEVICES**9**

Definition and scope, Synthetic Approaches: top-down versus bottom-up Approaches for Nano therapeutic Device Components, Applications for Nano therapeutic Devices.

UNIT IV-NANOSYSTEMS FOR HEALTHCARE MONITORING – I**9**

Single-Molecule Detection Techniques for Monitoring Cellular Activity at the Nano scale Level, Nano probes, Integrated Cantilever-Based Biosensors for the Detection of Chemical and Biological Entities.

UNIT V -NANOSYSTEMS FOR HEALTHCARE MONITORING – II**9**

Nano pore Methods for DNA Detection and Sequencing, Nano tube Based Membrane Systems, micro/nano fluidic systems for bio-object sorting, single chip electrophoresis system.

Total Periods : 45**TEXT BOOK:**

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Tuan Vo-Dinh	Nanotechnology in Biology and Medicine: Methods, Devices and Applications	CRC press	2006

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Chala Kumar	Bio functionalization of nanomaterials	Wiley	2005
2	Charles Pooles, Frank J. Ownes	Introduction to Nanotechnology	Wiley	2003
3	Bharat Bhushan	Handbook of Nanotechnology	Springer	2003

WEBSITES:

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| 1. https://nptel.ac.in/
2. https://ocw.mit.edu/ |
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LIST OF OPEN ELECTIVES OFFERED BY
DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

B.E Electronics and Communication Engineering **2022-2023**

22BEECOE01 **NEURAL NETWORKS AND ITS APPLICATIONS** **3H-3C**

Instruction Hours/week: L:3 T:0 P:0 **Marks: Internal:40 External:60 Total:100**

End Semester Exam:3 Hours

Course Objectives

The goal of this course for students is :

- To introduce the basic concepts of neural networks and its applications in various domain
- To educate about supervised and unsupervised learning process
- To gain a solid understanding of various neural network model
- To study about annealing technique
- To learn the concepts of Self-Organizing Map (SOM) algorithm
- To understand steps involved in ballistic arm movements.

Course Outcomes

At the end of the course the students will be able to

- Explore the basic concepts of neural networks and its applications in various domains
- Explain about learning process in Neural Networks
- Design using Adaptive Resonance Theory (ART) technique
- Describe steps in annealing process
- Apply knowledge on SOM concepts
- Explain ballistic arm movements.

UNIT I -INTRODUCTION TO NEURAL NETWORKS

Introduction-biological neurons and their artificial models-learning, adaptation and neural network's learning rules-types of neural networks-single layer, multiple layer-feed forward, feedback networks

UNIT II - LEARNING PROCESS

Error- correction learning- memory based learning- hebbian learning-competitive learning- Boltzmann learning-supervised and unsupervised learning-adaptation-statistical learning theory.

UNIT III- PERCEPTION

Single layer Perception-Adaptive filtering-unconstrained Optimization-Least-mean square algorithm- Leaning Curve-Annealing Technique-perception convergence Theorem-Relationship between perception and Baye's Classifier-Back propagation algorithm

UNIT IV- ATTRACTOR NEURAL NETWORK AND ART

Hopfield model-BAM model -BAM Stability-Adaptive BAM -Lyapunov function-effect of gain- Hopfield Design-Application to TSP problem-ART-layer 1-layer 2-orienting subsystem- ART algorithm-ARTMAP.

UNIT V- SELF ORGANIZATION

Self-organizing map-SOM Algorithm-properties of the feature map-LVQ-Hierarchical Vector Quantization. Applications of self-organizing maps: The Neural Phonetic Type Writer Learning Ballistic Arm Movements.

Suggested Readings

1. Simon Haykin Neural Networks and Learning Machines 3rd Edition Pearson/Prentice Hall 2009
2. Satish Kumar Neural Networks: A Classroom Approach TMH 2008
3. Rajasekaran.S, Vijayalakshmi Pai. G.A Neural Networks, Fuzzy Logic and Genetic Algorithms, Synthesis and Applications PHI, New Delhi 2003.
4. Laurene Fausett , Fundamentals of Neural Networks: Architectures, Algorithms, and Applications Pearson/Prentice Hall 1994
5. Wasserman P.D, Neural Computing Theory & Practice Van Norstrand Reinhold,1989.
6. Freeman J.A, S kapura D.M Neural networks, algorithms, applications, and programming techniques AdditionWesley2005.

Web links

- <https://nptel.ac.in/courses/117105084/>
- <https://www.geeksforgeeks.org/adaptive-resonance-theory-art/>

Course Objectives

The goal of this course for students is :

- To provide clear knowledge on evolution of communication systems
- To understand uplink and downlink concepts in mobile phone
- To make aware of mobile communication generations
- To deliver knowledge on wireless communication standards
- To enable students to have a better understanding on launching of satellite
- To study the concept of radar communication

Course Outcomes

At the end of the course the students will be able to

- Understand past, present and future trends in mobile communication.
- Explain how uplink and downlink is done in mobile phone
- Distinguish various standards in use for wireless communication
- Demonstrate some basic application of GPS.
- Appreciate launching mechanism of satellite
- Gain knowledge about RADAR working and its applications

UNIT I - THE EVOLUTION OF ELECTRONIC COMMUNICATION

From smoke signals to smart phones - History of communications: Theoretical Foundations, Development & Applications - Frequencies for communication - Frequency regulations - Overview of communication transmitter and receiver.

UNIT II - MOBILE CELLULAR COMMUNICATIONS

Evolution to cellular networks – Cellular systems generations and standards: 1G, 2G, 3G, 4G - Cellular network components - Components of a mobile phone - setting up a call process - Making a call process - Receiving a call process - Spectrum allocation: Policies and strategies, Role of TRAI.

UNIT III - WIRELESS COMMUNICATION

Introduction - Bluetooth - Infrared communication - IEEE Wireless LANs (Wi-Fi) - IEEE 802.16 (WiMaX) - Future mobile and wireless networks: Introduction to 5G- device to device communication- IoT.

UNIT IV - SATELLITE COMMUNICATION

History of Satellite communication, Basics of Satellites, Types of Satellites, Capacity Allocation - Launch Vehicles and Orbits: Introduction to launching vehicles, Important Orbits, working of rocket, Three Pioneers of Rocketry - Basics of Global Positioning System (GPS) - Applications of GPS.

UNIT V - RADAR & NAVIGATION

Introduction, Radar Block diagram and Operation, Radar Frequencies, Applications of Radar. Navigation Systems: Introduction & methods of navigation, Instrument Landing System, Microwave landing system- Modern Navigation systems.

Suggested Readings

1. S.Haykin, —Communication Systems, 4/e, John Wiley 2007
2. B.P.Lathi, —Modern Digital and Analog Communication Systems, 3/e, Oxford University Press, 2007
3. Rappaport Theodore S - Wireless Communications: Principles and Practice, 2/E, Pearson Education India, 2010
4. Vijay. K. Garg, —Wireless Communication and Networking, Morgan Kaufmann Publishers, 2007.
5. T.Pratt, C. Bostian and J.Allnutt; —Satellite Communications, John Wiley and Sons, Second Edition., 2003
6. M. I .Skolnik —Introduction to Radar Systems, Tata McGraw Hill 2006.
7. Myron Kyton and W.R.Fried —Avionics Navigation Systems, John Wiley & Sons 1997.

WEBSITES:

- | |
|---|
| <ol style="list-style-type: none">1. https://nptel.ac.in/2. https://ocw.mit.edu/ |
|---|

LIST OF OPEN ELECTIVES OFFERED BY
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING

B.E Computer Science and Engineering

2022-2023

22BEC SOE01

INTERNET PROGRAMMING

3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for the students:

- To understand the basics of Internet of Things
- To get an idea of some of the application areas where Internet of Things can be applied
- To understand the middleware for Internet of Things
- To understand the concepts of Web of Things
- To understand the concepts of Cloud of Things with emphasis on Mobile cloud computing
- To understand the IOT protocols

COURSE OUTCOMES:

Upon completion of the course, the students will be able to:

- Identify and design the new models for various applications using IoT
- Explain the underlying architectures and models in IoT.
- Analyse different connectivity technologies for IoT.
- Develop simple applications using Arduino /RaspberryPi.
- Analyze and design different models for network dynamics
- Apply data analytics techniques to IoT.

UNIT I - ARCHITECTURES AND MODELS

9

Introduction to IoT – IoT Architectures – Core IoT Functional Stack, Sensors and Actuators Layer, Communications Network Layer, Applications and Analytics Layer – IoT Data Management and Compute Stack, Fog Computing, Edge Computing, Cloud Computing – Sensors, Actuators, Smart Objects, Sensor networks. Middleware for IoT: Overview – Communication middleware for IoT –IoT Information Security

UNIT II- CONNECTIVITY

9

Communications Criteria – Access Technologies – IP as IoT Network Layer – Business case – Optimization – Profiles and compliances – Application Protocols – Transport Layer – Application Transport Methods.

UNIT III - SYSTEM DEVELOPMENT

9

Design Methodology – Case study – Basic blocks of IoT device – Raspberry Pi – Board, Interfaces, Linux, Setting up, Programming–Arduino–Other IoT Devices.

UNIT IV- DATA ANALYTICS AND IOT SECURITY**9**

Data Analytics for IoT – Big Data Analytics Tools and Technology – Edge Streaming Analytics –Network Analytics Applications. Security history, challenges, variations – Risk Analysis Structures –Application in Operational Environment.

UNIT V - IOT IN INDUSTRY**9**

Manufacturing, Architecture, Protocols – Utilities, Grid Blocks - Smart Cities, Architecture, Use cases– Transportation, Architecture, Use cases.

Total Hours: 45**TEXTBOOKS:**

1. Honbo Zhou, The Internet of Things in the Cloud: A Middleware Perspective, CRC Press, 2013
2. Dieter Uckelmann; Mark Harrison, Florian Michahelles, Architecting the Internet of Things, Springer, 2011
3. David Easley and Jon Kleinberg, Networks, Crowds, and Markets: Reasoning About a Highly Connected World, Cambridge University Press-2010

REFERENCES:

1. Olivier Hersent, Omar Elloumi and David Boswarthick, The Internet of Things: Applications to the Smart Grid and Building Automation, Wiley-2012
2. Olivier Hersent, David Boswarthick, Omar Elloumi, The Internet of Things–Key applications and Protocols, Wiley, 2012

WEBSITES:

https://nptel.ac.in/content/storage2/nptel_data3/html/mhrd/ict/text/106105166/lec1.pdf
<https://nptel.ac.in/courses/106105166/>
<https://nptel.ac.in/courses/108108098/>

COURSE OBJECTIVES

The goal of this course is for the students

- To introduce the basic concepts and techniques of Machine Learning.
- To have a complete understanding of the Supervised and Unsupervised learning techniques
- To study the various probability based learning techniques
- To learn Dimensionality Reduction Techniques.
- To understand Evolutionary Models and Graphical models of machine learning algorithms
- To design appropriate machine learning algorithms for problem solving

COURSE OUTCOMES

Upon completion of this course, the students will be able to:

- Distinguish between, supervised, unsupervised and semi-supervised learning
- Apply the appropriate machine learning strategy for any given problem
- Suggest supervised, unsupervised or semi-supervised learning algorithms for any given problem
- Design systems that uses the appropriate graph models of machine learning
- Modify existing machine learning algorithms to improve classification accuracy/ efficiency
- Analyse and suggest appropriate machine learning approaches for various types of problems

UNIT I- INTRODUCTION**9**

Learning Problems – Perspectives and Issues – Concept Learning – Version Spaces and Candidate Eliminations – Inductive bias – Decision Tree learning – Representation – Algorithm – Heuristic Space Search.

UNIT II- NEURAL NETWORKS AND GENETIC ALGORITHMS**9**

Neural Network Representation – Problems – Perceptrons – Multilayer Networks and Back Propagation Algorithms – Advanced Topics – Genetic Algorithms – Hypothesis Space Search – Genetic Programming – Models of Evaluation and Learning.

UNIT III - BAYESIAN AND COMPUTATIONAL LEARNING**9**

Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model.

UNIT IV- INSTANT BASED LEARNING**9**

K- Nearest Neighbour Learning – Locally weighted Regression – Radial Basis Functions – Case Based Learning.

UNIT V - ADVANCED LEARNING**9**

Learning Sets of Rules – Sequential Covering Algorithm – Learning Rule Set – First Order Rules – Sets of First Order Rules – Induction on Inverted Deduction – Inverting Resolution – Analytical Learning – Perfect Domain Theories – Explanation Base Learning – FOCL Algorithm – Reinforcement Learning – Task – Q-Learning – Temporal Difference Learning

Total Periods : 45**TEXT BOOKS**

- 1 Michael Bowles, “Machine Learning in Python-Essential Techniques for Predictive Analysis”, Second Edition, Wiley Publication, 2015.
3. Stephen Marsland, “Machine Learning – An Algorithmic Perspective”, Second Edition, Chapman and Hall/CRC Machine Learning and Pattern Recognition Series, 2014.
4. Jason Bell, “Machine learning – Hands on for Developers and Technical Professionals”, First Edition, Wiley, 2014.

REFERENCES

1. Ethem Alpaydin, “Introduction to Machine Learning”, 3e (Adaptive Computation and Machine Learning Series), Third Edition, MIT Press, 2014
2. Tom M Mitchell, “Machine Learning”, First Edition, McGraw Hill Education, 2013.
3. Peter Flach, “Machine Learning: The Art and Science of Algorithms that Make Sense of Data”, First Edition, Cambridge University Press, 2012.

WEBSITES:

<https://nptel.ac.in/courses/106106139/>

<https://ocw.mit.edu/courses/electrical-engineering-and-computer-science/6-867-machine-learning-fall-2006/>

<https://www.dataquest.io/blog/machine-learning-python/>

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

COURSE OBJECTIVES:

The goal of this course is for the students to:

- Understand how blockchain systems (mainly Bitcoin and Ethereum) work
- To securely interact with them
- Design, build, and deploy smart contracts and distributed applications,
- Integrate ideas from blockchain technology into their projects.
- Introduce application areas, current practices, and research activity

COURSE OUTCOMES

Upon completion of the course, the student will be able to:

- Understand the design principles of Bitcoin and Ethereum.
- Understand the Nakamoto consensus.
- Understand the Simplified Payment Verification protocol.
- List and describe differences between proof-of-work and proof-of-stake consensus.
- Interact with a blockchain system by sending and reading transactions.
- Design, build and deploy a distributed application.
- Evaluate the security, privacy, and efficiency of a given blockchain system.

UNIT I- BASICS**9**

Distributed Database, Two General Problem, Byzantine General problem and Fault Tolerance, Hadoop Distributed File System, Distributed Hash Table, ASIC resistance, Turing Complete. **Cryptography:** Hash function, Digital Signature - ECDSA(Elliptic Curve Digital Signature Algorithm), Memory Hard Algorithm, Zero Knowledge Proof.

UNIT II- BLOCKCHAIN**9**

Introduction, Advantage over the conventional distributed database, Blockchain Network, Mining Mechanism, Distributed Consensus, Merkle Patricia Tree, Gas Limit, Transactions and Fee, Anonymity, Reward, Chain Policy, Life of Blockchain application, Soft & Hard Fork, Private and Public blockchain.

UNIT III- DISTRIBUTED CONSENSUS**9**

Nakamoto consensus, Proof of Work, Proof of Stake, Proof of Burn, Difficulty Level, Sybil Attack, Energy utilization, and alternate.

UNIT IV- CRYPTOCURRENCY**9**

History, Distributed Ledger, Bitcoin protocols - Mining strategy and rewards, Ethereum - Construction, DAO(Decentralized Autonomous Organization), Smart Contract, GHOST(Greedy Heaviest Observed Subtree), Vulnerability, Attacks, Sidechain, Namecoin

UNIT V- CRYPTOCURRENCY REGULATION

9

Stakeholders, Roots of Bitcoin, Legal Aspects-Crypto currency Exchange, Black Market, and Global Economy. Applications: Internet of Things, Medical Record Management System, Domain Name Service and future of Blockchain.

Tutorial & Practical: Naive Blockchain construction, Memory Hard algorithm - Hashcash implementation, Direct Acyclic Graph, Play with Go-ethereum, Smart Contract Construction, Toy application using Blockchain, Mining puzzles

Total Hours:45

TEXT BOOK:

1. Arvind Narayanan, Joseph Bonneau, Edward Felten, Andrew Miller, and Steven Goldfeder, "Bitcoin and Cryptocurrency Technologies: A Comprehensive Introduction", Princeton University Press (2016).
2. Andreas M. Antonopoulos, "Mastering Bitcoin: Programming the Open Blockchain" O'Reilly, Media;2 edition(2017).

REFERENCE BOOKS :

1. Andreas M. Antonopoulos, Gavin Wood "Mastering Ethereum: Building Smart Contracts and DApps" O'Reilly Media; 1 edition (13 November 2018).
2. Don Tapscott, Alex Tapscott "Blockchain Revolution: How the Technology Behind Bitcoin and Other Cryptocurrencies" Penguin; 01 edition (10 May 2016)
3. Satoshi Nakamoto, Bitcoin: A Peer-to-Peer Electronic Cash System.
4. Dr. Gavin Wood, "ETHEREUM: A Secure Decentralized Transaction Ledger", Yellow paper.2014.
5. Nicola Atzei, Massimo Bartoletti, and Tiziana Cimoli, A survey of attacks on Ethereum smart contracts.

WEBSITES:

https://swayam.gov.in/nd1_noc20_cs01/preview.
<https://hyperledger.github.io/composer/latest/introduction/introduction.html>.
<https://ethereumbuilders.gitbooks.io/guide/content/en/index.html>.

**LIST OF OPEN ELECTIVES OFFERED BY
DEPARTMENT OF BIOTECHNOLOGY**

B.Tech BIOTECHNOLOGY

2022-2023

22BTBTOE01

BIOREACTOR DESIGN

3H-3C

Instruction Hours/week: L:3 T:0 P:0

Marks: Internal:40 External:60 Total:100

End Semester Exam:3 Hours

Course Objectives:

The goal of this course is for the students to,

- Impart basic knowledge in bioprocess Engineering
- Design the bioreactors for various operations.
- Discuss the principle and working of heat transfer equipments.
- Extend the knowledge in principle of heat transfer inside a bioreactor
- Construct the equipments used in mass transfer operations.
- Illustrate the equipments used in separation process.

Course Outcomes

After completing the course, the students will be able to

- Summarize the basic concepts in bioprocess Engineering.
- Design the bioreactors for various operations.
- Develop the heat transfer equipments for Bioprocess Engineering.
- Construct the equipments used in mass transfer operations.
- Categorize the equipments used in separation process.
- Describe the applications of bioreactors.

UNIT I- INTRODUCTION TO BIOPROCESS ENGINEERING

Introduction – Biotechnology and Bioprocess Engineering- Biologists and Engineers Differ in their approach to research-How Biologists and Engineers work Together- Bioprocesses: Regulatory constraints.

UNIT II- REACTOR DESIGN

Design of Airlift fermentor, Bubble column reactor and Continuous stirred tank reactor.

UNIT III- HEAT TRANSFER EQUIPMENTS

Design of Shell and tube Heat exchanger, Double pipe heat exchanger, long tube vertical evaporator and forced circulation evaporator.

UNIT IV - MASS TRANSFER EQUIPMENTS

Design of Bollmann extractor, fractionating column, packed tower and spray tray absorber

UNIT V- SEPARATION EQUIPMENTS

Design of plate and frame filter press, leaf filter, rotary drum filter, disc bowl centrifuge, rotary drum drier and Swenson –walker crystallizer.

SUGGESTED READINGS

1. James Edwin Bailey, David F.Ollis (2015) Biochemical Engineering Fundamentals, Second Edition. McGraw-Hill Education (India) private limited.
2. DonW.Green, Robert H.Perry (2008).Chemical Engineer Hand book. The Mc Graw-Hill Companies, Inc.
3. Pauline.M.Doran (2015). Bioprocess Engineering Principles Second Edition. Academic Press.

Course Objectives

The goal of this course is for the students to:

- Discuss the scope and importance of food processing.
- Impart basic knowledge in different food processing methods carried out in the food tech companies.
- Explain the methods of food preservation by cooling.
- Tell the concepts of preservation methods for fruits.
- Create deeper understanding on preservation methods for vegetables.
- Extend the brief knowledge in food conservation operations and packaging methodologies.

Course Outcomes

After completing the course, the students will be able to

- Describe the scope and importance of food processing.
- Outline the various processing methods for foods.
- Describe the methods of food preservation by cooling.
- Summarize the preservation methods for fruits and vegetables.
- Extend the knowledge in food conservation operations.
- Understand the types and materials used for packaging.

UNIT I- SCOPEANDIMPORTANCEOFFOODPROCESSING

Properties of food - Physical, thermal, mechanical, sensory. Raw material Preparation -Cleaning, sorting, grading, peeling.

UNIT II- PROCESSINGMETHODS

Heating-BlanchingandPasteurization.Freezing-Dehydration-canning-additives-fermentation-extrusion cooking-hydrostatic pressure cooking-dielectric heating-microwave processing and aseptic processing-Infrared radiation processing-Concepts and equipment used.

UNIT III - FOODCONVERSION OPERATIONS AND FOOD PACKAGING

Sizereduction – Fibrous foods, dry foods andliquidtheoryandfoods–equipments-membraneseparation-filtration-equipment and application. Basic packaging materials, types of packaging, packaging design, packaging for different types of foods, retort pouch packing, costs of packaging and recycling of materials.

UNIT IV – FOOD PRESERVATION BY COOLING

Refrigeration, Freezing-Theory, freezing time calculation, methods of freezing and freezing equipments, freeze drying, freeze concentration, thawing, effect of low temperature on food. Water activity, methods to control water activity.

UNIT V - PRESERVATION METHODS FOR FRUITS AND VEGETABLES

Preprocessing operations- preservation by reduction of water content: drying /dehydration and concentration – chemical preservation – preservation of vegetables by acidification, preservation with sugar-Heat preservation– Food irradiation-Combined preservation techniques.

SUGGESTED READINGS:

1. R. Paul Singh, Dennis R. Heldman (2014). Introduction to food engineering. Academic press.
2. Fellows. (2017). Food processing technology principles and practice, Fourth Edition. Woodhead publishing Ltd.
3. M. A. Rao, Syed S. H. Rizvi, Ashim K. Datta. (2014). Engineering properties of foods. CRC press.
4. B. Sivasankar. (2002). Food processing and preservation. PHI learning Pvt. Ltd.
5. Ranganna, S. (2000). Handbook of canning and aseptic packaging. Tata McGraw-Hill Publishing Company.
6. Sharma, M., Goyal, M. R., & Birwal, P. (Eds.). (2021). Handbook of Research on Food Processing and Preservation Technologies: Volume 5: Emerging Techniques for Food Processing, Quality, and Safety Assurance. CRC Press.

LIST OF OPEN ELECTIVES OFFERED TO OTHER DEPARTMENT

B.E Biomedical Engineering

2022-2023

22BEBMEOE01

HUMAN ANATOMY AND PHYSIOLOGY

3H-3C

Instruction Hours /week:L:3 T:0 P:0

Marks:Internal:40External:60Total:100

EndSemesterExam:3Hours

COURSE OBJECTIVES:

The goal of this course is for students

- To discuss all the organelles of an animal cell and their function.
- To perceive structure and functions of the various types of systems of human body.
- To outline about eye, ear and Endocrine glands of human
- To learn organs and structures involving in system formation and functions.
- To understand all systems in the human body.
- To infer basic understanding of the inter connection of various organ systems in human body

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explain basic structure and functions of cells and its organelles
- Demonstrate about anatomy and physiology of various organ systems
- Compare the functions of eye, ear and Endocrine glands of human
- Explain the interconnect of various organ systems in human body
- Interpret organs and structures involving in system formation and functions.
- Explain special senses in the human body.

UNIT I -CELL

9

Structure of Cell– Organelles and description–Function of each component of the cell– Membrane potential–Action Potential–Generation and Conduction –Electrical Stimulation. Blood Cell–Composition –Origin of RBC–Blood Groups–Estimation of RBC, WBC and Platelet–Tissues and its functions-.Homeostasis - Tissue: Types – Specialized tissues – functions.

UNIT II - CARDIACANDNERVOUSSYSTEM

9

Heart, Major blood vessels– Cardiac Cycle – ECG-Conducting system of heart--importance of blood groups – identification of blood groups- Nervous Control of Heart-Cardiac output–Coronary and Peripheral Circulation–Structure and function of Nervous tissue– Neuron-Synapse-Reflexes-Receptors-Brain-Brainstem-Spinalcord–Reflexaction.

UNIT III - RESPIRATORY SYSTEM AND MUSCULOSKELETAL SYSTEM 9

Physiological aspects of respiration–Trachea and lungs -Exchange of gases–Regulation of Respiration -Disturbance of respiration function -Pulmonary function test-Types of respiration - Oxygen and carbon dioxide transport and acid base regulation-Muscles-tissue-types-structure of skeletal muscle-types of muscle and joints.

UNIT IV - DIGESTIVE, EXCRETORY AND LYMPHATIC SYSTEM 9

Organisation of GI System, Digestion and absorption –Movements of GI tract–Intestine-Liver-Pancreas- Structure of Nephron–Mechanism of Urine formation–Urine Reflex–Skin and Sweat Gland–Temperature regulation, Lymphatic: Parts and Functions of Lymphatic systems– Types of Lymphatic organs and vessels.

UNIT V - EYE, EAR & ENDOCRINE GLANDS 9

Optics of Eye–Retina–Photochemistry of Vision–Accommodation-Neurophysiology of vision–EOG, Physiology of internal ear–Mechanism of Hearing–Auditory Pathway, Hearing Tests–Endocrine-Pituitary and thyroid glands.

Total periods :45

TEXTBOOKS:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	Textbook Equity Edition	Anatomy and Physiology : Volume 2 of 3.	Lulu.com	2014

REFERENCES:

S.NO.	Author(s) Name	Title of the book	Publisher	Year of publication
1	William F. Ganong	Review of Medical Physiology	Mc Graw Hill, New Delhi	2019
2	Arthur C. Guyton,	Text book of Medical Physiology	Elsevier Saunders	2011

WEB SITES:

- <https://dth.ac.in/medical/course.php>
- https://onlinecourses.swayam2.ac.in/cec20_bt19/preview

22BEBMEOE02**ARTIFICIAL ORGANS AND IMPLANTS****3H-3C****Instruction Hours /week:L:3 T:0 P:0****Marks:Internal:40External:60Total:100
EndSemesterExam:3Hours****COURSE OBJECTIVES**

The goal of this course is for students

- To have an overview of artificial organs & transplants
- To describe the principles of implant design with a case study
- To explain the implant design parameters and solution
- To study about various blood interfacing implant
- To study about soft tissue replacement and hard tissue replacement
- To learn about various implants

COURSE OUTCOMES:

Upon completion of this course, students will be able to:

- Explain the implant design parameters and solution in use
- Analyze about various blood interfacing implants
- Evaluate response of biomaterials in living system
- Perceive knowledge about artificial organs & transplants
- Demonstrate different types of soft tissue replacement and hard tissue replacement
- Assess biocompatibility of artificial organs

UNIT I- ARTIFICIAL ORGANS & TRANSPLANTS

ARTIFICIAL ORGANS:-Introduction, Outlook for organ replacements, Design consideration – Evaluation process. TRANSPLANTS:-Overview, Immunological considerations, Blood transfusions, Individual organs –kidney, liver, heart and lung, bone marrow, cornea.

UNIT II- PRINCIPLES OF IMPLANT DESIGN

Principles of implant design - body response to implants, Clinical problems requiring implants for solution, The missing organ and its replacement, Tissue engineering, scaffolds, Biomaterials, Regenerative medicine & Stem cells.

UNIT III - IMPLANT DESIGN PARAMETERS AND ITS SOLUTION

Biocompatibility, Local and systemic effects of implants, Design specifications for tissue bonding and modulus matching, Degradation of devices, natural and synthetic polymers, corrosion, wear and tear, Implants for Bone, Devices for nerve regeneration. Limb prosthesis, Externally Powered limb Prosthesis.

UNIT IV- BLOOD INTERFACING IMPLANTS

Neural and neuromuscular implants, Heart valve implants, heart and lung assist devices, artificial heart, cardiac pacemakers, Prosthetic cardiac valves, Artificial kidney-dialysis membrane and artificial blood.

UNIT V - IMPLANTABLE MEDICAL DEVICES AND ORGANS

Gastrointestinal system, Dentistry, Soft tissue replacement & Hard tissue replacement –sutures, surgical tapes, adhesive, percutaneous implants, internal fracture fixation devices, joint replacements. Maxillofacial and craniofacial replacement, Recent advancement and future directions.

Total periods :45

TEXT BOOKS:

S.NO.	Author(s)Name	Title of the book	Publisher	Year of publication
1	Park J.B	Biomaterials Science and Engineering	Plenum Press	2011

REFERENCES :

S.NO.	Author(s)Name	Title of the book	Publisher	Year of publication
1	J D Bronzino	Biomedical Engineering handbook Volume II	CRC Press	2010
2	RS Khandpur	Hand book of Biomedical Instrumentation	Tata McGraw Hill	2016

WEB SITES:

- <https://ocw.mit.edu/courses/mechanical-engineering/2-782j-design-of-medical-devices-and-implants-spring-2006/>

LIST OF VALUE ADDED COURSE

- Circuit Simulation and PCB designing
- Biosignal Processing using MATLAB
- Data science and Bigdata Analytics
- 3D Printing
- Hands on training on Artificial Organs and Implants.